

SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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1. GUIDE TO THE SUPPLEMENT

The opening sections prove the costate, performance, and observation results used in the main article and report all primary and fixed-work outcomes. The retained sections give the complete application-specific arguments and earlier numerical evidence. A finite-game, finite-grid, or scalar-reference statement retains its stated domain; each continuous-time capital statement uses its own transfer account.

2. ADDITIONAL PROOFS AND NUMERICAL CONTRACTS

2.1. Costate error and action quality

Let $\hat{m}$ maximize $H_{\hat{q}}$ on the stated convex feasible set, and let $e = q - \hat{q}$. Strong concavity and the variational first-order condition at $\hat{m}$ imply

$$H_{\hat{q}}(\hat{m}) - H_{\hat{q}}(m) \geq \frac{\mu}{2} \|m - \hat{m}\|_d^2$$

for every feasible m , including boundary maximizers. Therefore $\|\tilde{m} - \hat{m}\|_d \leq \sqrt{2\delta/\mu}$. Decomposing the true Hamiltonian difference yields

$$\begin{aligned} H_q(m_q) - H_q(\tilde{m}) &\leq \delta - \frac{\mu}{2} \|m_q - \hat{m}\|_d^2 + \langle e, \tilde{m} - m_q \rangle_d \\ &\leq \delta + \|e\|_d \sqrt{2\delta/\mu} + \frac{\|e\|_d^2}{2\mu}. \end{aligned}$$

The last step completes the square in $\|m_q - \hat{m}\|_d$. This is (49). The Hessian of H_q is the negative diagonal matrix with entries $1/(dm_i^2)$ minus the positive semidefinite mean-penalty matrix. It is bounded above by $-\mu I/d$, proving the strong-concavity premise.

For (50), write $Z - g(X) = [Z - \mathbb{E}(Z | X)] + [\mathbb{E}(Z | X) - g(X)]$. The cross term has zero expectation by conditioning on X . Integrability justifies this orthogonal

decomposition. If the conditional Euler costate differs from the continuous policy costate by $b(X)$, that term remains an additional bias; it is not removed by the projection identity. Neither the sampled costate loss nor a successful optimizer return is a uniform upper bound on this bias.

2.2. Cancellation, clipping and finite-population inference

For each candidate, the exact payoff identity used in Proposition 8 consists of a policy production term, its consumption term, its terminal dispersion, and the corresponding reference terms. When two candidates use the identical reference path, subtraction cancels every exact reference term. Comparing the two policy terms with their numerical approximations therefore requires the two policy transfer errors and the statistic-roundoff accounts, but no reference-state transfer error. The implementation retains conservative original statistic budgets even where their reference quadrature components also cancel.

For any real x, y and positive thresholds b_a, b_b ,

$$(|x - y| - b_a - b_b)_+ \leq (|x| - b_a)_+ + (|y| - b_b)_+.$$

Taking expectations proves the summed clipping-tail allowance without requiring independence between policies. Applying the empirical Bernstein inequality to the clipped difference and then to its negative gives the two one-sided bounds. A union bound covers all declared tests. Our allocation counts both directions and every method, profile design and auxiliary evaluation; it does not spend new confidence mass for deterministic transformations such as (58).

For the finite population, draw I with the fixed probabilities p_j , independently of each future innovation path. The pair (I, W) is then independent across paths. Conditional transfer gives

$$\left| \mathbb{E}[X_{a,h}] - \sum_j p_j \{J_{y_j}(a) - J_{y_j}(m_0)\} \right| \leq \sum_j p_j E_{a,h}(y_j).$$

The clipping threshold is the maximum of the profile-specific thresholds; its expectation-tail allowance is bounded by the maximum of their allowances. The

prescribed implementation uses equal probabilities over nine fixed profiles. Both averaging and the final subtraction are interval-enclosed. Random empirical profile frequencies are not substituted for the population weights when calculating deterministic error budgets. They are part of the sampling variation of the final statistic.

For the paired method difference, the same profile is assigned to both policies on each path. Common random numbers across different methods or training seeds do not invalidate a finite-family union bound. They also do not create independent replications of the optimizer. The ten fixed training streams are reported individually and descriptively; no confidence assertion about an unspecified distribution of future hyperparameter choices is inferred.

2.3. Observed-history implementation

Under the observation assumptions, M is a continuous martingale with bracket $\langle M \rangle_t = Qt$. Let $A = \Sigma^\top Q^{-1}$. The matrix $A\Sigma$ is the orthogonal projector onto the row space of Σ , and $nn^\top$ is the projector onto its orthogonal complement. Independence of the private Brownian motion gives

$$\langle AM + nB^0 \rangle_t = (AQA^\top + nn^\top)t = I_{d+1}t.$$

The multivariate Lévy characterization proves the Brownian assertion. Also $\Sigma A = I_d$ and $\Sigma n = 0$, so $\Sigma \tilde{W} = M$. On each decision cell, the action is determined at its left endpoint by the stored internal state and past recovered increments. With that held action, the physical state equation has a unique strong solution because the production drift is globally Lipschitz and bounded. Induction over the cells establishes uniqueness in law for the joint physical and internal process. Driven by the Brownian motion just constructed, it therefore has the same law as the controller originally defined using directly supplied innovations. Applying the specified componentwise clipping to the recovered vector preserves this distributional equivalence.

The proposition uses only private randomization and observed state history; it does not reveal the original null-space economic driver. Randomized controls are admissible in the enlarged filtration. Since the original deterministic performance

envelope holds against all such adapted controls and independent randomization cannot improve an unconstrained expected-payoff supremum beyond the best deterministic randomization realization, the comparison with the original optimum is preserved. The result is not an implementability theorem for discrete noisy state measurements. The exact drift integral is stated explicitly, and errors in approximating it must be treated separately.

2.4. Independent scalar HJB approximation

For $d = 1$, the terminal dispersion penalty vanishes but production remains nonlinear in log capital. The reference solves the same primitive action-box problem, not a Riccati or homothetic reduction. Put $f(y) = \kappa \tanh(By)$ and $v = (\sigma_L^2 + \sigma_C^2)/2$. On a uniform grid of width Δy , define the positive rates

$$\ell_i(m) = \frac{v}{\Delta y^2} + \frac{(m - c - f(y_i))_+}{\Delta y}, \quad r_i(m) = \frac{v}{\Delta y^2} + \frac{(c + f(y_i) - m)_+}{\Delta y}.$$

The implicit Bellman step solves

$$\frac{v_i - v_i^+}{h} + \rho v_i = \max_m \{ \log m + y_i - \zeta m^2/2 + \ell_i(m)(v_{i-1} - v_i) + r_i(m)(v_{i+1} - v_i) \}.$$

For a fixed action vector the coefficient matrix is strictly diagonally dominant with positive diagonal and nonpositive off-diagonal entries. Its inverse is nonnegative. Howard updates maximize over each drift-sign interval; on either interval the positive root of $1/m - \zeta m = p$ is clipped to its feasible endpoints, and the two candidates are compared. This is an independent monotone matrix computation, not a trained critic used as its own reference.

At the two outer boundaries, the prescribed affine asymptotes replace production by its limiting saturation value. If $w(t)$ is the coefficient on initial log capital and $m_0(t)$ the common schedule, their intercept is the numerical integral of the corresponding deterministic discounted flow. The reference reports the grid, time step, boundary distance, linear-system residual, policy-iteration convergence and domain sensitivity. Fitted actors are evaluated as Markov feedbacks on this finite grid. These comparisons do not conflate that controller with the innovation-driven

implementation. Grid differences and asymptotic boundary values are not asserted to be rigorous continuous-domain error enclosures.

2.5. Reproducibility and cost accounting

The primary fitting budget includes model initialization, optimizer creation, roll-outs, derivative calculations, updates, validation and checkpoint writes. A process completes its current iteration and records the resulting overshoot. The common numerical-runtime warm-up precedes the clock for every method. Method order is rotated across the ten independent streams. Checkpoint frontiers choose saved objects by elapsed fitting work, before any final simulation result is inspected. Every final frontier row receives the same full path count and transfer calculation as a selected primary policy.

Wall-clock stopping is hardware-dependent. The ledger consequently retains the realized iteration count, simulator-state visits, optimization steps, peak process memory, all checkpoints, selected checkpoint and source hash. Re-executing the time-budget protocol reproduces the experimental design, not a promise of bitwise identical weights on a different processor. A fixed-iteration replay can use the saved counts and the same random streams to investigate numerical discrepancies. The numerical compiler, random generator and machine arithmetic remain subject to the inherited conditional contract.

3. FINITE-OBSERVATION TRANSFER PROOF

3.1. Proof of Proposition 15

All processes are coupled using the same economic Brownian motion and initial profile; the sensor noise uses the independent randomizer. The real-map Lipschitz bound, uniform implementation error, and nonexpansiveness of the common guard give

$$\|\hat{\pi}_k(\bar{Y}_{t_k} + \eta_k) - \hat{\pi}_k(Z_k)\|_{2,d} \leq L\{\|\bar{Y}_{t_k} - Z_k\|_{2,d} + \nu_{\text{obs}}\} + 2\delta_{\pi}.$$

The common action tube also bounds this difference by 2ϵ .

Apply Itô's formula to $f(\bar{Y})$ on each cell, where the sensor action has already been chosen. The generator and derivative bounds used in the R11 proof remain

valid for this adapted held action. Stochastic Fubini yields

$$\begin{aligned} \int_{t_j}^{t_{j+1}} \{f(\bar{Y}_s) - f(\bar{Y}_{t_j})\} ds &= \int_{t_j}^{t_{j+1}} (t_{j+1} - v) \mathcal{A}^{\bar{a}_j} f(\bar{Y}_v) dv \\ &\quad + \int_{t_j}^{t_{j+1}} (t_{j+1} - v) Df(\bar{Y}_v) \Sigma dW_v. \end{aligned}$$

The L^2 norm of the accumulated first term through t_k is at most $Kht_k/2$. The stochastic integrals are martingale increments over disjoint cells. Itô's isometry bounds their accumulated normalized L^2 norm by $Hh\sqrt{t_k/3}$.

Before representation error, every component of $\Delta W_j - \sqrt{h} \text{clip}(\Delta W_j / \sqrt{h}, -z, z)$ has zero mean and variance at most $h v_{\text{clip}}$. Components and cells are independent. Therefore the normalized second moment of the accumulated projected clipping error is at most

$$\frac{1}{d} \text{tr}(\Sigma \Sigma^\top) t_k v_{\text{clip}} = \bar{\sigma}^2 t_k v_{\text{clip}}.$$

The clipping error and the preceding local-drift martingale need not be independent: the triangle inequality between their already accumulated norms is sufficient. The implementation error contributes at most $k\zeta_h$.

Let $E_k = \|\bar{Y}_{t_k} - Z_k\|_{2,d}$. Subtracting the internal recursion from the integrated sensor state equation thus gives

$$E_k \leq \beta h \sum_{j < k} E_j + h \sum_{j < k} \|\bar{a}_j - a_j\|_{2,d} + b_k.$$

The recursion in (56) is equivalent to $e_k = \beta h \sum_{j < k} e_j + h \sum_{j < k} a_j^\Delta + b_k$. Induction, using the action bound already proved, establishes $E_k \leq e_k$ and the asserted action inequalities.

The two physical diffusions have the same Brownian term. Subtracting their integral equations and applying the β -Lipschitz production bound gives

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq \beta \int_0^t \|\bar{Y}_s - Y_s\|_{2,d} ds + \int_0^t a_{\lfloor s/h \rfloor}^\Delta ds.$$

The integral form of Gronwall's inequality proves the bound by $D(t)$. Projection away from the common-state direction eliminates common drift, the common

schedule, and common diffusion. Bounded production, the action tube, and Itô's isometry give, for both physical states, $\sup_{t \leq T} \|\Pi Y_t\|_{2,d} \leq Q_T$.

It remains to compare payoffs. Subtract the two exact identities (44); the common schedule terms cancel. Production contributes at most $\beta \int_0^T W(t)D(t)dt$. For the consumption deficit R_t in (43), its gradient in the normalized inner product has components

$$-\frac{1}{m_i} + w(t) + \zeta \bar{m} = \frac{1}{m_0(t)} - \frac{1}{m_i} + \zeta \{\bar{m} - m_0(t)\}.$$

Every point on the segment between the two held actions remains in their common coordinate tube at t_k . Its distance from $m_0(t)$ is at most $r_k(t)$, and its coordinates are at least $m_{-,k}$. Thus the normalized gradient norm is bounded by $C_k(t)$. The mean-value inequality and Cauchy–Schwarz bound the absolute expected deficit difference by $C_k(t)a_k^\Delta$.

Finally,

$$\left| \mathbb{E} \{ \|\Pi \bar{Y}_T\|_d^2 - \|\Pi Y_T\|_d^2 \} \right| \leq \|\bar{Y}_T - Y_T\|_{2,d} \{ \|\Pi \bar{Y}_T\|_{2,d} + \|\Pi Y_T\|_{2,d} \} \leq 2Q_T D(T).$$

Adding these three contributions proves (57). Adding and subtracting the parent's payoff on the existing confidence event proves the interval transfer.

An outward finite computation. One need not infer a scalar integral error from a refinement plot. The function D is nondecreasing and obeys the exact cell recursion

$$D(t_{k+1}) = e^{\beta h} D(t_k) + a_k^\Delta \int_0^h e^{\beta s} ds.$$

An interval enclosure of m_0 on each cell gives an upper bound C_k^+ for $C_k(t)$. With the already enclosed weights A_k, B_k , a computable upper allowance is

$$\beta \sum_k B_k D(t_{k+1}) + \sum_k A_k C_k^+ a_k^\Delta + 2\chi e^{-\rho T} Q_T D(T),$$

evaluated outward. In the forcing recurrence, use nondecreasing saved upper endpoints $b_k^+ \geq b_k$ and the differences of those exact saved scalars. This preserves the telescoping majorant. The network and recurrence arithmetic assumptions are the same conditional binary64 assumptions used for the protected parent.

4. COMPLETE R12 SEED AND AUXILIARY RESULTS

All rows below use the same stated conditional inference and are retained irrespective of sign. The raw arrays and hashes provide the full precision.

TABLE I: Primary seed-level endpoints

d	Seed	Method	State	Statistic	Lower	Upper
10	15401	DPO	0	0.001910	0.001112	0.002708
10	15401	DPO	P	0.001389	0.000549	0.002229
20	15401	DPO	0	0.001504	0.000619	0.002390
20	15401	DPO	P	0.001257	0.000336	0.002178
50	15401	DPO	0	0.001657	0.000751	0.002563
50	15401	DPO	P	0.001519	0.000575	0.002464
10	15401	LINEAR	0	0.001879	0.001081	0.002677
10	15401	LINEAR	P	0.001356	0.000516	0.002196
20	15401	LINEAR	0	0.001476	0.000591	0.002361
20	15401	LINEAR	P	0.001234	0.000314	0.002155
50	15401	LINEAR	0	0.001596	0.000691	0.002501
50	15401	LINEAR	P	0.001481	0.000538	0.002424
10	15401	NBO	0	0.001980	0.001182	0.002778
10	15401	NBO	P	0.001469	0.000630	0.002309
20	15401	NBO	0	0.001588	0.000703	0.002473
20	15401	NBO	P	0.001349	0.000428	0.002269
50	15401	NBO	0	0.001755	0.000849	0.002660
50	15401	NBO	P	0.001610	0.000666	0.002555
10	22807	DPO	0	0.001898	0.001100	0.002696
10	22807	DPO	P	0.001374	0.000534	0.002214
20	22807	DPO	0	0.001482	0.000597	0.002367
20	22807	DPO	P	0.001246	0.000326	0.002167
50	22807	DPO	0	0.001650	0.000744	0.002555
50	22807	DPO	P	0.001512	0.000568	0.002457
10	22807	LINEAR	0	0.001864	0.001067	0.002661
10	22807	LINEAR	P	0.001356	0.000517	0.002196
20	22807	LINEAR	0	0.001452	0.000568	0.002336

	<i>d</i>	Seed	Method	State	Statistic	Lower	Upper	
1								1
2								2
3	20	22807	LINEAR	P	0.001229	0.000310	0.002149	3
4	50	22807	LINEAR	0	0.001485	0.000580	0.002389	4
5	50	22807	LINEAR	P	0.001423	0.000479	0.002367	5
6	10	22807	NBO	0	0.001982	0.001184	0.002780	6
7	10	22807	NBO	P	0.001470	0.000631	0.002310	7
8	20	22807	NBO	0	0.001585	0.000700	0.002470	8
9	20	22807	NBO	P	0.001345	0.000425	0.002266	9
10	50	22807	NBO	0	0.001755	0.000849	0.002661	10
11	50	22807	NBO	P	0.001608	0.000664	0.002553	11
12	10	30223	DPO	0	0.001908	0.001110	0.002707	12
13	10	30223	DPO	P	0.001388	0.000548	0.002227	13
14	20	30223	DPO	0	0.001490	0.000605	0.002376	14
15	20	30223	DPO	P	0.001254	0.000333	0.002174	15
16	50	30223	DPO	0	0.001655	0.000749	0.002561	16
17	50	30223	DPO	P	0.001521	0.000577	0.002466	17
18	10	30223	LINEAR	0	0.001880	0.001083	0.002678	18
19	10	30223	LINEAR	P	0.001360	0.000520	0.002200	19
20	20	30223	LINEAR	0	0.001464	0.000580	0.002349	20
21	20	30223	LINEAR	P	0.001228	0.000307	0.002148	21
22	50	30223	LINEAR	0	0.001543	0.000638	0.002448	22
23	50	30223	LINEAR	P	0.001456	0.000512	0.002399	23
24	10	30223	NBO	0	0.001982	0.001184	0.002780	24
25	10	30223	NBO	P	0.001470	0.000631	0.002309	25
26	20	30223	NBO	0	0.001584	0.000699	0.002469	26
27	20	30223	NBO	P	0.001345	0.000425	0.002266	27
28	50	30223	NBO	0	0.001755	0.000849	0.002661	28
29	50	30223	NBO	P	0.001612	0.000668	0.002556	29
30	10	37633	DPO	0	0.001911	0.001112	0.002709	30
31	10	37633	DPO	P	0.001383	0.000543	0.002223	31
32	20	37633	DPO	0	0.001491	0.000606	0.002376	32
	20	37633	DPO	P	0.001251	0.000331	0.002172	
	50	37633	DPO	0	0.001657	0.000751	0.002562	
	50	37633	DPO	P	0.001517	0.000572	0.002461	

	<i>d</i>	Seed	Method	State	Statistic	Lower	Upper	
1								1
2								2
3	10	37633	LINEAR	0	0.001861	0.001064	0.002658	3
4	10	37633	LINEAR	P	0.001354	0.000515	0.002193	4
5	20	37633	LINEAR	0	0.001443	0.000559	0.002328	5
6	20	37633	LINEAR	P	0.001221	0.000301	0.002142	6
7	50	37633	LINEAR	0	0.001487	0.000582	0.002391	7
8	50	37633	LINEAR	P	0.001421	0.000478	0.002365	8
9	10	37633	NBO	0	0.001982	0.001184	0.002780	9
10	10	37633	NBO	P	0.001469	0.000630	0.002308	10
11	20	37633	NBO	0	0.001586	0.000701	0.002471	11
12	20	37633	NBO	P	0.001344	0.000424	0.002265	12
13	50	37633	NBO	0	0.001755	0.000849	0.002661	13
14	50	37633	NBO	P	0.001610	0.000666	0.002554	14
15	10	45061	DPO	0	0.001907	0.001109	0.002705	15
16	10	45061	DPO	P	0.001386	0.000547	0.002226	16
17	20	45061	DPO	0	0.001486	0.000601	0.002371	17
18	20	45061	DPO	P	0.001248	0.000327	0.002168	18
19	50	45061	DPO	0	0.001653	0.000747	0.002559	19
20	50	45061	DPO	P	0.001520	0.000575	0.002465	20
21	10	45061	LINEAR	0	0.001863	0.001066	0.002660	21
22	10	45061	LINEAR	P	0.001355	0.000516	0.002195	22
23	20	45061	LINEAR	0	0.001444	0.000560	0.002329	23
24	20	45061	LINEAR	P	0.001222	0.000302	0.002142	24
25	50	45061	LINEAR	0	0.001490	0.000586	0.002395	25
26	50	45061	LINEAR	P	0.001426	0.000482	0.002369	26
27	10	45061	NBO	0	0.001980	0.001182	0.002779	27
28	10	45061	NBO	P	0.001469	0.000630	0.002309	28
29	20	45061	NBO	0	0.001584	0.000699	0.002469	29
30	20	45061	NBO	P	0.001345	0.000424	0.002265	30
31	50	45061	NBO	0	0.001755	0.000849	0.002661	31
32	50	45061	NBO	P	0.001610	0.000666	0.002555	32
	10	52453	DPO	0	0.001905	0.001106	0.002703	
	10	52453	DPO	P	0.001378	0.000538	0.002218	
	20	52453	DPO	0	0.001500	0.000615	0.002385	

	<i>d</i>	Seed	Method	State	Statistic	Lower	Upper	
1								1
2								2
3	20	52453	DPO	P	0.001252	0.000331	0.002173	3
4	50	52453	DPO	0	0.001657	0.000751	0.002563	4
5	50	52453	DPO	P	0.001520	0.000575	0.002464	5
6	10	52453	LINEAR	0	0.001856	0.001059	0.002653	6
7	10	52453	LINEAR	P	0.001355	0.000516	0.002194	7
8	20	52453	LINEAR	0	0.001446	0.000562	0.002331	8
9	20	52453	LINEAR	P	0.001224	0.000305	0.002144	9
10	50	52453	LINEAR	0	0.001472	0.000567	0.002376	10
11	50	52453	LINEAR	P	0.001415	0.000471	0.002358	11
12	10	52453	NBO	0	0.001981	0.001182	0.002779	12
13	10	52453	NBO	P	0.001468	0.000629	0.002308	13
14	20	52453	NBO	0	0.001584	0.000699	0.002469	14
15	20	52453	NBO	P	0.001344	0.000424	0.002265	15
16	50	52453	NBO	0	0.001754	0.000848	0.002659	16
17	50	52453	NBO	P	0.001608	0.000664	0.002553	17
18	10	59879	DPO	0	0.001903	0.001105	0.002702	18
19	10	59879	DPO	P	0.001376	0.000536	0.002215	19
20	20	59879	DPO	0	0.001495	0.000609	0.002380	20
21	20	59879	DPO	P	0.001248	0.000328	0.002169	21
22	50	59879	DPO	0	0.001667	0.000761	0.002573	22
23	50	59879	DPO	P	0.001528	0.000583	0.002472	23
24	10	59879	LINEAR	0	0.001855	0.001058	0.002653	24
25	10	59879	LINEAR	P	0.001349	0.000510	0.002189	25
26	20	59879	LINEAR	0	0.001445	0.000561	0.002329	26
27	20	59879	LINEAR	P	0.001223	0.000304	0.002143	27
28	50	59879	LINEAR	0	0.001482	0.000577	0.002386	28
29	50	59879	LINEAR	P	0.001426	0.000482	0.002369	29
30	10	59879	NBO	0	0.001981	0.001183	0.002779	30
31	10	59879	NBO	P	0.001468	0.000629	0.002308	31
32	20	59879	NBO	0	0.001584	0.000699	0.002470	32
	20	59879	NBO	P	0.001344	0.000424	0.002264	
	50	59879	NBO	0	0.001755	0.000850	0.002661	
	50	59879	NBO	P	0.001611	0.000666	0.002555	

	<i>d</i>	Seed	Method	State	Statistic	Lower	Upper	
1								1
2								2
3	10	67307	DPO	0	0.001905	0.001107	0.002704	3
4	10	67307	DPO	P	0.001383	0.000543	0.002222	4
5	20	67307	DPO	0	0.001498	0.000613	0.002384	5
6	20	67307	DPO	P	0.001259	0.000338	0.002179	6
7	50	67307	DPO	0	0.001663	0.000757	0.002569	7
8	50	67307	DPO	P	0.001521	0.000577	0.002466	8
9	10	67307	LINEAR	0	0.001881	0.001084	0.002679	9
10	10	67307	LINEAR	P	0.001357	0.000517	0.002197	10
11	20	67307	LINEAR	0	0.001473	0.000588	0.002357	11
12	20	67307	LINEAR	P	0.001231	0.000310	0.002151	12
13	50	67307	LINEAR	0	0.001599	0.000694	0.002504	13
14	50	67307	LINEAR	P	0.001487	0.000543	0.002431	14
15	10	67307	NBO	0	0.001981	0.001183	0.002779	15
16	10	67307	NBO	P	0.001467	0.000628	0.002307	16
17	20	67307	NBO	0	0.001587	0.000702	0.002472	17
18	20	67307	NBO	P	0.001348	0.000428	0.002268	18
19	50	67307	NBO	0	0.001758	0.000852	0.002664	19
20	50	67307	NBO	P	0.001614	0.000669	0.002558	20
21	10	74719	DPO	0	0.001917	0.001119	0.002715	21
22	10	74719	DPO	P	0.001393	0.000553	0.002232	22
23	20	74719	DPO	0	0.001503	0.000617	0.002388	23
24	20	74719	DPO	P	0.001260	0.000339	0.002180	24
25	50	74719	DPO	0	0.001658	0.000753	0.002564	25
26	50	74719	DPO	P	0.001525	0.000580	0.002469	26
27	10	74719	LINEAR	0	0.001878	0.001081	0.002676	27
28	10	74719	LINEAR	P	0.001356	0.000515	0.002196	28
29	20	74719	LINEAR	0	0.001470	0.000586	0.002355	29
30	20	74719	LINEAR	P	0.001233	0.000313	0.002153	30
31	50	74719	LINEAR	0	0.001595	0.000690	0.002500	31
32	50	74719	LINEAR	P	0.001486	0.000542	0.002430	32
	10	74719	NBO	0	0.001984	0.001186	0.002782	
	10	74719	NBO	P	0.001471	0.000631	0.002310	
	20	74719	NBO	0	0.001588	0.000703	0.002473	

<i>d</i>	Seed	Method	State	Statistic	Lower	Upper
20	74719	NBO	P	0.001349	0.000429	0.002270
50	74719	NBO	0	0.001758	0.000853	0.002664
50	74719	NBO	P	0.001613	0.000668	0.002557
10	7919	DPO	0	0.001891	0.001093	0.002690
10	7919	DPO	P	0.001372	0.000532	0.002211
20	7919	DPO	0	0.001497	0.000612	0.002383
20	7919	DPO	P	0.001255	0.000334	0.002175
50	7919	DPO	0	0.001657	0.000751	0.002562
50	7919	DPO	P	0.001514	0.000569	0.002459
10	7919	LINEAR	0	0.001860	0.001063	0.002657
10	7919	LINEAR	P	0.001352	0.000513	0.002192
20	7919	LINEAR	0	0.001444	0.000559	0.002328
20	7919	LINEAR	P	0.001220	0.000300	0.002140
50	7919	LINEAR	0	0.001482	0.000577	0.002386
50	7919	LINEAR	P	0.001418	0.000475	0.002362
10	7919	NBO	0	0.001982	0.001184	0.002781
10	7919	NBO	P	0.001470	0.000631	0.002309
20	7919	NBO	0	0.001586	0.000701	0.002471
20	7919	NBO	P	0.001347	0.000426	0.002267
50	7919	NBO	0	0.001754	0.000848	0.002660
50	7919	NBO	P	0.001609	0.000665	0.002553

Individual fitted-policy endpoints, with outward decimal presentation; no seed is replaced.

TABLE II: Fresh-radius and critic ablations

<i>d</i>	Method	Variant	Statistic	Lower	Upper
10	DPO	r0.05	0.001310	0.000754	0.001866
10	DPO	r0.15	0.001376	-0.000006	0.002758
50	DPO	r0.05	0.001349	0.000700	0.001999
50	DPO	r0.15	0.001532	0.000033	0.003031
10	NBO	costate only	0.001470	0.000630	0.002309

<i>d</i>	Method	Variant	Statistic	Lower	Upper
10	NBO	r0.05	0.001400	0.000844	0.001955
10	NBO	r0.15	0.001470	0.000089	0.002852
10	NBO	raw costate	0.001472	0.000632	0.002311
10	NBO	updates10	0.001469	0.000630	0.002309
10	NBO	updates1	0.001451	0.000612	0.002291
10	NBO	value only	0.000390	-0.000443	0.001223
10	NBO	width16	0.001462	0.000623	0.002301
10	NBO	width64	0.001468	0.000628	0.002307
50	NBO	costate only	0.001611	0.000667	0.002556
50	NBO	r0.05	0.001431	0.000782	0.002080
50	NBO	r0.15	0.001626	0.000128	0.003124
50	NBO	raw costate	0.001616	0.000671	0.002560
50	NBO	updates10	0.001615	0.000671	0.002559
50	NBO	updates1	0.001593	0.000648	0.002537
50	NBO	value only	-0.003482	-0.004479	-0.002486
50	NBO	width16	0.001607	0.000662	0.002551
50	NBO	width64	0.001612	0.000668	0.002556

Every radius variant is freshly trained. The raw-costate variant has no fitted critic; the stored method identifier retains loader compatibility and is not a claim of Bellman evaluation.

TABLE III: Full-path work-selected checkpoint frontier

<i>d</i>	Method	Iteration	Statistic	Lower	Upper
10	DPO	160	0.001359	0.000519	0.002199
10	DPO	320	0.001370	0.000530	0.002209
20	DPO	140	0.001217	0.000297	0.002138
20	DPO	280	0.001231	0.000310	0.002152
50	DPO	120	0.001503	0.000559	0.002448
50	DPO	220	0.001509	0.000564	0.002454
10	LINEAR	200	0.001111	0.000276	0.001947
10	LINEAR	420	0.001290	0.000453	0.002128

d	Method	Iteration	Statistic	Lower	Upper
20	LINEAR	180	0.001057	0.000139	0.001975
20	LINEAR	360	0.001176	0.000258	0.002095
50	LINEAR	140	0.001206	0.000262	0.002151
50	LINEAR	260	0.001315	0.000372	0.002259
10	NBO	100	0.001447	0.000608	0.002286
10	NBO	200	0.001465	0.000625	0.002304
20	NBO	100	0.001326	0.000406	0.002247
20	NBO	200	0.001338	0.000418	0.002259
50	NBO	160	0.001603	0.000658	0.002547
50	NBO	80	0.001594	0.000649	0.002538

Checkpoints are chosen by elapsed fitting work, not by final economic results. All frontier rows receive the full final path count; corresponding clocks, simulator visits and verification costs are saved.

TABLE IV: Every direct primary method comparison

d	Seed	State	Other	Mean	Lower	Upper
10	7919	0	DPO	0.000091	-0.001112	0.001294
10	7919	0	LINEAR	0.000122	-0.001081	0.001326
10	15401	0	DPO	0.000070	-0.001133	0.001273
10	15401	0	LINEAR	0.000101	-0.001102	0.001304
10	22807	0	DPO	0.000084	-0.001118	0.001287
10	22807	0	LINEAR	0.000118	-0.001085	0.001322
10	30223	0	DPO	0.000074	-0.001129	0.001276
10	30223	0	LINEAR	0.000102	-0.001102	0.001305
10	37633	0	DPO	0.000072	-0.001131	0.001274
10	37633	0	LINEAR	0.000121	-0.001082	0.001325
10	45061	0	DPO	0.000073	-0.001129	0.001276
10	45061	0	LINEAR	0.000118	-0.001086	0.001321
10	52453	0	DPO	0.000076	-0.001127	0.001279
10	52453	0	LINEAR	0.000125	-0.001079	0.001328

	<i>d</i>	Seed	State	Other	Mean	Lower	Upper	
1								1
2								2
3	10	59879	0	DPO	0.000078	-0.001125	0.001280	3
4	10	59879	0	LINEAR	0.000125	-0.001078	0.001329	4
5	10	67307	0	DPO	0.000076	-0.001127	0.001278	5
6	10	67307	0	LINEAR	0.000100	-0.001104	0.001303	6
7	10	74719	0	DPO	0.000067	-0.001136	0.001269	7
8	10	74719	0	LINEAR	0.000105	-0.001098	0.001309	8
9	10	7919	P	DPO	0.000098	-0.001155	0.001351	9
10	10	7919	P	LINEAR	0.000118	-0.001136	0.001371	10
11	10	15401	P	DPO	0.000080	-0.001173	0.001333	11
12	10	15401	P	LINEAR	0.000113	-0.001141	0.001366	12
13	10	22807	P	DPO	0.000097	-0.001157	0.001350	13
14	10	22807	P	LINEAR	0.000114	-0.001139	0.001367	14
15	10	30223	P	DPO	0.000082	-0.001171	0.001335	15
16	10	30223	P	LINEAR	0.000110	-0.001144	0.001363	16
17	10	37633	P	DPO	0.000086	-0.001167	0.001339	17
18	10	37633	P	LINEAR	0.000115	-0.001138	0.001368	18
19	10	45061	P	DPO	0.000083	-0.001170	0.001336	19
20	10	45061	P	LINEAR	0.000114	-0.001139	0.001367	20
21	10	52453	P	DPO	0.000090	-0.001163	0.001343	21
22	10	52453	P	LINEAR	0.000114	-0.001140	0.001367	22
23	10	59879	P	DPO	0.000093	-0.001160	0.001346	23
24	10	59879	P	LINEAR	0.000119	-0.001135	0.001372	24
25	10	67307	P	DPO	0.000085	-0.001168	0.001338	25
26	10	67307	P	LINEAR	0.000110	-0.001143	0.001364	26
27	10	74719	P	DPO	0.000078	-0.001175	0.001331	27
28	10	74719	P	LINEAR	0.000115	-0.001139	0.001369	28
29	20	7919	0	DPO	0.000088	-0.001223	0.001400	29
30	20	7919	0	LINEAR	0.000142	-0.001170	0.001454	30
31	20	15401	0	DPO	0.000084	-0.001228	0.001395	31
32	20	15401	0	LINEAR	0.000112	-0.001200	0.001424	32
	20	22807	0	DPO	0.000103	-0.001209	0.001414	
	20	22807	0	LINEAR	0.000133	-0.001179	0.001445	
	20	30223	0	DPO	0.000094	-0.001218	0.001405	

	<i>d</i>	Seed	State	Other	Mean	Lower	Upper	
1								1
2								2
3	20	30223	0	LINEAR	0.000120	-0.001192	0.001432	3
4	20	37633	0	DPO	0.000095	-0.001216	0.001407	4
5	20	37633	0	LINEAR	0.000143	-0.001169	0.001455	5
6	20	45061	0	DPO	0.000098	-0.001214	0.001410	6
7	20	45061	0	LINEAR	0.000140	-0.001172	0.001452	7
8	20	52453	0	DPO	0.000084	-0.001227	0.001396	8
9	20	52453	0	LINEAR	0.000138	-0.001174	0.001450	9
10	20	59879	0	DPO	0.000090	-0.001222	0.001402	10
11	20	59879	0	LINEAR	0.000139	-0.001173	0.001452	11
12	20	67307	0	DPO	0.000089	-0.001223	0.001400	12
13	20	67307	0	LINEAR	0.000114	-0.001198	0.001426	13
14	20	74719	0	DPO	0.000086	-0.001226	0.001397	14
15	20	74719	0	LINEAR	0.000118	-0.001194	0.001430	15
16	20	7919	P	DPO	0.000092	-0.001272	0.001455	16
17	20	7919	P	LINEAR	0.000127	-0.001237	0.001491	17
18	20	15401	P	DPO	0.000092	-0.001272	0.001455	18
19	20	15401	P	LINEAR	0.000114	-0.001250	0.001478	19
20	20	22807	P	DPO	0.000099	-0.001265	0.001462	20
21	20	22807	P	LINEAR	0.000116	-0.001249	0.001480	21
22	20	30223	P	DPO	0.000092	-0.001272	0.001455	22
23	20	30223	P	LINEAR	0.000118	-0.001246	0.001482	23
24	20	37633	P	DPO	0.000093	-0.001271	0.001456	24
25	20	37633	P	LINEAR	0.000123	-0.001241	0.001487	25
26	20	45061	P	DPO	0.000097	-0.001266	0.001461	26
27	20	45061	P	LINEAR	0.000123	-0.001241	0.001487	27
28	20	52453	P	DPO	0.000092	-0.001271	0.001456	28
29	20	52453	P	LINEAR	0.000120	-0.001244	0.001484	29
30	20	59879	P	DPO	0.000096	-0.001268	0.001459	30
31	20	59879	P	LINEAR	0.000121	-0.001244	0.001485	31
32	20	67307	P	DPO	0.000089	-0.001274	0.001453	32
	20	67307	P	LINEAR	0.000117	-0.001247	0.001482	
	20	74719	P	DPO	0.000090	-0.001274	0.001453	
	20	74719	P	LINEAR	0.000116	-0.001248	0.001480	

	<i>d</i>	Seed	State	Other	Mean	Lower	Upper	
1								1
2								2
3	50	7919	0	DPO	0.000098	-0.001238	0.001433	3
4	50	7919	0	LINEAR	0.000272	-0.001064	0.001609	4
5	50	15401	0	DPO	0.000097	-0.001238	0.001433	5
6	50	15401	0	LINEAR	0.000158	-0.001178	0.001494	6
7	50	22807	0	DPO	0.000105	-0.001230	0.001441	7
8	50	22807	0	LINEAR	0.000270	-0.001066	0.001606	8
9	50	30223	0	DPO	0.000100	-0.001236	0.001435	9
10	50	30223	0	LINEAR	0.000212	-0.001124	0.001548	10
11	50	37633	0	DPO	0.000099	-0.001237	0.001434	11
12	50	37633	0	LINEAR	0.000269	-0.001068	0.001605	12
13	50	45061	0	DPO	0.000102	-0.001234	0.001437	13
14	50	45061	0	LINEAR	0.000264	-0.001072	0.001601	14
15	50	52453	0	DPO	0.000097	-0.001239	0.001432	15
16	50	52453	0	LINEAR	0.000282	-0.001054	0.001618	16
17	50	59879	0	DPO	0.000088	-0.001247	0.001423	17
18	50	59879	0	LINEAR	0.000274	-0.001063	0.001610	18
19	50	67307	0	DPO	0.000095	-0.001240	0.001431	19
20	50	67307	0	LINEAR	0.000159	-0.001176	0.001495	20
21	50	74719	0	DPO	0.000100	-0.001235	0.001435	21
22	50	74719	0	LINEAR	0.000163	-0.001172	0.001499	22
23	50	7919	P	DPO	0.000095	-0.001293	0.001483	23
24	50	7919	P	LINEAR	0.000191	-0.001198	0.001580	24
25	50	15401	P	DPO	0.000091	-0.001297	0.001479	25
26	50	15401	P	LINEAR	0.000129	-0.001259	0.001518	26
27	50	22807	P	DPO	0.000096	-0.001291	0.001484	27
28	50	22807	P	LINEAR	0.000186	-0.001204	0.001575	28
29	50	30223	P	DPO	0.000091	-0.001297	0.001478	29
30	50	30223	P	LINEAR	0.000156	-0.001232	0.001545	30
31	50	37633	P	DPO	0.000094	-0.001294	0.001481	31
32	50	37633	P	LINEAR	0.000189	-0.001200	0.001578	32
	50	45061	P	DPO	0.000090	-0.001297	0.001478	
	50	45061	P	LINEAR	0.000185	-0.001205	0.001574	
	50	52453	P	DPO	0.000088	-0.001299	0.001476	

d	Seed	State	Other	Mean	Lower	Upper
50	52453	P	LINEAR	0.000194	-0.001196	0.001583
50	59879	P	DPO	0.000083	-0.001304	0.001471
50	59879	P	LINEAR	0.000185	-0.001205	0.001574
50	67307	P	DPO	0.000093	-0.001295	0.001480
50	67307	P	LINEAR	0.000127	-0.001261	0.001515
50	74719	P	DPO	0.000088	-0.001300	0.001476
50	74719	P	LINEAR	0.000127	-0.001262	0.001515

NBO minus the named comparator on the same initial profiles and innovations. Every declared stream is retained. A positive lower endpoint establishes a positive fitted-policy difference; a negative upper endpoint establishes a negative difference; the remaining intervals are inconclusive. P is the nine-profile population and 0 is the origin. No optimizer-population inference is made.

TABLE V: All extension policy endpoints and verification work

OS	d	Seed	Method	k	Mean	Lower	Upper	Verify sec.
24	10	7919	NBO	20	0.001379	0.000539	0.002218	17.13
24	10	7919	NBO	80	0.001443	0.000604	0.002283	15.42
24	10	7919	DPO	20	0.000996	0.000157	0.001835	16.25
24	10	7919	DPO	80	0.001347	0.000506	0.002188	16.22
24	10	7919	Affine	20	0.000524	-0.000301	0.001349	9.31
24	10	7919	Affine	80	0.000862	0.000030	0.001693	9.33
24	10	7919	Raw	20	0.001436	0.000597	0.002275	16.70
24	10	7919	Raw	80	0.001456	0.000617	0.002296	17.17
24	10	15401	NBO	20	0.001384	0.000543	0.002224	16.50
24	10	15401	NBO	80	0.001443	0.000604	0.002282	15.72
24	10	15401	DPO	20	0.001001	0.000163	0.001839	15.72
24	10	15401	DPO	80	0.001346	0.000504	0.002187	15.81
24	10	15401	Affine	20	0.000535	-0.000290	0.001361	8.91
24	10	15401	Affine	80	0.000854	0.000022	0.001686	8.90

	OS	d	Seed	Method	k	Mean	Lower	Upper	Verify sec.	
1										1
2										2
3	24	10	15401	Raw	20	0.001434	0.000595	0.002274	14.34	3
4	24	10	15401	Raw	80	0.001455	0.000616	0.002295	14.16	4
5	24	20	7919	NBO	20	0.001277	0.000356	0.002197	25.96	5
6	24	20	7919	NBO	80	0.001322	0.000402	0.002242	25.60	6
7	24	20	7919	DPO	20	0.000929	0.000009	0.001849	24.59	7
8	24	20	7919	DPO	80	0.001220	0.000300	0.002141	24.04	8
9	24	20	7919	Affine	20	0.000665	-0.000257	0.001587	16.61	9
10	24	20	7919	Affine	80	0.000876	-0.000042	0.001795	16.43	10
11	24	20	7919	Raw	20	0.001312	0.000391	0.002232	23.15	11
12	24	20	7919	Raw	80	0.001335	0.000415	0.002255	23.46	12
13	24	20	15401	NBO	20	0.001289	0.000368	0.002210	25.11	13
14	24	20	15401	NBO	80	0.001320	0.000400	0.002241	24.62	14
15	24	20	15401	DPO	20	0.000930	0.000009	0.001851	24.79	15
16	24	20	15401	DPO	80	0.001217	0.000295	0.002139	24.01	16
17	24	20	15401	Affine	20	0.000665	-0.000257	0.001588	17.09	17
18	24	20	15401	Affine	80	0.000882	-0.000036	0.001800	17.15	18
19	24	20	15401	Raw	20	0.001311	0.000390	0.002231	24.68	19
20	24	20	15401	Raw	80	0.001333	0.000413	0.002253	23.98	20
21	24	50	7919	NBO	20	0.001539	0.000595	0.002484	52.73	21
22	24	50	7919	NBO	80	0.001586	0.000641	0.002530	51.20	22
23	24	50	7919	DPO	20	0.001167	0.000221	0.002113	52.28	23
24	24	50	7919	DPO	80	0.001435	0.000491	0.002379	52.57	24
25	24	50	7919	Affine	20	0.000874	-0.000081	0.001829	44.92	25
26	24	50	7919	Affine	80	0.001090	0.000142	0.002038	44.55	26
27	24	50	7919	Raw	20	0.001554	0.000610	0.002499	51.12	27
28	24	50	7919	Raw	80	0.001595	0.000650	0.002539	50.07	28
29	24	50	15401	NBO	20	0.001524	0.000580	0.002469	50.51	29
30	24	50	15401	NBO	80	0.001579	0.000634	0.002523	49.57	30
31	24	50	15401	DPO	20	0.001212	0.000267	0.002157	48.06	31
32	24	50	15401	DPO	80	0.001436	0.000492	0.002381	48.56	32
	24	50	15401	Affine	20	0.000874	-0.000081	0.001829	42.04	
	24	50	15401	Affine	80	0.001087	0.000141	0.002034	41.96	
	24	50	15401	Raw	20	0.001559	0.000615	0.002504	48.93	

OS	d	Seed	Method	k	Mean	Lower	Upper	Verify sec.
24	50	15401	Raw	80	0.001594	0.000650	0.002539	47.42
22	10	7919	NBO	20	0.001379	0.000539	0.002218	13.73
22	10	7919	NBO	80	0.001443	0.000604	0.002283	12.62
22	10	7919	DPO	20	0.000996	0.000157	0.001835	12.72
22	10	7919	DPO	80	0.001347	0.000506	0.002188	12.92
22	10	7919	Affine	20	0.000524	-0.000301	0.001349	7.39
22	10	7919	Affine	80	0.000862	0.000030	0.001693	7.41
22	10	7919	Raw	20	0.001436	0.000597	0.002275	12.58
22	10	7919	Raw	80	0.001456	0.000617	0.002296	12.58

All 56 full-bank policy rows are retained. Each verification has 8,192 paired paths and 1,024 time cells, hence 16,777,216 actor-plus-anchor state transitions. The saved verification clock excludes model loading and interval-account construction. OS denotes Ubuntu 22.04 or 24.04; the repeated environment does not add an independent fitted-seed observation.

TABLE VI: Every fixed-work NBO-minus-comparator end-point

OS	d	Seed	k	Comparator	Mean	Lower	Upper
24	10	7919	20	DPO	0.000383	-0.000876	0.001642
24	10	7919	20	Affine	0.000855	-0.000422	0.002131
24	10	7919	20	Raw	-0.000058	-0.001312	0.001196
24	10	7919	80	DPO	0.000096	-0.001158	0.001351
24	10	7919	80	Affine	0.000582	-0.000682	0.001846
24	10	7919	80	Raw	-0.000013	-0.001266	0.001240
24	10	15401	20	DPO	0.000383	-0.000877	0.001643
24	10	15401	20	Affine	0.000849	-0.000429	0.002126
24	10	15401	20	Raw	-0.000051	-0.001305	0.001204
24	10	15401	80	DPO	0.000097	-0.001158	0.001352
24	10	15401	80	Affine	0.000589	-0.000676	0.001853
24	10	15401	80	Raw	-0.000013	-0.001265	0.001240
24	20	7919	20	DPO	0.000348	-0.001020	0.001716

	OS	d	Seed	k	Comparator	Mean	Lower	Upper	
1									1
2									2
3	24	20	7919	20	Affine	0.000612	-0.000770	0.001994	3
4	24	20	7919	20	Raw	-0.000035	-0.001399	0.001329	4
5	24	20	7919	80	DPO	0.000102	-0.001263	0.001467	5
6	24	20	7919	80	Affine	0.000446	-0.000926	0.001818	6
7	24	20	7919	80	Raw	-0.000012	-0.001376	0.001351	7
8	24	20	15401	20	DPO	0.000359	-0.001009	0.001727	8
9	24	20	15401	20	Affine	0.000623	-0.000759	0.002006	9
10	24	20	15401	20	Raw	-0.000022	-0.001386	0.001342	10
11	24	20	15401	80	DPO	0.000103	-0.001262	0.001468	11
12	24	20	15401	80	Affine	0.000438	-0.000934	0.001810	12
13	24	20	15401	80	Raw	-0.000013	-0.001376	0.001350	13
14	24	50	7919	20	DPO	0.000372	-0.001023	0.001768	14
15	24	50	7919	20	Affine	0.000665	-0.000746	0.002076	15
16	24	50	7919	20	Raw	-0.000015	-0.001403	0.001372	16
17	24	50	7919	80	DPO	0.000151	-0.001238	0.001539	17
18	24	50	7919	80	Affine	0.000496	-0.000905	0.001896	18
19	24	50	7919	80	Raw	-0.000009	-0.001396	0.001378	19
20	24	50	15401	20	DPO	0.000312	-0.001081	0.001705	20
21	24	50	15401	20	Affine	0.000650	-0.000761	0.002062	21
22	24	50	15401	20	Raw	-0.000035	-0.001423	0.001353	22
23	24	50	15401	80	DPO	0.000142	-0.001246	0.001531	23
24	24	50	15401	80	Affine	0.000491	-0.000909	0.001892	24
25	24	50	15401	80	Raw	-0.000016	-0.001403	0.001372	25
26	22	10	7919	20	DPO	0.000383	-0.000876	0.001642	26
27	22	10	7919	20	Affine	0.000855	-0.000422	0.002131	27
28	22	10	7919	20	Raw	-0.000058	-0.001312	0.001196	28
29	22	10	7919	80	DPO	0.000096	-0.001158	0.001351	29
30	22	10	7919	80	Affine	0.000582	-0.000682	0.001846	30
31	22	10	7919	80	Raw	-0.000013	-0.001266	0.001240	31
32									32

All 36 Ubuntu 24.04 and six repeated Ubuntu 22.04 direct comparisons. Endpoints are reconstructed from common-path differences within each environment; neither differences

of separate lower endpoints nor counts of training seeds constitute a method-population inference.

TABLE VII: Accounted work to each declared certified-gain target

OS	d	Seed	Method	Target	k	Prefix sec.	Executed sec.	Calls
24	10	7919	NBO	0.0000	20	18.68	21.20	1
24	10	7919	NBO	0.0005	20	18.68	21.20	1
24	10	7919	NBO	0.0010	–	–	–	–
24	10	7919	DPO	0.0000	20	16.80	18.46	1
24	10	7919	DPO	0.0005	80	34.69	34.69	2
24	10	7919	DPO	0.0010	–	–	–	–
24	10	7919	Affine	0.0000	80	20.23	20.23	2
24	10	7919	Affine	0.0005	–	–	–	–
24	10	7919	Affine	0.0010	–	–	–	–
24	10	7919	Raw	0.0000	20	17.34	19.28	1
24	10	7919	Raw	0.0005	20	17.34	19.28	1
24	10	7919	Raw	0.0010	–	–	–	–
24	10	15401	NBO	0.0000	20	18.12	20.87	1
24	10	15401	NBO	0.0005	20	18.12	20.87	1
24	10	15401	NBO	0.0010	–	–	–	–
24	10	15401	DPO	0.0000	20	16.35	18.23	1
24	10	15401	DPO	0.0005	80	34.04	34.04	2
24	10	15401	DPO	0.0010	–	–	–	–
24	10	15401	Affine	0.0000	80	19.66	19.66	2
24	10	15401	Affine	0.0005	–	–	–	–
24	10	15401	Affine	0.0010	–	–	–	–
24	10	15401	Raw	0.0000	20	15.06	17.22	1
24	10	15401	Raw	0.0005	20	15.06	17.22	1
24	10	15401	Raw	0.0010	–	–	–	–
24	20	7919	NBO	0.0000	20	27.73	30.86	1
24	20	7919	NBO	0.0005	–	–	–	–
24	20	7919	NBO	0.0010	–	–	–	–

	OS	d	Seed	Method	Target	k	Prefix sec.	Executed sec.	Calls	
1	24	20	7919	DPO	0.0000	20	25.28	27.45	1	1
2	24	20	7919	DPO	0.0005	–	–	–	–	2
3	24	20	7919	DPO	0.0010	–	–	–	–	3
4	24	20	7919	Affine	0.0000	–	–	–	–	4
5	24	20	7919	Affine	0.0005	–	–	–	–	5
6	24	20	7919	Affine	0.0010	–	–	–	–	6
7	24	20	7919	Raw	0.0000	20	23.95	26.33	1	7
8	24	20	7919	Raw	0.0005	–	–	–	–	8
9	24	20	7919	Raw	0.0010	–	–	–	–	9
10	24	20	15401	NBO	0.0000	20	26.88	30.06	1	10
11	24	20	15401	NBO	0.0005	–	–	–	–	11
12	24	20	15401	NBO	0.0010	–	–	–	–	12
13	24	20	15401	DPO	0.0000	20	25.54	27.79	1	13
14	24	20	15401	DPO	0.0005	–	–	–	–	14
15	24	20	15401	DPO	0.0010	–	–	–	–	15
16	24	20	15401	Affine	0.0000	–	–	–	–	16
17	24	20	15401	Affine	0.0005	–	–	–	–	17
18	24	20	15401	Affine	0.0010	–	–	–	–	18
19	24	20	15401	Raw	0.0000	20	25.53	28.09	1	19
20	24	20	15401	Raw	0.0005	–	–	–	–	20
21	24	20	15401	Raw	0.0010	–	–	–	–	21
22	24	50	7919	NBO	0.0000	20	54.60	58.08	1	22
23	24	50	7919	NBO	0.0005	20	54.60	58.08	1	23
24	24	50	7919	NBO	0.0010	–	–	–	–	24
25	24	50	7919	DPO	0.0000	20	53.12	55.66	1	25
26	24	50	7919	DPO	0.0005	–	–	–	–	26
27	24	50	7919	DPO	0.0010	–	–	–	–	27
28	24	50	7919	Affine	0.0000	80	92.37	92.37	2	28
29	24	50	7919	Affine	0.0005	–	–	–	–	29
30	24	50	7919	Affine	0.0010	–	–	–	–	30
31	24	50	7919	Raw	0.0000	20	52.08	54.94	1	31
32	24	50	7919	Raw	0.0005	20	52.08	54.94	1	32
	24	50	7919	Raw	0.0010	–	–	–	–	

OS	d	Seed	Method	Target	k	Prefix sec.	Executed sec.	Calls
24	50	15401	NBO	0.0000	20	52.48	56.16	1
24	50	15401	NBO	0.0005	20	52.48	56.16	1
24	50	15401	NBO	0.0010	–	–	–	–
24	50	15401	DPO	0.0000	20	48.93	51.55	1
24	50	15401	DPO	0.0005	–	–	–	–
24	50	15401	DPO	0.0010	–	–	–	–
24	50	15401	Affine	0.0000	80	86.84	86.84	2
24	50	15401	Affine	0.0005	–	–	–	–
24	50	15401	Affine	0.0010	–	–	–	–
24	50	15401	Raw	0.0000	20	49.93	52.87	1
24	50	15401	Raw	0.0005	20	49.93	52.87	1
24	50	15401	Raw	0.0010	–	–	–	–
22	10	7919	NBO	0.0000	20	14.93	16.87	1
22	10	7919	NBO	0.0005	20	14.93	16.87	1
22	10	7919	NBO	0.0010	–	–	–	–
22	10	7919	DPO	0.0000	20	13.22	14.70	1
22	10	7919	DPO	0.0005	80	27.63	27.63	2
22	10	7919	DPO	0.0010	–	–	–	–
22	10	7919	Affine	0.0000	80	16.26	16.26	2
22	10	7919	Affine	0.0005	–	–	–	–
22	10	7919	Affine	0.0010	–	–	–	–
22	10	7919	Raw	0.0000	20	13.16	14.88	1
22	10	7919	Raw	0.0005	20	13.16	14.88	1
22	10	7919	Raw	0.0010	–	–	–	–

A target is attained when the simultaneous lower endpoint strictly exceeds it. Every earlier verification is charged, including the 20-iteration verification when the first qualifying checkpoint is 80. Prefix seconds add the checkpoint fitting clock to cumulative saved verification clocks. Executed seconds charge the full 80-iteration fit, which precedes both evaluations in the executed program. These recorded components exclude interval-account construction and other untimed overhead. A dash means neither declared checkpoint attained that target.

TABLE VIII: Repeated environment: economic reproducibility and clock variation

Method	k	Max. gain diff.	Fit 24	Fit 22	Verify 24	Verify 22
NBO	20	1.63e-17	1.55	1.20	17.13	13.73
NBO	80	1.65e-17	4.07	3.14	15.42	12.62
DPO	20	1.25e-17	0.55	0.49	16.25	12.72
DPO	80	9.76e-18	2.22	1.98	16.22	12.92
Affine	20	4.23e-18	0.40	0.36	9.31	7.39
Affine	80	1.07e-17	1.59	1.45	9.33	7.41
Raw	20	1.28e-17	0.64	0.58	16.70	12.58
Raw	80	1.67e-17	2.58	2.30	17.17	12.58

Dimension 10, training seed 7919, identical final initial states and the same nominal innovation seed. Recorded innovation byte hashes differ between operating systems, so the realized innovation arrays are not asserted to be bitwise identical. Maximum differences are over individual paired-gain samples. Clocks are seconds for the recorded components and depend on the complete software and hardware environment. Both environments and tensor-storage differences are retained in the audit; this one-cell repeat is not a second economic inference sample.

5. INDEPENDENT SCALAR REFERENCE: EQUATIONS AND COMPLETE DIAGNOSTICS

5.1. *The scalar reference and its residual account*

The scalar reference isolates policy evaluation and policy improvement in a nonlinear member of the coupled-capital family. It uses the three stored scalar actors selected in the original experiment and the analytical anchor, without retraining. The four state–time–domain specifications are $(401, 256, 6)$, $(801, 512, 6)$, $(1601, 1024, 6)$ and $(1601, 1024, 8)$, where the last coordinate is the half-width L of the log-capital interval. All model coefficients, action limits, terminal values and boundary data remain those of the original reference. The diagnostic program checks the source and input hashes before computation and verifies the input hashes again afterwards.

Write $\eta = (\sigma_{\text{id}}^2 + \sigma_{\text{com}}^2)/2$, $c = \alpha - \eta$, and a for the adjustment-cost coefficient. The realized scalar coupling entry is $B_{11} = 1$. For scalar log capital y and consumption rate $m \in [\underline{m}, \bar{m}] = [0.02, 2]$, the running payoff and drift are

$$u(y, m) = y + \log m - \frac{a}{2}m^2, \quad \mu(y, m) = c + \beta \tanh y - m.$$

The terminal payoff is y ; the cross-sectional dispersion term vanishes in one dimension. Put $t_k = kh$, $h = T/n_t$ and let δ be the uniform state spacing. The original upwind generator is

$$(L_\delta^m v)_i = q_i^-(m)(v_{i-1} - v_i) + q_i^+(m)(v_{i+1} - v_i),$$

$$q_i^- = \frac{\eta}{\delta^2} + \frac{(-\mu_i)^+}{\delta},$$

$$q_i^+ = \frac{\eta}{\delta^2} + \frac{\mu_i^+}{\delta}.$$

Both coefficients are nonnegative. A fixed-policy backward step solves

$$\frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - (L_\delta^{m^k} v^k)_i - u(y_i, m_i^k) = 0$$

at every interior node. Boundary values enter neighboring differences exactly once. In particular, residuals are evaluated in this original equation, not by subtracting a boundary-adjusted right-hand side after reintroducing the boundary terms.

For Howard improvement, the Hamiltonian depending on m is

$$\log m - \frac{a}{2}m^2 + (d_i - m)^+ p_i^f + (d_i - m)^- p_i^b,$$

where

$$d_i = c + \beta \tanh y_i, \quad p_i^f = \frac{v_{i+1} - v_i}{\delta}, \quad p_i^b = \frac{v_i - v_{i-1}}{\delta},$$

where $z^- = \min(z, 0)$. On each of the intervals cut by $m = d_i$, the objective is strictly concave. Its unconstrained maximizer has the form $2/(p + \sqrt{p^2 + 4a})$, with the algebraically equivalent stable expression used when $p < 0$. Clipping this root to each nonempty interval and comparing the two attained values gives the global scalar action maximum for the upwind Hamiltonian. Thus the reported HJB residual

uses the full economic action interval, without action-grid enumeration or a neural critic.

The imposed boundary functions have the form $b_{\pm}(t) = \pm Lw(t) + Q_{\pm}(t)$, where

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad \bar{m}(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4a}},$$

and

$$Q_{\pm}(t) = \int_t^T e^{-\rho(s-t)} \left[\log \bar{m}(s) - \frac{a}{2} \bar{m}(s)^2 + w(s) \{c \pm \beta - \bar{m}(s)\} \right] ds.$$

The original cumulative-trapezoid rule and time interpolation evaluate these functions. They represent asymptotic boundary data shared by all policies; they are not independently verified exact boundary values for each learned policy in the unbounded economy. Their satisfaction at grid endpoints is an algebraic check, not a bound on economic boundary bias.

For the optimized grid value, define the original-equation HJB defect

$$r_i^k = \frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - \max_m \{u(y_i, m) + (L_{\delta}^m v^k)_i\}.$$

For a fixed policy, omit the maximization and use its stored feedback action. The diagnostic files retain the maximum residual at every time step, the linear residual of every Howard evaluation, every iterate difference, and the iteration count. The original stopping rule is an iterate difference below 10^{-11} , with a cap of 100 iterations. A separate Howard check evaluates $\|\mathcal{T}_k(v^k) - v^k\|_{\infty}/h$, where $\mathcal{T}_k$ solves the linear system for the greedy policy selected from v^k .

The significance of these residuals can be stated directly. Let $\tilde{v}$ be the exact solution of the same finite-grid backward equations, with the same frozen policy (or the same Bellman operator), terminal data and boundary data, and let v be the approximate sequence whose defect is r . In exact arithmetic, the fixed-policy matrix has positive diagonal, nonpositive off-diagonal entries, and row sums at least $h^{-1} + \rho$ after boundary elimination. The discrete maximum principle therefore yields

$$\|\tilde{v}^k - v^k\|_{\infty} \leq \frac{\|\tilde{v}^{k+1} - v^{k+1}\|_{\infty} + hR_k}{1 + \rho h}$$

whenever the boundary data agree and $\|r^k\|_\infty \leq R_k$. The same estimate holds for the Bellman step: at a positive maximum of the difference, select an action maximizing the Hamiltonian of the upper candidate and apply the fixed-action maximum principle; interchange the candidates for the negative part. With exact common terminal data this gives

$$\|\tilde{v}^0 - v^0\|_\infty \leq \sum_{k=0}^{n_t-1} \frac{hR_k}{(1+\rho h)^{k+1}} \leq \frac{\max_k R_k}{\rho} \{1 - (1+\rho h)^{-n_t}\}.$$

The archived propagation diagnostic substitutes the observed floating-point residuals in this formula. These residuals are not outward-rounded enclosures, so that substitution is a numerical diagnostic rather than a certified bound. The estimate concerns the finite-grid equations with the prescribed boundary data; it does not include consistency, state interpolation or boundary bias.

The tables report all four grids, the five initial states $y_0 \in \{-1, -0.5, 0, 0.5, 1\}$, the policies' action-bound frequencies, and changes in values, optimal actions and policy losses. Initial-state values are linearly interpolated from nodal time-zero values. Bound frequencies use an absolute tolerance of 10^{-12} and count equally weighted interior state-time nodes; they are not occupation probabilities. The first two grid comparisons halve both h and δ . Increasing L from 6 to 8 at fixed $(n_y, n_t) = (1601, 1024)$ changes δ from 0.0075 to 0.01. This comparison measures joint domain and spacing sensitivity, not boundary truncation error in isolation. Full common-domain changes and their locations accompany the central-region summaries.

The scalar calculation supplies an independent finite-grid benchmark for non-linear feedback, with a reproducible algebraic error account. In the 10-, 20- and 50-dimensional economies, cross-sectional dispersion, the coupling matrix and the covariance structure remain economically active. Moreover, this scalar benchmark evaluates Markov feedback, whereas the continuous-time deployment theorem specifies a randomized controller with innovation history. Scalar reference losses therefore have their stated one-dimensional meaning; multidimensional conclusions rely on the separate multidimensional policy comparisons and certification arguments.

TABLE IX

SCALAR FINITE-GRID POLICY LOSSES AT FIVE INITIAL STATES

Grid (n_y, n_t, L)	y_0	NBO	DPO	Affine	Anchor
(401,256,6)	-1	1.02×10^{-5}	6.78×10^{-5}	9.63×10^{-5}	1.16×10^{-4}
	-0.5	7.17×10^{-6}	8.61×10^{-5}	5.67×10^{-5}	5.64×10^{-4}
	0	2.55×10^{-6}	1.09×10^{-4}	1.93×10^{-4}	1.54×10^{-3}
	0.5	3.23×10^{-6}	1.04×10^{-4}	1.00×10^{-4}	1.68×10^{-3}
	1	9.79×10^{-5}	5.30×10^{-5}	1.69×10^{-4}	7.17×10^{-4}
(801,512,6)	-1	1.08×10^{-5}	6.93×10^{-5}	9.51×10^{-5}	1.17×10^{-4}
	-0.5	8.23×10^{-6}	8.96×10^{-5}	5.73×10^{-5}	5.70×10^{-4}
	0	3.12×10^{-6}	1.14×10^{-4}	2.00×10^{-4}	1.56×10^{-3}
	0.5	3.08×10^{-6}	1.04×10^{-4}	9.95×10^{-5}	1.67×10^{-3}
	1	1.02×10^{-4}	5.11×10^{-5}	1.72×10^{-4}	6.99×10^{-4}
(1601,1024,6)	-1	1.11×10^{-5}	7.00×10^{-5}	9.45×10^{-5}	1.17×10^{-4}
	-0.5	8.81×10^{-6}	9.13×10^{-5}	5.77×10^{-5}	5.74×10^{-4}
	0	3.44×10^{-6}	1.16×10^{-4}	2.03×10^{-4}	1.57×10^{-3}
	0.5	2.99×10^{-6}	1.03×10^{-4}	9.89×10^{-5}	1.67×10^{-3}
	1	1.05×10^{-4}	5.02×10^{-5}	1.74×10^{-4}	6.90×10^{-4}
(1601,1024,8)	-1	1.10×10^{-5}	6.98×10^{-5}	9.46×10^{-5}	1.17×10^{-4}
	-0.5	8.62×10^{-6}	9.08×10^{-5}	5.76×10^{-5}	5.72×10^{-4}
	0	3.32×10^{-6}	1.15×10^{-4}	2.02×10^{-4}	1.57×10^{-3}
	0.5	3.02×10^{-6}	1.04×10^{-4}	9.90×10^{-5}	1.67×10^{-3}
	1	1.04×10^{-4}	5.04×10^{-5}	1.73×10^{-4}	6.92×10^{-4}

Entries are the Howard grid value minus the value of the unchanged stored feedback policy, with common asymptotic Dirichlet data. Values at the five stated initial log-capital levels use linear interpolation of time-zero nodal values when necessary. No policy is refitted or chosen using this table. These are scalar finite-grid losses; they do not estimate the optimum in dimensions 10, 20 or 50.

TABLE X

SCALAR VALUE AND POLICY CHANGES UNDER REFINEMENT

Comparison	Policy	Core value	Core action	Core loss	Common value
$401 \rightarrow 801, L = 6$	Howard	6.59×10^{-4}	2.87×10^{-4}	0	6.59×10^{-4}
	NBO	6.57×10^{-4}	0	4.28×10^{-6}	6.57×10^{-4}
	DPO	6.61×10^{-4}	0	4.64×10^{-6}	6.61×10^{-4}
	Affine	6.60×10^{-4}	0	7.20×10^{-6}	6.60×10^{-4}
	Anchor	6.78×10^{-4}	0	1.91×10^{-5}	6.78×10^{-4}
$801 \rightarrow 1601, L = 6$	Howard	3.31×10^{-4}	1.45×10^{-4}	0	3.31×10^{-4}
	NBO	3.30×10^{-4}	0	2.22×10^{-6}	3.30×10^{-4}
	DPO	3.32×10^{-4}	0	2.40×10^{-6}	3.32×10^{-4}
	Affine	3.32×10^{-4}	0	3.72×10^{-6}	3.32×10^{-4}
	Anchor	3.40×10^{-4}	0	9.74×10^{-6}	3.40×10^{-4}
$L = 6 \rightarrow 8, n_y = 1601$	Howard	5.21×10^{-5}	4.72×10^{-5}	0	2.72×10^{-4}
	NBO	5.23×10^{-5}	7.59×10^{-7}	6.88×10^{-7}	5.03×10^{-4}
	DPO	5.28×10^{-5}	7.14×10^{-7}	8.13×10^{-7}	2.89×10^{-4}
	Affine	5.25×10^{-5}	1.02×10^{-8}	1.29×10^{-6}	7.37×10^{-3}
	Anchor	5.39×10^{-5}	0	3.63×10^{-6}	2.72×10^{-4}

Entries are maximum absolute changes in values, actions or policy losses. The first three numerical columns use time-zero nodes with $|y| \leq 1$; the final column uses the full common state interval. The second grid is linearly interpolated at nodes of the first. Policy loss is $\ell = V^H - V^m$. The first two comparisons halve both state spacing and time step. The last expands the domain at fixed n_y, n_t , changing state spacing from 0.0075 to 0.01; it is joint domain/spacing sensitivity and does not isolate boundary truncation error. Frozen-policy action changes arise only from interpolation and floating-point evaluation. Full common-interior action changes and their locations are archived.

6. FINITE-OBSERVATION DIAGNOSTIC ACCOUNTS

The experiment fixes the same saved NBO checkpoint before drawing the diagnostic bank. The sensor uses finite state measurements and independent private noise; the parent uses a clipped-innovation internal state. The deterministic allowance and the finite-Euler measurement answer different questions. Neither the observed mean nor its simulation standard error supplies the fine-Euler bias.

TABLE XI: Components of the finite-observation allowance

d	Cells	ν_{obs}	Production	Consumption	Terminal
10	64	0	0.000270	0.000833	0.000169
10	64	0.001	0.000687	0.001876	0.000361
10	64	0.01	0.004433	0.011263	0.002095
10	128	0	0.000137	0.000412	0.000087
10	128	0.001	0.000553	0.001432	0.000282
10	128	0.01	0.004295	0.010610	0.002037
20	64	0	0.000627	0.001756	0.000369
20	64	0.001	0.001535	0.003886	0.000787
20	64	0.01	0.009706	0.023047	0.004553
20	128	0	0.000320	0.000876	0.000191
20	128	0.001	0.001232	0.002973	0.000619
20	128	0.01	0.009442	0.021853	0.004471
50	64	0	0.001754	0.005424	0.001192
50	64	0.001	0.004613	0.013341	0.002877
50	64	0.01	0.022028	0.044889	0.008488
50	128	0	0.000912	0.002769	0.000634
50	128	0.001	0.003840	0.010758	0.002406
50	128	0.01	0.021850	0.043484	0.008503

Each component is displayed upward. The machine-readable records give the full recurrence, network Lipschitz and roundoff bounds, initial-profile bounds, seeds, pathwise diagnostic arrays, and the separate Monte Carlo standard errors.

7. EARLIER VERIFICATION AND APPLICATION RESULTS

The new policy-specific proofs appear first. The inherited supplement follows, with its historical iteration labels retained. Full application sections, prior proofs, and the R10 comparison are reproduced later without deleting their mathematical content. The corresponding main-text roots are archived by exact byte identity. Historical statements about the then-current revision are read in that temporal context; the R11 protocol, results, and response identify the present evidence.

8. PROOFS FOR POLICY-SPECIFIC VERIFICATION

This section proves the continuous-time transfer, clipping, and statistical statements used in the main text. Norms are normalized by dimension throughout. The internal controller and the economic state are distinct. This distinction is maintained in every comparison below.

8.1. Generator and stochastic local-error bounds

Let $f_i(y) = \kappa \tanh(B_i \cdot y)$, $b(y, m) = c\mathbf{1} + f(y) - m$, and let Σ have idiosyncratic columns $\sigma_I e_i$ and common column $\sigma_C \mathbf{1}$. For the ordinary spectral majorant $\beta \geq \kappa \|B\|_2$,

$$\|f(y) - f(z)\|_d \leq \beta \|y - z\|_d, \quad \|b(y, m)\|_d \leq M_\epsilon.$$

Each row of $Df(y)\Sigma$ has squared norm at most $\kappa^2 q_i$, so its normalized squared Hilbert–Schmidt norm is at most H^2 . Also

$$\left| \frac{1}{2} \text{tr}\{\Sigma \Sigma^\top D^2 f_i(y)\} \right| \leq \frac{\kappa}{2} q_i,$$

because $|\tanh''| \leq 1$. The generator of f under any held admissible action therefore satisfies

$$\|\mathcal{A}^m f(y)\|_d \leq \beta M_\epsilon + Q = K_\epsilon. \quad (1)$$

The controls used during the comparison are adapted, but fixed over each cell. The same bound applies to the reference’s deterministic, time-varying common control, with $\epsilon = 0$ and the time-dependent generator.

First take exact arithmetic and untruncated internal innovations. On cell k , Itô's formula and stochastic Fubini give

$$\begin{aligned} \int_{t_k}^{t_{k+1}} \{f(Y_s) - f(Y_{t_k})\} ds &= \int_{t_k}^{t_{k+1}} (t_{k+1} - v) \mathcal{A}^{a_k} f(Y_v) dv \\ &\quad + \int_{t_k}^{t_{k+1}} (t_{k+1} - v) Df(Y_v) \Sigma dW_v =: D_k + M_k. \end{aligned}$$

The deterministic term has L^2 norm at most $K_\epsilon h^2/2$. The stochastic terms are martingale increments over disjoint cells. Itô's isometry, not a sum of their absolute norms, gives

$$\left(\mathbb{E} \left\| \sum_{j < k} M_j \right\|_d^2 \right)^{1/2} \leq Hh \sqrt{t_k/3}.$$

Set $e_k = Y_{t_k} - Z_k$. The chosen actions cancel from the recursions. The triangle inequality yields

$$(\mathbb{E} \|e_k\|_d^2)^{1/2} \leq \beta h \sum_{j < k} (\mathbb{E} \|e_j\|_d^2)^{1/2} + h \{K_\epsilon t_k/2 + H \sqrt{t_k/3}\}.$$

The inhomogeneous term is nondecreasing in t_k . Discrete Gronwall and $(1 + \beta h)^k \leq e^{\beta t_k}$ prove (45). Importantly, no derivative of a_k with respect to its internal state appears in this recursion. Replacing a_k by a policy evaluated separately at Y_{t_k} and Z_k would destroy this cancellation and would be a different proposition.

For $t_k \leq t < t_{k+1}$, conditioning Itô's formula gives

$$\|\mathbb{E}\{f(Y_t) - f(Y_{t_k})\}\|_d \leq K_\epsilon(t - t_k).$$

Combining this inequality with the grid-point strong error bounds the absolute bias in the weighted mean-production sum by

$$\beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt.$$

The common components of drift, action, and Brownian noise vanish after applying II. The policy's coordinate deviations from its common schedule at grid dates are

at most ϵ . Therefore

$$(\mathbb{E}\|\Pi Y_T\|_d^2)^{1/2} \leq s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T} = S_\epsilon.$$

For any coupled Y, Z , Cauchy–Schwarz gives

$$\left| \mathbb{E}\{\|\Pi Y\|_d^2 - \|\Pi Z\|_d^2\} \right| \leq (\mathbb{E}\|Y - Z\|_d^2)^{1/2} \{2(\mathbb{E}\|\Pi Y\|_d^2)^{1/2} + (\mathbb{E}\|Y - Z\|_d^2)^{1/2}\}.$$

This proves the terminal term in (46). Applying the calculation once to the policy and once to the reference proves Proposition 8 in exact arithmetic.

8.2. Clipped innovations and numerical integration

Let g be a standard normal variable and $\text{clip}_z(g)$ its projection onto $[-z, z]$. Symmetry gives zero mean for $g - \text{clip}_z(g)$, and

$$\begin{aligned} \mathbb{E}\{g - \text{clip}_z(g)\}^2 &\leq \mathbb{E}[g^2 \mathbf{1}\{|g| > z\}] = 2\{z\varphi(z) + \Pr(g > z)\} \\ &\leq 2\varphi(z)(z + z^{-1}) =: v_z. \end{aligned}$$

Thus the accumulated internal-noise discrepancy has normalized L^2 norm at most $\sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$. It is added to the same Gronwall recurrence, giving the conservative enlargement $e^{\beta T} \sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$ at every grid point. This bounds a change in the controller’s internal update, not a replacement of the economic innovation law.

The scalar weights A_k, B_k are integrated from their exponential expressions. C_k , the reference action integral, and the within-cell production weight are enclosed on a dyadic subdivision with 64 panels per cell. Their midpoint errors are explicitly accumulated. If δ_M bounds the sum of the absolute errors in the integrated reference actions, $e^{\beta T} \delta_M$ enlarges the reference state error. The errors in A_k, B_k, C_k are multiplied by global bounds on consumption, its logarithm, and bounded production, and are added to the statistic error. These terms are not inferred from a difference between two grids.

8.3. Arithmetic contract and forward-error budget

All stored binary64 model coefficients denote exact real inputs to the economic problem. The computation assumes round-to-nearest elementary arithmetic, the

usual dot-product forward-error bound, and correctly rounded input conversion. It does not assume correct rounding of a platform exponential, logarithm, or hyperbolic tangent when proving a payoff endpoint. Network library functions produce proposals only; the interval schedule and inward projection enforce feasibility independently of proposal accuracy. The simulation drift and log utility use the polynomial kernels described next. The Python interpreter, numerical libraries, compiler, hardware, and random-number generator are not machine-formally verified.

For $|x| \leq 20$, write $r = x/256$ and approximate e^r by its degree-14 Taylor polynomial in Horner form. Nine squarings approximate e^{2x} . The Taylor remainder is bounded by $e^{|r|}|r|^{15}/15!$. The coefficient representation and Horner operations satisfy the standard γ_{30} forward bound, with $\gamma_j = ju/(1 - ju)$ and $u = 2^{-52}$, a conservative choice relative to binary64 unit roundoff. Propagating relative error through nine squarings bounds the exponential relative error by less than 2^{-35} on this range. The derivative of $1 - 2/(1 + z)$ with respect to $\log z$ is $2z/(1 + z)^2 \leq 1/2$. Including the final arithmetic and saturation outside $[-20, 20]$ therefore bounds absolute tanh error by 2^{-34} . The saturation error itself is below $2e^{-40}$.

For positive consumption, binary exponent extraction writes $x = 2^{e-1}(2m)$, with $m \in [1/2, 1)$. Then

$$\log x = 2 \sum_{j=0}^{\infty} \frac{q^{2j+1}}{2j+1} + (e-1) \log 2, \quad q = \frac{2m-1}{2m+1} \in [0, 1/3].$$

The implementation retains 18 terms and evaluates $\log 2$ with the same series at $q = 1/3$. The geometric remainder and Horner forward error, including the bounded exponents on the primitive consumption interval, are below 2^{-34} . No empirical comparison with the platform logarithm serves as this error bound. The optional compiled kernels use identical binary64 coefficients, operation order, and disabled fast-math transformations; the array implementation is retained as a fallback.

Clipped innovations yield a deterministic internal state cap. A conservative cap for the implemented ranges is

$$S_{\text{cap}} = 2\{\|y\|_{\infty} + M_{\epsilon}T + (\sigma_I + \sigma_C)10T/\sqrt{h} + 1\}.$$

Using $\gamma_{d+2} S_{\text{cap}} \max_i \sum_j |B_{ij}|$ for the matrix-product error, the code propagates drift, innovation-combination, and state-addition errors through the same discrete Gronwall bound. It verifies the bootstrap condition that this enlargement is less than one. This condition, together with the doubled cap, bounds the entire numerical path. Separately, it accumulates the errors of all statistic sums, log evaluations, and terminal squares. The exact formulas and intermediate constants are returned by `constants`; each final record saves the resulting state cap, forward-roundoff bound, scalar-integration bound, and total bias endpoint. Scalar transcendental evaluations used only to assemble conservative budgets receive additional 10^{-12} bias and 10^{-10} clipping cushions on the declared bounded parameter range; they do not replace the enclosed spectral or integral calculations.

The sampling contract is ideal independent standard-normal innovations rounded to binary64 before the specified clipping, with the forward-error allowance for their conversion. Pseudorandom draws are a reproducible implementation of this contract, not a proof of ideal independence. The resulting certificate is conditional numerical mathematics and conditional statistical inference. It must not be described as a machine-formal guarantee of a particular hardware random stream.

8.4. A global clipping envelope

Let Z^0 denote the internal reference trajectory. In exact arithmetic, synchronous coupling gives

$$\|Z_k^\phi - Z_k^0\|_d \leq D_k, \quad D_k = \epsilon_{\text{eff}} \{e^{\beta t_k} - 1\} / \beta,$$

where ϵ_{eff} also covers schedule movement within one cell, integrated-action error, and both forward-rounding increments. For the retained parameters, $|m'_0| \leq 1$, so $\epsilon + h$ suffices before these numerical enlargements. Define D_T analogously. Production differences have absolute sum at most $\beta \sum_k B_k D_k$. The consumption deficit is at most

$$\frac{\epsilon_{\text{eff}}^2}{2} \{[m_0(0) - \epsilon_{\text{eff}}]^{-2} + \zeta\} \int_0^T e^{-\rho t} dt.$$

The code verifies the strictly positive denominator. Terminal differences are bounded by $\chi e^{-\rho T} \{D_T^2 + 2D_T \|\Pi Z_T^0\|_d\}$.

The projected reference state satisfies

$$\|\Pi Z_T^0\|_d \leq s_0 + \kappa T + e_{\text{round}} + \frac{\sigma_I \sqrt{T}}{\sqrt{d}} H_0,$$

where H_0 is the Euclidean norm of the projected sum of the standardized clipped idiosyncratic innovations, normalized to unit total variance. As a function of all underlying independent standard Gaussians, H_0 is one-Lipschitz. Independence, symmetry, and the variance reduction under clipping give $\mathbb{E}H_0 \leq \sqrt{d-1}$. Gaussian concentration therefore gives $\Pr\{H_0 > \sqrt{d-1} + v\} \leq e^{-v^2/2}$. Collecting the deterministic terms defines C_ϕ and setting $A_\phi = 2\chi e^{-\rho T} D_T \sigma_I \sqrt{T/d}$ gives the claimed tail envelope. Integrating that envelope beyond $v_* > 0$ gives

$$\mathbb{E}[(|X_{\phi,h}| - b_\phi)_+] \leq A_\phi \int_{v_*}^{\infty} e^{-v^2/2} dv \leq A_\phi e^{-v_*^2/2} / v_*.$$

The construction does not require the learned network to be Lipschitz as a function of shocks; only its feasible action range is used in the deterministic separation bound.

For completeness, the Gaussian concentration fact can be obtained from the Gaussian logarithmic Sobolev inequality. If P_t is the Ornstein–Uhlenbeck semigroup, entropy differentiation and the identity $\nabla P_t = e^{-t} P_t \nabla$ imply $\text{Ent}(e^g) \leq \frac{1}{2} \mathbb{E}[e^g \|\nabla g\|^2]$ by Cauchy–Schwarz and integration of e^{-2t} . For $g = \lambda H_0$, the one-Lipschitz bound gives $\log \mathbb{E} e^{\lambda(H_0 - \mathbb{E}H_0)} \leq \lambda^2/2$ by integrating the entropy inequality in λ . Chernoff optimization yields the stated one-sided tail. Smooth approximation and truncation justify the calculation for the norm and clipping functions.

8.5. Finite families, selection, and interpretation

Conditioning on the fitted weights and their validation selection makes each final-test statistic a fixed function of one fresh path. Each row is therefore an independent sample across paths, even when its Brownian bank is shared with other policies. Applying [Maurer and Pontil \(2009, Theorem 4\)](#) to both signs and taking a union bound proves the simultaneous statement. The implementation allocates 2,000 one-sided statements, so at most 1,000 two-sided rows may be reported under this allocation. The audit rejects an overrun. It also checks source weights, path counts,

raw-array hashes, every recomputed mean and variance, and all displayed table inputs.

A certificate may fail to establish improvement. That is distinct from an exception in the numerical code. Both outcomes are retained. A zero correction, a deliberately saturated poor correction, and trained corrections are compared under the same accounting. Small training loss or successful completion is never substituted for a positive lower improvement endpoint. If a candidate is chosen after inspecting several final rows, the simultaneous guarantee continues to hold only within the predeclared finite family; an undeclared search requires a new allocation and independent data.

9. RELATION TO THE MAIN TEXT

This supplement expands the models and algorithms of the revised main manuscript. It does not introduce an alternative total loss or a different equilibrium definition. Policy evaluation freezes all actor outputs. Policy improvement freezes the critic's value and derivative channels. For games, only the acting player's objective is differentiated. For temporal selves, both exponential components evaluate the same future policy. The earlier main and supplementary sources are preserved in the revision archive for comparison; they are not alternative implementations of the revised method.

The inherited portfolio and quadratic laboratories retain their declared exact-structure classes. R6 trains generic multilayer policies in stopped consumption, preference adjustment, and dense nonlinear capital economies. R7 adds successful finite preference actors, neural game policies with full dynamic best responses, common-noise coverage records, and an independent arithmetic replay. R8 integrates those results and adds continuous-action training without maximizing labels, complete-action enclosures, direct and projection comparisons, and a separate continuous-transfer theorem. These approximation and verification classes are distinguished throughout the supplement.

10. POLICY EVALUATION AND THE RECURSIVE POLICY GRADIENT

Let a_ϕ be differentiable in a scalar parameter direction, and suppose V^{a_ϕ} is sufficiently differentiable for the following calculation. The terminal payoff is independent of the policy parameter. Assume differentiation under the expectation is justified, that the differentiated state and utility equations are integrable, and that the boundary conditions of the differentiated problem are zero. Denote the directional derivative of the policy value by w .

Differentiating the fixed-policy equation gives

$$0 = w_t + \mathcal{L}^{a_\phi} w + f_y(t, x, a_\phi, V^{a_\phi}) w + D_a \mathcal{H}^{a_\phi}[V^{a_\phi}] D_\phi a_\phi.$$

The operator derivative includes the dependence of both drift and covariance on the action. The derivative of the generator acting on the changing value is contained in $\mathcal{L}^{a_\phi} w$, while the action derivative of the generator acting on the fixed value is contained in $D_a \mathcal{H}$. No Hessian-action channel is discarded.

Applying the linear Feynman–Kac formula to the differentiated equation gives

$$w(t, x) = \mathbb{E}_{t,x}^{a_\phi} \int_t^\tau \exp \left(\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv \right) D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds.$$

This formula concerns the performance derivative from the stated initial state. A domain-uniform Hamiltonian objective uses a different weighting. Such an objective is useful for approximate pointwise improvement, but it becomes a lifetime policy gradient only with the appropriate occupation and adjoint weights. Trajectories sampled by the current actor are detached in the implemented domain objective. Differentiating through their sampling law would define a different gradient estimator and must be analyzed separately.

The critic’s residual derivative is the derivative of the entire fixed-policy equation. In particular, its level is determined by the recursive evaluation equation together with boundary data. A separate negative-Hamiltonian gradient in the critic is not an innocuous level anchor: in the scalar stationary example it moves the minimizer away from the Bellman root.

11. REPRESENTATION, BOUNDARY CONDITIONS, AND ERROR MEASUREMENT

11.1. *Smooth approximation and what it establishes*

A derivative-approximation result concerns a smooth target on a compact domain and a family of networks whose capacity can increase. It does not imply that a fixed network attains zero residual, that optimization finds its best approximation, or that a nonsmooth target has classical derivatives everywhere. In a smooth problem, increasing the representation class can reduce approximation error in the value and derivatives. In a nonsmooth problem, NBO instead evaluates a consistent positive-weight Bellman equation, or solves an explicitly regularized equation before taking a justified limit.

The hard-terminal architecture is $v(t, x) = G(x) + (T - t)N(t, x)$. It exactly enforces $v(T, x) = G(x)$ independently of training. A lateral Dirichlet face needs an additional compatible lifting and vanishing factor. For a reflected state, the Neumann condition is a different constraint. A loss on only interior points leaves all of these boundary questions unanswered. The regression test evaluates the terminal architecture at randomly chosen states and checks exact equality, not merely a small mean terminal error.

The recursive Epstein–Zin domain is $bV > 0$. An exponential output ensures positivity, while an interval transformation can keep bV between declared positive barriers. If $bG > 0$, one terminal-compatible positive representation is

$$bv(t, x) = \exp [\log(bG(x)) + (T - t)N(t, x)].$$

Its terminal value is exact. It does not by itself impose a lateral exit payoff or a two-sided barrier. These additional conditions must be built into the architecture or measured explicitly. Clipping an invalid utility argument inside a fractional power changes the equation and is not an acceptable hidden numerical repair.

11.2. *From residual samples to a uniform bound*

Let r be K -Lipschitz on a domain in $\mathbb{R}^d$. If every point is within distance h_V of a validation point, the triangle inequality gives $\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kh_V$. If K

is estimated from samples rather than bounded, the resulting expression is itself a diagnostic, not a certified uniform bound.

An integral bound needs more assumptions. Suppose the sampling density is at least $p_0 > 0$, and each ball of radius r_0 or less centered in the domain intersects the domain in volume at least $c_D r^d$. Let $M = \|r\|_\infty$ and assume $M/(2K) \leq r_0$. Near a point approaching the maximum, Lipschitz continuity gives $|r| \geq M/2$ on a ball of radius $M/(2K)$. Hence

$$\mathbb{E}r^2 \geq \frac{p_0 c_D}{2^{d+2} K^d} M^{d+2}, \quad M \leq \left(\frac{2^{d+2} K^d \mathbb{E}r^2}{p_0 c_D} \right)^{1/(d+2)}.$$

For larger M , use the capped radius r_0 and the corresponding two-case inequality. A concentration argument is additionally needed to replace the population mean by an empirical one. This calculation explains why a dimension-free uniform certificate cannot be read directly from an average training loss.

The action gap is treated similarly. If an action net has covering radius h_A and the Hamiltonian is K_A -Lipschitz in the action, the unobserved supremum exceeds the net maximum by at most $K_A h_A$. Thus the net gap plus this term is an upper bound. A collection of successful deviations without a covering or duality argument is only a lower bound on exploitability.

12. BENCHMARK DERIVATIONS

12.1. Merton consumption and transversality

Write $b = 1 - \gamma$ and $V = KX^b/b$. Then $V_X = KX^{-\gamma}$ and $V_{XX} = -\gamma KX^{-\gamma-1}$. Substituting $c = mX$ into the stationary time-additive equation and dividing by the positive number KX^b gives

$$0 = \frac{m^b}{bK} - \frac{\rho}{b} + r + p(\mu - r) - m - \frac{1}{2}\gamma p^2 \sigma^2.$$

Consumption maximization gives $m^{-\gamma} = K$; portfolio maximization gives $p = (\mu - r)/(\gamma\sigma^2)$ when interior. With a no-short-sale constraint and negative excess return, the maximizing share is zero. Let $R_* = r + p(\mu - r) - \gamma p^2 \sigma^2/2$. After substitution, $\gamma m = \rho + (\gamma - 1)R_*$.

For the optimal constant shares, $e^{-\rho t} \mathbb{E}[X_t^{1-\gamma}]$ decays as e^{-mt} . Thus $m > 0$ gives the candidate's additive transversality condition. Verification against a larger policy class also requires the relevant localization and integrability conditions for the competitors, as specified in the main text. Policies with minus-infinite utility are not preferred to the finite candidate. The numerical laboratory checks the candidate formulas and does not silently apply a stationary constant to an unspecified finite-horizon terminal problem.

12.2. Recursive coefficient and the utility interval

For $f(c, V) = \rho[c^a(bV)^{1-a/b} - bV]/a$, the same homothetic substitution gives

$$0 = \frac{\rho}{a}[m^a K^{-a/b} - 1] + R_* - m.$$

Since $f_c = V_X$ implies $\rho m^{a-1} K^{-a/b} = 1$, the first term becomes $(m - \rho)/a$. Consequently, $m = \psi\rho + (1 - \psi)R_*$ and $K = [\rho m^{a-1}]^{b/a}$. The formula assumes the stationary candidate has positive consumption. It is not an infinite-horizon utility-selection theorem.

For the finite-horizon matching-bequest test at $a > 0 > b$, define $Y = bV$. The consumption domain is compact and bounded away from zero. At sufficiently small positive Y , $c^a Y^{-a/b} < 1$ uniformly in consumption, so $f < 0$. At sufficiently large Y , the reverse inequality holds and $f > 0$. Because dividing by negative b reverses the order of the utility endpoints, the large- Y constant is a lower utility barrier and the small- Y constant is an upper utility barrier. The matching bequest lies between them after enlarging the interval. Lipschitz scalar comparison then keeps each policy utility inside the interval, on which the aggregator and its utility derivative are bounded. This verifies the finite-horizon interpretation without claiming an unrestricted infinite-horizon result at those parameters.

13. ENDOGENOUS-PREFERENCE DETAILS

13.1. State restrictions and the reference experiment

In the baseline application, the risk-aversion interval lies strictly above one and the process is reflected. The reference wealth problem is stopped at positive finite

bounds. Consumption shares are bounded away from zero, adjustment efforts are bounded, and the horizon is finite. These choices bound the flow utility, its relevant derivatives away from the terminal-boundary compatibility issue, and the bequest. They provide a well-defined finite reference calculation. The numerical reflection is a folded reflection of Brownian cubature endpoints; the wealth boundary is absorbing with the prescribed payoff. The same terminal and boundary values are imposed for both adjustment-cost parameters.

The grid reference operates in $y = \log X$. Its drift is

$$b_y = r + p(\mu - r) - m - \frac{1}{2}p^2\sigma_X^2,$$

and its covariance with u is $q\sigma_u p\sigma_X$. The two Brownian-factor columns can be written $(\sigma_u, qp\sigma_X)^\top$ and $(0, \sqrt{1 - q^2}p\sigma_X)^\top$. Four symmetric cubature directions reproduce this covariance. The correction $-p^2\sigma_X^2/2$ in log wealth is essential; omitting it would change the economic diffusion.

For each action, the backward update is current utility times the time step plus the exponentially discounted positive-weight interpolation of next-period value. Feasible actions are enumerated on the declared 27-element grid. Because the terminal/exit payoff is fixed, raising k weakly lowers each action's payoff and, by backward induction, lowers value at every state and time. The recorded arrays permit a direct check of this order. The reported grid-refinement difference is not an estimate of unrestricted neural approximation error.

13.2. Policy conditions and comparative statics

The consumption marginal utility in reference units is $U_c = \bar{c}^{-1}(c/\bar{c})^{-u}$. Equating it to V_X yields $c = \bar{c}(\bar{c}V_X)^{-1/u}$. Adjustment maximizes $\theta V_u - k\theta^2/2$ and hence uses the projection of V_u/k under a box. Portfolio choice maximizes a quadratic with coefficient $X^2\sigma_X^2 V_{XX}/2$. The projected first-order solution is valid only when this coefficient is negative. These conditions clarify why numerical curvature and feasibility must be checked independently.

The cross derivative V_{uX} is an endogenous object. Its sign cannot be inferred from the label "risk aversion" or from a negative shock correlation. Nor can the sign of

$\partial_k \theta^k$ be inferred by differentiating $1/k$ while keeping V_u^k fixed. In contrast, the value monotonicity in k holds policy by policy. At a differentiable optimum the envelope formula follows because k appears only in the quadratic running cost. Changing the normalization function $\kappa(u)$ can change both adjustment and portfolio policies even while leaving utility curvature in consumption unchanged.

An observation model is needed for structural estimation. With latent u , uncertainty in κ , and unknown adjustment or shock parameters, a sensitivity plot for one policy component does not prove identification. A practical estimation exercise would compute moment derivatives jointly, inspect their rank after normalization restrictions, and propagate numerical approximation error into the implied moment map. Those are distinct tasks from solving the controlled economy.

14. TEMPORAL-SELF DEVIATIONS

For the two-exponential kernel, the component value under constant consumption share m is $W_j(X) = m^b X^b / [b\lambda_j(m)]$, where $\lambda_j(m) = \rho_j - b(r - m)$. A deviation to share z for duration ϵ , followed by m , has component value at $X = 1$

$$J_j(z, \epsilon) = \frac{1}{b} \left[z^b \frac{1 - e^{-\lambda_j(z)\epsilon}}{\lambda_j(z)} + e^{-\lambda_j(z)\epsilon} \frac{m^b}{\lambda_j(m)} \right].$$

The continuous extension is used if $\lambda_j(z) = 0$. The deviation gain is $\sum_j w_j J_j(z, \epsilon) - \sum_j w_j W_j(1)$, with weights $(\beta, 1 - \beta)$. This is the quantity independently maximized in the experiment, rather than a restatement of the equilibrium first-order condition.

At an equilibrium share, the first-order gain is nonpositive for every feasible instantaneous share and equals zero at the equilibrium share. A finite-duration self can generally choose a slightly different share, so the maximized gain need not be exactly zero for positive duration. The required diagnostic is its gain divided by duration as duration tends to zero. For $\beta = 1$ the ordinary exponential solution is recovered.

As κ grows, the kernel becomes increasingly concentrated near the present before approaching its long-run component. Taking the instantaneous-deviation limit and taking $\kappa \rightarrow \infty$ are conceptually different operations. The finite- κ model in this paper explicitly specifies the order: the spike condition is derived for fixed κ , and

any duration-limit comparative static is taken afterward. This avoids assigning an undefined continuous-time meaning to a discrete present-bias multiplier.

15. GAMES AND INDEPENDENT BEST RESPONSES

A fixed rival profile turns each player's problem into an ordinary controlled Markov problem, so the main verification results apply player by player. The derivative graph must implement this reduction. During player i 's actor step, every rival policy and every derivative of player i 's critic is detached. During player i 's critic step, all players' actions are detached. A sum of these reporting losses can be logged, but differentiation of a joint-profit objective is not equivalent to those block steps.

The two-firm static diagnostic has best response $q_i = (1 - q_j)/2$ when interior. At the joint-profit allocation $(1/4, 1/4)$, the deviating firm chooses $3/8$ and improves profit from $1/8$ to $9/64$, a gain of $1/64$. The own-action derivative is $1/4$ at the joint optimum. These numbers verify a graph-level distinction; they are not a computation of the dynamic capital game.

For the dynamic problem, a reported equilibrium should include an independent solution of each fixed-rival best-response problem, or an upper bound on the unilateral Hamiltonian gap over the entire verification domain. A finite list of unsuccessful deviations does not supply such an upper bound. Multiple initializations, asymmetric profiles, boundary actions, and the market-size normalization are part of the same experiment. The specification in the main text retains the dynamic Cournot application but does not invent a dynamic Nash table from the static arithmetic.

16. STOCHASTIC HESSIAN OBJECTIVES AND COMPLEXITY

Let $B = \Sigma^\top D^2 v \Sigma$, a symmetric matrix in Brownian-factor space. For Gaussian probes, Isserlis' identity gives $\mathbb{E}(z^\top B z) = \text{tr } B$ and $\text{Var}(z^\top B z) = 2 \text{tr}(B^2) = 2 \|B\|_F^2$. Averaging K independent probes divides the variance by K . Rademacher probes instead have variance $2 \sum_{i \neq j} B_{ij}^2 / K$. The variance depends on the chosen covariance factor and probe law, both of which must be specified.

Write $\hat{q}_j = q + \epsilon_j$, with conditionally mean-zero independent errors. Then

$$\mathbb{E}[(d - \hat{q}_1/2)(d - \hat{q}_2/2)] = (d - q/2)^2 + \frac{1}{4}\mathbb{E}[\epsilon_1\epsilon_2] = (d - q/2)^2.$$

Using one bank twice would retain its variance term. In the scalar Gaussian example, a single-bank squared residual has expectation $1 - h + 3h^2/4$, minimized at $h = 2/3$, whereas the exact residual $(1 - h/2)^2$ is minimized at $h = 2$. The independent product restores the latter population objective. The saved Monte Carlo diagnostic uses 400,000 pairs and records both empirical estimates; it does not claim that the sampling error is identically zero.

The core implementation uses exact Hessians to make the covariance contraction independently testable. A Hessian-vector implementation can replace it without changing the conditional-independence requirement. The cost of a Hessian-vector product depends on the network graph, not just the state dimension. Total cost also includes sampling, optimization, domain coverage, action maximization, and stopping at an agreed error level. A quadratic-neural representation with a Lyapunov solve has different approximation and optimization costs from a generic multi-layer network. The inherited quadratic laboratory is explicitly the former; the new dense nonlinear experiments actually train generic networks and Hessian-vector estimators.

17. REPRODUCTION AND FURTHER COMPARISONS

From the repository root, execute the experiment program and then the regression suite. The dependency versions and CPU setting are recorded with the results. The experiment output distinguishes acceptance of a calibration or identity test from execution of a reference computation. A separate reviewer-check output reproduces the original twelve diagnostics against the archived original manuscript, not against the revised article.

A broader method comparison should use one problem specification and evaluate value error, policy error, boundary error, and the independent action gap at matched accuracy. A direct neural-HJB comparator eliminates the actor with the same feasible maximization routine. A temporal-regression comparator uses the same recursive discretization and utility domain, without adding discount twice. A separability-

aware classical comparator exploits all available economic structure. Network capacity, sampling, covariance treatment, random seeds, failures, wall time, and memory should all be reported. R6 executes the architecture-matched direct-HJB and finite-Bellman comparisons. The broader deep-BSDE comparison remains unexecuted and is not counted as a measured advantage. R7 neural-game evidence and the additional R8 comparisons are reported in the integrated sections below.

The inherited read-only workflow is preserved. The separately scoped R6 replication workflow commits materialized source and raw recomputation outputs only to its designated candidate branch, without editing the review or earlier revision branches. Hashes and exact source commits distinguish these artifacts. No workflow rewrites a failed numerical accuracy target as a pass.

18. IMPLEMENTED ERROR ACCOUNTS AND ARITHMETIC

The implemented map returns a candidate and a verification record, rather than assigning authority to an optimizer's exit code. In the differential consumption economy, the record identifies the binary64 weights, both stopping faces, all dyadic leaves, global consumption maximization, and the error bounds for the continuous diffusion. In NDU it instead identifies the finite state-action-time model, its positive transition weights, and time-indexed evaluation and improvement errors. The dynamic-game record fixes the entire rival policy before an independent backward best-response solve. An unchecked continuous-action or time-discretization term is represented as unavailable in the machine-readable output, not as zero.

For finite time levels, let $\delta_t = \|v_t - T_t^a v_{t+1}\|_\infty$ and $\epsilon_t = \|T_t v_{t+1} - T_t^a v_{t+1}\|_\infty$. If both maps are β_t -Lipschitz, the triangle inequality gives

$$\|v_t - V_t^a\| \leq \delta_t + \beta_t \|v_{t+1} - V_{t+1}^a\|, \quad \|v_t - V_t^*\| \leq \delta_t + \epsilon_t + \beta_t \|v_{t+1} - V_{t+1}^*\|.$$

Summing the two inequalities yields the regret account. The same recurrence with a player's unilateral gap bounds the gain from a dynamic deviation when that player's policy evaluation is exact. Accounts must be checked at each time level: a time-zero bound need not dominate a later-time bound when discounting and errors differ. The coarse-game regression exercises this distinction and verifies that a missing pure equilibrium is reported rather than silently treated as exact.

18.1. *Full-domain interval implementation*

An interval $[l, u]$ encloses an exact real quantity. Each elementary operation expands its rounded endpoints with the adjacent binary64 number. Affine layers split positive and negative weights and include a conservative dot-product forward-error allowance. The arithmetic contract requires binary64 round-to-nearest arithmetic, the stated dot-product error bound, and absence of exceptional overflow or flush-to-zero behavior. Stored parameters denote their exact binary64 real values; the verifier does not assert an unrecorded decimal model identity.

For an exponential argument x , range reduction gives $r = x/2^s$ with $|r| \leq 1/8$. A degree-fourteen Taylor polynomial has remainder bounded by

$$e^{|r|}|r|^{15}/15! < 2(1/8)^{15}/15! < 2^{-80}.$$

Interval Horner evaluation and repeated squaring enclose e^x . The identity $\tanh x = 1 - 2/(e^{2x} + 1)$ then supplies $\tanh$ values and derivative channels $1 - \tanh^2 x$ and $-2 \tanh x(1 - \tanh^2 x)$. Powers of two and the range-reduction domain are checked in the code.

For logarithms, exact binary decomposition gives $x = 2^k m$, with $1 \leq m < 2$. Writing $z = (m - 1)/(m + 1)$,

$$\log m = 2 \sum_{j=0}^{24} \frac{z^{2j+1}}{2j+1} + R, \quad |R| \leq \frac{2|z|^{51}}{51(1-z^2)} < 2^{-72}$$

for the outward-enclosed range $|z| < .334$. The same expansion encloses $\log 2$. Neither construction presumes correctly rounded library $\tanh$ or logarithm functions. The saved test suite checks transcendental endpoints against eighty-digit arithmetic, differentiates the trained network independently, and checks all certificate hashes and leaf adjacencies. These tests supplement, but do not replace, the arithmetic contract.

The scalar action objective $h(c) = \log c - pc$ has second derivative $-1/c^2$. If C encloses both the actor and its maximizing alternative, strong concavity on their joining segment gives

$$h(c^*) - h(c) \leq h'(c)(c^* - c) - (c^* - c)^2/(2C^2) \leq C^2 h'(c)^2/2.$$

The argument includes constrained endpoints. Taking the minimum with the natural interval action-gap enclosure markedly reduces verification cost, without using a sampled gradient as a bound. Residual targets are decided on every leaf; reaching a maximum subdivision depth does not change a failed target into a pass.

18.2. *Application-specific hypotheses*

The stopped consumption diffusion is uniformly nondegenerate on its compact domain; controls and coefficients are bounded and its endpoint data are continuous. The smooth networks and strictly positive consumption map meet the regularity and log-domain requirements of the verification argument. Both stopping-face errors and the full action interval are checked. The upwind reference error is not a premise of its continuous certificate.

The NDU computation is a positive-weight, finite-action, finite-horizon economy with prescribed discrete stopping and folded preference transitions. Positivity, all terminal and wealth-face values, and every finite action are evaluated. The continuous reflected PDE's corner compatibility, continuous-action coverage error, and diffusion approximation error are not inferred from these tests. Expanding a feasible box changes the constrained economy; densifying a fixed box does not. The sensitivity record keeps these operations separate.

The dense capital computation uses bounded nonlinear drift in log coordinates, bounded positive consumption fractions, constant diffusion, and finite horizon. Its flow has at most linear log-state growth and its terminal payoff at most quadratic growth; these are integrable under the declared bounded-drift diffusion. The fitted box is a sampling domain, not a state constraint. Its sampled differential residuals do not check comparison barriers or uniform errors on $\mathbb{R}^d$. Euler payoff estimates carry both sampling error and unbounded-state/time-discretization issues not covered by the reported Monte Carlo standard error.

The dynamic capital game is a fully enumerated finite approximation with stopped joint boundaries. Stage equilibria and dynamic fixed-rival best responses are numerical statements about that model. The continuous-control best-response comparison assumptions in the main theorem remain logically distinct. The re-

tained homothetic recursive-utility and temporal-self examples use their stated utility domains, bequests, and integrability conditions; R6 does not relabel them as unrestricted neural experiments.

19. DESIGN, BUDGETS, AND REPRODUCIBILITY

The stopped-consumption study uses float64 and the existing generic derivative core. Identical seed labels fix initialization and collocation construction across methods. L-BFGS iteration caps are matched, but line-search closure counts need not match. Those counts, fit times, and separate verification times are saved. The grid reference uses exact maximization within each upwind drift region, not the learned actor. Reference comparisons omit four endpoint-adjacent nodes only for finite-difference derivative comparisons; all endpoints enter the boundary check and the interval certificate.

The NDU actor predicts one of the finite feasible triples. Its normalized Bellman-loss objective gives economically costly action mistakes more weight than ordinary classification would. Absorbing boundary rows are excluded only from actor fitting, because their actions have no effect; they remain in policy evaluation and verification. The failed initial classification implementation is archived with its actual outputs. Principal seed 11 misses the declared policy-regret target even after the improvement; its result is not discarded. The direct finite-Bellman comparator uses the same critic class and total inner-iteration cap.

The dense-capital primary study uses exact Hessians; the trace follow-up actually backpropagates through Hessian-vector products. Product and single-bank implementations use four probes in total, but only the product splits them into conditionally independent banks. The Monte Carlo regression separately checks the value and gradient of the product objective with tolerances .012 and .025. An unbiased objective can have higher variance, and no claim of superior finite-sample accuracy follows merely from its expectation. The covariance in the retained experiment is modest; the reported results reflect that scale.

The initial high-dimensional state box was too narrow to certify the realized policy paths. The wider-box design is therefore identified as a follow-up, not a preregistered experiment. Forty- and eighty-step policy simulations share the same

seed convention across methods at a given resolution. They are not a coupled Brownian extrapolation across resolutions. Raw per-path payoffs allow paired policy comparisons, but neither a Monte Carlo confidence interval nor a critic-payoff agreement proves closeness to the optimal value.

Each executed program records all declared seeds, its budget, residual or deviation arrays, and an explicit scope. Process peak resident memory is a high-water mark of the process, potentially including earlier runs. It is not an isolated memory-complexity measurement for an individual method. CPU timings are execution-specific and do not reproduce the historical accelerator claims. The independent raw-output workflow binds its source commit and file hashes; the source archive keeps the R2 manuscript and supplement byte-for-byte for comparison.

20. RUN-LEVEL NUMERICAL ACCOUNTS

20.1. *Consumption fitting and verification*

Method	Seed	Width/depth	Fit seconds	Verify seconds	Pass
nbo	11	16/2	2.1790	1.0822	True
nbo	29	16/2	2.0422	1.4045	True
nbo	47	16/2	2.0550	1.3100	True
direct	11	16/2	1.9767	0.4974	True
direct	29	16/2	0.8018	0.4875	True
direct	47	16/2	2.1912	0.5481	True
joint	11	16/2	0.2578	–	False
joint	29	16/2	2.5836	–	False
joint	47	16/2	2.5935	–	False
nbo	11	8/2	2.5767	–	True
nbo	11	32/2	1.6599	–	True
nbo	11	16/1	1.6198	–	True
nbo	11	16/3	2.3456	–	True
nbo	11	16/2	2.1977	–	True
nbo	11	16/2	1.8599	–	True

Pass in this table means the prespecified sampled diagnostic. Continuous certification is reported separately and only for the primary NBO and direct-HJB runs.

Every closure count, pointwise derivative array, parameter, and iteration trace is in the corresponding raw result.

20.2. Action bounds and numerical domains

Actions	Grid/time	k	Center value	Upper m freq.	Upper p freq.
r2	17x25/20	1.0000	-11.9152	0.9994	0.7885
r2	17x25/20	5.0000	-11.9570	0.9992	0.7890
adjustment	17x25/20	1.0000	-10.6081	1.0000	0.8364
adjustment	17x25/20	5.0000	-10.9202	1.0000	0.8405
expanded	17x25/20	1.0000	-4.8410	0.9898	0.5440
expanded	17x25/20	5.0000	-4.9243	0.9855	0.5515
refined	17x25/20	1.0000	-4.8406	0.9737	0.5301
refined	17x25/20	5.0000	-4.9214	0.9688	0.5353
expanded_domain	17x25/20	1.0000	-2.7504	0.9698	0.5665
expanded_domain	17x25/20	5.0000	-2.8452	0.9499	0.6068
expanded	17x25/20*	1.0000	-5.0185	0.9997	0.3164
expanded	17x25/20*	5.0000	-5.1095	0.9971	0.3251
expanded	25x37/40	1.0000	-4.8395	0.9902	0.5738
expanded	25x37/40	5.0000	-4.9228	0.9853	0.5787
expanded	33x49/80	1.0000	-4.8302	0.9872	0.6050
expanded	33x49/80	5.0000	-4.9143	0.9823	0.6102

An asterisk marks wealth interval $[\cdot 25, 4]$, rather than $[\cdot 5, 2]$. Frequencies use all nonabsorbing state–time nodes. The raw record includes both bounds for each of the three controls. “Expanded has 175 actions; “refined has 1,053 actions in the same box; “expanded_domain changes the box itself.

20.3. Dense neural runs, trace estimators, and coverage

Method	d/seed	Steps	RMS	Gap	Seconds	Pass
exact	2/11	600	0.0125	1.44×10^{-4}	2.7329	True
direct	2/11	600	0.0123	0	3.4870	True

	Method	d/seed	Steps	RMS	Gap	Seconds	Pass	
1								1
2	exact	2/29	600	0.0161	2.28×10^{-4}	2.6554	True	2
3	direct	2/29	600	0.0161	0	3.4759	True	3
4	exact	2/47	600	0.0132	2.42×10^{-4}	2.6361	True	4
5	direct	2/47	600	0.0136	0	3.5572	True	5
6	exact	5/11	600	0.0101	0.0012	4.1518	True	6
7	direct	5/11	600	0.0101	0	5.0391	True	7
8	exact	5/29	600	0.0131	8.38×10^{-4}	4.2150	True	8
9	direct	5/29	600	0.0129	0	4.9992	True	9
10	exact	5/47	600	0.0122	0.0020	4.1258	True	10
11	direct	5/47	600	0.0118	0	5.0485	True	11
12	exact	10/11	600	0.0163	0.0052	6.8652	True	12
13	direct	10/11	600	0.0138	0	7.6139	True	13
14	exact	10/29	600	0.0188	0.0035	6.9393	True	14
15	direct	10/29	600	0.0168	0	7.7251	True	15
16	exact	10/47	600	0.0141	0.0050	6.8241	True	16
17	direct	10/47	600	0.0139	0	7.6988	True	17
18	exact	20/11	600	0.0148	0.0113	12.8180	False	18
19	direct	20/11	600	0.0135	0	13.5001	True	19
20	exact	20/29	600	0.0138	0.0090	12.6025	True	20
21	direct	20/29	600	0.0129	0	13.4929	True	21
22	exact	20/47	600	0.0320	0.0114	12.9103	False	22
23	direct	20/47	600	0.0170	0	13.6205	True	23
24	product	10/11	600	0.0163	0.0052	3.9801	True	24
25	single	10/11	600	0.0162	0.0052	3.9075	True	25
26	product	10/29	600	0.0187	0.0035	3.9346	True	26
27	single	10/29	600	0.0186	0.0034	3.9263	True	27
28	product	10/47	600	0.0141	0.0050	3.9233	True	28
29	single	10/47	600	0.0140	0.0050	3.9264	True	29
30	exact*	10/11	1000	0.0236	0.0019	11.1414	True	30
31	direct*	10/11	1000	0.0278	0	12.4295	False	31
32	exact*	10/29	1000	0.0249	0.0022	10.9333	True	32
	direct*	10/29	1000	0.0276	0	12.3312	False	
	exact*	10/47	1000	0.0281	0.0026	11.0150	False	
	direct*	10/47	1000	0.0293	0	12.3768	False	
	exact*	20/11	1000	0.0200	0.0040	20.8834	True	
	direct*	20/11	1000	0.0211	0	22.9271	True	

Method	d/seed	Steps	RMS	Gap	Seconds	Pass
exact*	20/29	1000	0.0200	0.0034	20.0811	True
direct*	20/29	1000	0.0227	0	21.9897	True
exact*	20/47	1000	0.0216	0.0071	19.9588	True
direct*	20/47	1000	0.0244	0	21.7639	True

An asterisk marks the follow-up fitting domain $[-1.5, .5]^d$ and 1,000 updates. Product uses two independent two-probe banks; single squares a four-probe trace. Every row retains its own history and exact-trace diagnostic set. Runtime is not a hardware-independent complexity bound.

20.4. Policy-payoff diagnostics

Method	d/seed	Critic at zero	Euler mean	MC s.e.	Critic minus mean
exact	10/11	-1.1925	-1.3045	0.0041	0.1120
direct	10/11	-1.2054	-1.2729	0.0036	0.0675
exact	10/29	-1.1781	-1.2999	0.0040	0.1218
direct	10/29	-1.1899	-1.2701	0.0036	0.0802
exact	10/47	-1.2080	-1.2967	0.0040	0.0887
direct	10/47	-1.2077	-1.2698	0.0036	0.0621
exact	20/11	-1.1809	-1.3458	0.0041	0.1649
direct	20/11	-1.1986	-1.2834	0.0035	0.0848
exact	20/29	-1.1920	-1.2895	0.0035	0.0975
direct	20/29	-1.2071	-1.2754	0.0034	0.0684
exact	20/47	-1.0343	-1.5052	0.0036	0.4709
direct	20/47	-1.1661	-1.3051	0.0036	0.1390
exact*	10/11	-1.2974	-1.2521	0.0035	-0.0453
direct*	10/11	-1.2992	-1.2527	0.0035	-0.0465
exact*	10/29	-1.2933	-1.2521	0.0035	-0.0412
direct*	10/29	-1.2871	-1.2521	0.0035	-0.0350
exact*	10/47	-1.2939	-1.2527	0.0035	-0.0412
direct*	10/47	-1.2959	-1.2527	0.0035	-0.0433
exact*	20/11	-1.2788	-1.2646	0.0034	-0.0142
direct*	20/11	-1.2787	-1.2656	0.0034	-0.0132

Method	d/seed	Critic at zero	Euler mean	MC s.e.	Critic minus mean
exact*	20/29	-1.2712	-1.2646	0.0034	-0.0066
direct*	20/29	-1.2563	-1.2689	0.0034	0.0126
exact*	20/47	-1.2772	-1.2645	0.0034	-0.0127
direct*	20/47	-1.2821	-1.2652	0.0034	-0.0169

These estimates use 2,048 paths and 80 Euler steps from zero log capital. Forty-step samples are also saved. Identical random seeds pair policy comparisons at the same step size; the forty- and eighty-step simulations are not a Brownian-coupled refinement estimator. The standard error measures sampling error only. The initial critic is not an external optimal-value reference.

21. PROOF AND IMPLEMENTATION OF THE CONTINUOUS-ACTION ENVELOPE

We use the notation of the main text’s budget-safe envelope proposition. The nodal economy is fixed before verification. Its terminal and stopping payoffs, primitive coefficients, interpolation abscissae, and compact action box are part of the object being bounded. This is stronger than checking many action samples and weaker than verifying the continuous-state diffusion without a discretization account.

21.1. *Complete-cover invariant and backward proof*

At a given state and time, the initial action box covers the entire feasible set. Bisection replaces one closed box by two closed boxes with the same union. Their shared face is harmless. Induction on the number of subdivisions preserves coverage. A pruned box is retained logically through the inequality $u_B \leq c_i + \tau$, where u_B is an outward upper enclosure on that whole box, and c_i is a lower enclosure at a feasible incumbent. If the incumbent later increases, the previous pruning inequality remains true. Consequently every feasible action is either in an unresolved box or in a pruned box with payoff at most the current threshold. Taking the maximum of the threshold and all unresolved upper bounds dominates the action supremum, whether or not the supremum is attained. Termination on a numerical budget does not change this argument.

At N , the terminal inequalities are given. If $V_{n+1}^* \leq U_{n+1}$, positivity gives $T_n V_{n+1}^* \leq T_n U_{n+1} \leq U_n$. If $L_{n+1} \leq V_{n+1}^\pi$, the same argument gives $L_n \leq T_n^\pi L_{n+1} \leq T_n^\pi V_{n+1}^\pi = V_n^\pi$. Feasibility implies $V_n^\pi \leq V_n^*$. This proves the main envelope inequality. Substochasticity further gives

$$\|T_n w - T_n z\|_\infty \leq q_n \|w - z\|_\infty.$$

Applying this inequality to $U_n - T_n V_{n+1}^*$ yields the stated excess recursion and its product-sum expansion. The proof uses $q_n > 0$, not a stationary contraction assumption. Outward addition is used for the pruning threshold and final subtraction. The implementation records a requested tolerance and a retained optimization bracket separately. A budget-exhausted bracket is never replaced by that tolerance. *Q.E.D.*

21.2. Interpolation, reflection, stopping, and arithmetic

For a cell with coordinates $(u_0, u_1) \times (y_0, y_1)$, the continuation is

$$Iw(u, y) = (1 - s)(1 - t)w_{00} + s(1 - t)w_{10} + (1 - s)tw_{01} + stw_{11},$$

$$s = \frac{u - u_0}{u_1 - u_0}, \quad t = \frac{y - y_0}{y_1 - y_0}.$$

For fixed u , this is affine in y , so its extremes on a subrectangle occur at the two y endpoints. At either endpoint it is affine in u , so the extremes occur at the two u endpoints. Applying this argument to every intersected cell proves the corner enclosure. Nonnegative interpolation weights also permit interval nodal inputs: evaluating the corner expressions outward encloses every admissible nodal realization. A separate valid enclosure evaluates the rectangle center and adds cell-slope bounds over every intersected cell. Slopes are computed from stored nodal differences, not estimated from random derivative samples.

Preference reflection is piecewise affine. Its extrema on an interval occur at the interval endpoints or a crossed folding point. The allowed one-step movement is checked to exclude unhandled multiple folds. For log wealth, boxes fully inside the domain use interpolation, boxes fully outside use the prescribed clipped exit payoff, and boxes crossing an exit face use the union of both. On the stored grid the boundary nodes equal the exact prescribed payoff; at an off-grid exit the analytic

1 payoff need not equal its nodal interpolant. The union construction avoids an 1
 2 unjustified global smoothness or Lipschitz assumption across that switch. 2

3 The implementation inherits the archived outward elementary and transcen- 3
 4 dental routines. Products, quotients, powers through logarithm and exponential, 4
 5 and endpoint rounding are enclosed. The logarithm's input is bounded below by 5
 6 .02; the preference denominator is separated from zero because $u \in [1.2, 3]$. The 6
 7 recorded control model treats the stored binary64 primitive coefficients and ab- 7
 8 scissae as exact real inputs. The evidence also tests agreement with the inherited 8
 9 training evaluator on the finite action grid. Such a regression check is useful for 9
 10 implementation consistency but is not the proof of global action coverage. That 10
 11 proof is the maintained union invariant and the interval inclusion property. The 11
 12 arithmetic account is conditional on the specified floating-point execution model, 12
 13 not a formal verification of hardware, libraries, or the Python interpreter. 13

14 21.3. *A common upper envelope for multiple policies* 15

16 The optimal upper array depends on the economy, not on which feasible incum- 16
 17 bent produced the lower search values. Thus the completed upper array can be 17
 18 reused for any frozen feasible policy on the same state, time, and continuous-action 18
 19 domains. We separately propagate each policy's lower and upper payoff arrays and 19
 20 subtract its lower array from the common optimal upper array. Costs are reported 20
 21 both as shared verification and as the cost of an individual policy evaluation. Shar- 21
 22 ing a verifier does not make its cost vanish. It also does not justify transferring the 22
 23 upper array to a changed wealth box, action constraint, utility normalization, time 23
 24 step, or interpolation rule. 24

25 22. SEPARATING NUMERICAL AND CONTINUOUS-ECONOMY ERROR 26

27 The following result specifies the transfer work needed in addition to a nodal cer- 27
 28 tificate. It is not used to set an uncomputed transfer term to zero. Let $\mathcal{T}_n$ be the exact 28
 29 economic one-step optimality operator with its continuously monitored stopping 29
 30 and reflection rule, and let $\mathcal{T}_n^\pi$ be the operator for a declared continuous-state im- 30
 31 plementation of the numerical policy. Let I_n be a positive, sup-norm-nonexpansive 31
 32 32

interpolation from grid values to the economic state domain. Suppose the exact operators are q_n -Lipschitz on the comparison class.

PROPOSITION 1—Transfer with separate optimal and policy defects: *Suppose terminal interpolation error is at most B and, along the relevant nodal continuations,*

$$\|\mathcal{T}_n I_{n+1} v_{n+1}^* - I_n T_n v_{n+1}^*\|_\infty \leq d_n^*,$$

$$\|\mathcal{T}_n^\pi I_{n+1} v_{n+1}^\pi - I_n T_n^\pi v_{n+1}^\pi\|_\infty \leq d_n^\pi.$$

Define $D_N^* = D_N^\pi = B$ and $D_n^j = d_n^j + q_n D_{n+1}^j$ for $j \in \{*, \pi\}$. Then the exact economic values satisfy

$$0 \leq \mathcal{V}_n^* - \mathcal{V}_n^\pi \leq I_n(U_n - L_n) + D_n^* + D_n^\pi.$$

The action-domain approximation belongs in d_n^* , while policy implementation, boundary monitoring, state interpolation, and time/cubature errors may affect both defects.

PROOF: Insert $\mathcal{T}_n I_{n+1} v_{n+1}^*$ between the exact and interpolated nodal Bellman values. The Lipschitz property bounds the propagated value difference by $q_n D_{n+1}^*$, and the displayed defect bounds the remaining difference by d_n^* . Backward induction gives $\|\mathcal{V}_n^* - I_n v_n^*\| \leq D_n^*$. The same argument gives the fixed-policy bound. Subtract the two decompositions and use positivity of I_n and $v_n^* - v_n^\pi \leq U_n - L_n$. Feasibility gives the lower inequality. Q.E.D.

The proposition can use local regularity, coupling, barrier, or weak-approximation estimates to bound its defects; it does not claim that those hypotheses hold automatically for a fitted critic near a stopped boundary. In the present two-state computations, action coverage is operationally bounded on the nodal economy. The continuous-state/time defects remain explicitly uncomputed. Nested grids and common frozen policies diagnose sensitivity but do not provide the one-sided inequalities required by this proposition.

TABLE XVI: Frozen continuous-policy refinement, common economic domain

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	11	17×25	20	-4.843442	-4.841008
Actor	11	25×37	40	-4.848086	-4.839462
Actor	11	33×49	80	-4.842578	-4.830165
Actor	29	17×25	20	-4.841652	-4.841008
Actor	29	25×37	40	-4.846754	-4.839462
Actor	29	33×49	80	-4.840763	-4.830165
Actor	47	17×25	20	-4.844135	-4.841008
Actor	47	25×37	40	-4.848828	-4.839462
Actor	47	33×49	80	-4.843387	-4.830165
Direct	11	17×25	20	-4.843773	-4.841008
Direct	11	25×37	40	-4.857445	-4.839462
Direct	11	33×49	80	-4.852920	-4.830165
Direct	29	17×25	20	-4.841889	-4.841008
Direct	29	25×37	40	-4.854488	-4.839462
Direct	29	33×49	80	-4.848755	-4.830165
Direct	47	17×25	20	-4.844185	-4.841008
Direct	47	25×37	40	-4.854178	-4.839462
Direct	47	33×49	80	-4.848680	-4.830165

Notes. Policies are frozen on the original time schedule and bilinearly interpolated in state. References still use 175 actions. Differences are sensitivity diagnostics, not continuous-state/time upper bounds.

23. STRATEGIC DEVIATIONS AND MISSING PURE STAGES

For a fixed Markov rival profile in a finite-horizon finite-state/action game, condition on any history. The own player's conditional payoff from a randomized current action is a convex combination of pure-action payoffs with the same continuation best-response values. It cannot exceed the largest such payoff. Backward induction therefore bounds every randomized and history-dependent unilateral policy by the

Markov pure-action best-response recursion. Outward propagation then provides a bound over this full deviation class, not merely the particular local deviations observed during actor training.

Existence of a pure equilibrium in a fitted one-step game is a different matter. If none exists, minimum-defect selection produces a feasible profile, not an exact equilibrium. The independent dynamic best response still evaluates that profile correctly. The following records give every missing pure stage in the six principal R7 neural runs, including time index, state, and minimum maximum stage defect. Indices begin at zero. The six runs have $6 \times 30 \times 225 = 40,500$ active state-time nodes; the five recorded exceptions are not excluded from their final dynamic bounds. The profit targets are absolute units of the displayed per-firm payoff. Raw

TABLE XVII

EVERY MISSING PURE FITTED ONE-STEP EQUILIBRIUM IN R7

Market	Seed	Time index	K_1	K_2	Minimum defect
1	11	1	0.140625	0.865625	6.492e-06
1	11	1	0.684375	0.140625	4.018e-07
1	11	10	0.321875	0.140625	1.538e-05
2	11	16	0.231250	1.046875	1.534e-05
2	29	13	0.140625	0.956250	1.394e-06

Minimum defect is the smallest maximum unilateral one-step improvement over all joint finite actions. These nodes remain in full dynamic verification; they are not called exact stage equilibria.

network policy, shortlist policy, and corrected finite lookup policy are different deployment objects. The principal R7 guard fraction is zero, but the full joint-action oracle was nevertheless constructed during training. A mixed-strategy *profile* was not estimated; the preceding argument explains why mixed unilateral deviations against the fixed pure profile are already covered by the finite best-response bound.

24. PAIRED PAYOFFS, COSTS, AND ECONOMIC SCALING

Each row below compares the two frozen methods on the same stored Brownian paths, at the indicated dimension, seed, fitting domain, and Euler resolution. Intervals are pointwise Student intervals across 512 pathwise differences. They are

not independent across step sizes, because the Brownian increments are nested; no simultaneous-coverage claim is made. The narrow and wide policies were separately fitted, so their comparison also changes the training domain and update budget.

TABLE XVIII: All method-paired policy-payoff comparisons

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
10	11	Narrow	40	-0.03043	-0.03234	-0.02852
10	11	Narrow	80	-0.03237	-0.03438	-0.03036
10	11	Narrow	160	-0.03338	-0.03544	-0.03133
10	11	Wide	40	0.00061	0.00058	0.00064
10	11	Wide	80	0.00055	0.00052	0.00057
10	11	Wide	160	0.00052	0.00049	0.00054
10	29	Narrow	40	-0.02800	-0.02946	-0.02655
10	29	Narrow	80	-0.02993	-0.03146	-0.02840
10	29	Narrow	160	-0.03094	-0.03250	-0.02938
10	29	Wide	40	-0.00007	-0.00008	-0.00006
10	29	Wide	80	-0.00004	-0.00005	-0.00003
10	29	Wide	160	-0.00003	-0.00004	-0.00002
10	47	Narrow	40	-0.02522	-0.02695	-0.02349
10	47	Narrow	80	-0.02675	-0.02856	-0.02493
10	47	Narrow	160	-0.02755	-0.02941	-0.02569
10	47	Wide	40	0.00004	0.00003	0.00005
10	47	Wide	80	0.00001	0.00001	0.00002
10	47	Wide	160	0.00000	-0.00001	0.00001
20	11	Narrow	40	-0.05798	-0.06054	-0.05543
20	11	Narrow	80	-0.06131	-0.06396	-0.05866
20	11	Narrow	160	-0.06302	-0.06572	-0.06033
20	11	Wide	40	0.00123	0.00118	0.00128
20	11	Wide	80	0.00097	0.00092	0.00101
20	11	Wide	160	0.00084	0.00080	0.00088
20	29	Narrow	40	-0.01308	-0.01346	-0.01269
20	29	Narrow	80	-0.01401	-0.01442	-0.01361

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
20	29	Narrow	160	-0.01449	-0.01490	-0.01408
20	29	Wide	40	0.00480	0.00463	0.00496
20	29	Wide	80	0.00432	0.00417	0.00448
20	29	Wide	160	0.00409	0.00394	0.00424
20	47	Narrow	40	-0.19715	-0.19765	-0.19666
20	47	Narrow	80	-0.20011	-0.20061	-0.19962
20	47	Narrow	160	-0.20161	-0.20211	-0.20111
20	47	Wide	40	0.00084	0.00078	0.00089
20	47	Wide	80	0.00066	0.00061	0.00071
20	47	Wide	160	0.00058	0.00053	0.00062

Notes. Pointwise 95 percent Monte Carlo intervals, 512 paired paths per row. They do not bound optimality, continuous-time bias, or simultaneous coverage. The additional representation comparisons below use the identical NDU finite model. Gated regression has more parameters and higher per-update work than the width-32 tanh critic. Tensor projection exploits low state dimension. Both facts are explicit rather than hidden by matching only nominal iteration counts. The R7 NDU actor’s

TABLE XIX

ADDITIONAL CONTINUATION REPRESENTATIONS, IDENTICAL FINITE PREFERENCE ECONOMY

Method	Seed	Parameters	Policy bound	Value error	Train (s)	Verify (s)
Chebyshev	11	121	0.0121	0.0775	0.14	4.41
Chebyshev	11	225	0.0039	0.0165	0.15	4.48
Chebyshev	11	49	0.0373	0.2325	0.14	4.36
Gated	11	9345	0.0079	0.0284	41.28	4.12
Gated	29	9345	0.0098	0.0200	42.59	4.49
Gated	47	9345	0.0088	0.0306	38.99	4.30

Gated denotes a DGM-style architecture fitted to finite Bellman targets, not a differential DGM solver. Chebyshev degrees are 6, 10, and 14. All actions are enumerated; construction and reference costs are separately retained in JSON. Deterministic projections are not replicated under fictitious random seeds.

correction map stores a small number of changed action indices, but its complete policy table occupies 68,000 bytes with the archived int64 representation. A minimal pair of int64 state-time and replacement indices requires 16 bytes per correction, excluding indexing and container overhead. A stored-table deployment avoids actor evaluation at existing nodes; a raw-network deployment requires its network weights and argmax calculation. Neither observation establishes off-grid policy or guard correctness. The continuous hybrid additionally requires local action queries using the stored continuation representation.

In the continuous-action study, training-query counts exclude verification queries by construction. All-action references are created only after policies and critics have been frozen. The raw actor, direct-gradient, multi-head, and smoke failures are retained with their full configurations. The complete action verifier’s tighter and looser implementations have separate records, including unresolved boxes and declared budgets. A passing regression suite means that arithmetic, scope, and accounting checks passed; it does not change a failed economic accuracy target into a success.

TABLE XX
RETAINED CONTINUOUS-ACTION DEVELOPMENT FAILURES

Configuration	Seed	Deviation gain	Train (s)	Target .1
Actor smoke	11	0.1872	0.53	Fail
Raw actor	11	0.2596	15.77	Fail
Four-head reset	11	1.1033	16.83	Fail
Direct gradient	11	0.1997	16.07	Fail

All executed pilot sources, raw arrays, and histories remain. The smoke grid and budget differ from the principal grid; it is not a principal-seed replicate. Pilot selection and subsequent improvements are explicitly post-review development.

25. HISTORICAL NUMERICAL RECORD AND SOURCE PRESERVATION

The preceding manuscript and supplement are archived byte-for-byte under the R8 revision directory. Every earlier economic equation, application, numerical source, raw array, failed run, and referee report remains in the repository history

and the candidate tree. The main text now presents the successful R7 finite-policy computations and the new continuous-action study at the point where the corresponding economic model is introduced. Earlier R6 policy failures and non-neural game results are not relabelled as current neural results. Their original discussions are reproduced below as dated historical material.

25.1. *R6 preference-policy discussion (historical)*

25.2. *Neural solution and action-domain sensitivity*

We now apply multilayer NBO to the same two-state economy, with the original cardinal utility, correlation, reflection rule, and stopping payoff unchanged. At each backward time level a two-hidden-layer, width-thirty-two tanh critic approximates its own Bellman continuation; a multilayer categorical actor selects among feasible actions. The reference solution is computed separately and is not used as a training target. The actor is fitted before policy evaluation at that time level, with the next-period critic fixed. The direct comparator eliminates the actor by enumeration of the same action set. Both methods have a cap of 160 L-BFGS inner iterations per time level, divided 100/60 between critic and actor for NBO.

An initial unweighted classification implementation produced a large policy loss. Absorbing wealth-face labels, for which every action has identical value, contaminated the policy fit; classification error also ignored the economic cost of a mistaken low-consumption action. The retained implementation excludes those faces from actor training and minimizes expected normalized Bellman action loss, with a small cross-entropy term. The pilot source, weights, and failed output are preserved. This change repairs an implementation failure rather than redefining the economic objective: the deployed action is still evaluated against *all* finite feasible alternatives.

For the principal comparison, $(n_u, n_x, N_t) = (17, 25, 20)$ and $k = 1$. The expanded action set contains 175 triples, with $c/X \in \{.02, .05, .10, .20, .30\}$, $p \in \{0, .375, .75, 1.125, 1.5\}$, and $\theta \in \{-.45, -.30, -.15, 0, .15, .30, .45\}$. Every time-state-action combination enters the error accounts in the main-text Bellman error accounts. Independently solving the fixed learned policy gives its actual regret relative to the

TABLE XXI

NEURAL PREFERENCE ADJUSTMENT ON THE SAME FINITE ECONOMY

Method	Seed	Value error	Policy regret	Bound R_0	Midpoint RMS
NBO	11	0.1679	0.1656	3.3954	0.0438
NBO	29	0.1495	0.1487	2.8503	0.0413
NBO	47	0.1572	0.1014	3.0605	0.0445
Direct Bellman	11	0.1162	0.0261	1.7944	0.0477
Direct Bellman	29	0.1061	0.0261	1.6930	0.0400
Direct Bellman	47	0.1342	0.0269	1.8224	0.0424

The first three error columns use every state and time of the $17 \times 25 \times 21$ grid with 175 actions; R_0 is the time-zero certificate, while the first two columns maximize their independently computed errors over all times. Midpoints compare the time-zero neural critic with the finite-reference interpolant. Continuous-action and controlled-diffusion discretization errors are not included. The accuracy targets were value error .2 and policy regret .1.

finite reference. Midpoint value errors and derivative comparisons are additional diagnostics, not continuous-domain certificates. The three-seed record contains a failed NBO accuracy target and reports it without exclusion. The direct comparator is more accurate at this budget. A categorical actor therefore has a measurable approximation cost even when the critic class is identical. Neural representation alone is not a reason to replace an inexpensive two-state grid solver; the grid here supplies an unusually complete audit of the neural computation.

Action bounds are part of the economy, not innocuous numerical constants. The sensitivity study reproduces the referee's action-set comparison and extends it to 1,053 actions in the same expanded box, a wider consumption–portfolio–adjustment box, a wider wealth domain, and two finer space–time grids, for both $k = 1$ and $k = 5$. The table in the supplement records lower- and upper-bound frequencies for every control. Consumption and portfolio bounds remain frequently active after the first expansion. Thus state-grid convergence of the old 27-action model does not establish convergence to an unconstrained control optimum. The wider sets describe different constrained economies; refinement within a fixed box is a different comparison. No continuous-action coverage or controlled-diffusion discretization

TABLE XXII

DYNAMIC COURNOT: INDEPENDENT FIXED-RIVAL BEST RESPONSES

Grid	Steps	Actions	M_2	$I_1(0)$	$V_1(0)$	Max. gain
17^2	30	11	1	0	1.3852	0
17^2	30	11	2	0.4800	2.6737	0
25^2	60	17	1	0	1.3905	0
25^2	60	17	2	0.5500	2.6610	0

Center state is $(K_1, K_2) = (.775, .775)$. Horizon is three, capital lies in $[.05, 1.5]^2$, and investment in $[0, .8]$.

The final column maximizes independently computed dynamic unilateral gains over both firms and every finite-model state and time. Zero denotes the displayed floating-point precision; it is not a continuous-game error bound. Six principal cases additionally include a high-index equilibrium tie rule and asymmetric adjustment costs.

error is set to zero in the finite certificate. This preserves the economic application while making its constraint dependence observable.

25.3. R6 dynamic-game discussion (historical)

25.4. A dynamic capital-game computation

For two firms, we compute the finite-horizon capital game above with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[.05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. The baseline quadratic adjustment coefficients are $b_1 = b_2 = 1$; an asymmetric experiment uses $b_2 = 1.2$. Independent diffusion cubature and positive interpolation define the finite dynamic game. Investment nets include both bounds.

Backward induction constructs the two continuation-payoff matrices at every state and time. The solver enumerates pure stage equilibria; alternative deterministic tie rules are retained. If no pure equilibrium exists, the solver records this fact and the smallest maximum stage deviation it finds, rather than declaring equilibrium. After fixing the entire computed rival policy, a *separate backward optimization* permits every own action at every state and time. Its value minus the profile payoff is the unilateral exploitability map. A one-period check with a frozen continuation value alone would not perform this test. All six principal finite-game cases have

pure equilibria at every stage and zero detected exploitability to the reported arithmetic precision. A deliberately coarse regression case has a missing pure stage equilibrium and is reported as approximate, with time-indexed deviation bounds. The table includes fixed-market $M_2 = 1$ and growing-market $M_2 = 2$ normalizations as distinct economies; the resulting investment differences do not prove an $N \rightarrow \infty$ limit. Binding-investment frequencies, alternative equilibrium selections, and the asymmetric case are in the raw record. This is an actual dynamic best-response computation, not the static $1/64$ diagnostic. It is a finite-grid game result; a trained neural game and a continuous-state equilibrium bound are different objects and are not inferred from it.

26. COMPLETE R9 REFINEMENT AND VERIFICATION ACCOUNTS

This section retains every configuration used to assess the new signed cover and guarded neural map. The preceding R6–R8 sections are historical evidence, not overwritten runs. The new common and fine-grid envelopes must not be interchanged: they concern different state-time discretizations. Continuous-state/time defects remain separately unspecified for the stopped preference diffusion.

TABLE XXIII

FROZEN-POLICY ERRORS OVER EVERY REFINEMENT NODE

Policy	Grid/dates	Max.	95th	99th	t	u	X
A11	17/25/20	0.0612	0.0023	0.0073	0.000	1.99	0.530
A11	25/37/40	0.5360	0.2255	0.3885	0.000	3.00	0.540
A11	33/49/80	0.6550	0.2647	0.5264	0.000	3.00	0.530
A29	17/25/20	0.0143	0.0013	0.0038	0.550	1.20	1.782
A29	25/37/40	0.4959	0.2178	0.4154	0.000	1.27	1.587
A29	33/49/80	0.7011	0.2552	0.5493	0.000	1.26	1.587
A47	17/25/20	0.0467	0.0048	0.0073	0.000	1.54	0.530
A47	25/37/40	0.5371	0.2040	0.3549	0.000	3.00	0.540
A47	33/49/80	0.6562	0.2484	0.4698	0.000	3.00	0.530
D11	17/25/20	0.0230	0.0026	0.0047	0.400	1.20	1.782
D11	25/37/40	0.9249	0.3603	0.5999	0.000	2.85	0.540
D11	33/49/80	1.2997	0.4389	0.7763	0.000	2.89	0.530
D29	17/25/20	0.0157	0.0013	0.0025	0.550	1.20	1.782
D29	25/37/40	1.5662	0.3699	0.5710	0.000	3.00	0.520
D29	33/49/80	2.1234	0.4423	0.7166	0.163	3.00	0.515
D47	17/25/20	0.0231	0.0034	0.0057	0.800	1.20	0.595
D47	25/37/40	0.7631	0.3158	0.4675	0.250	2.92	0.540
D47	33/49/80	0.9957	0.3882	0.6275	0.000	2.94	0.530

A denotes actor-plus-search and D direct-plus-search. Quantiles concern the positive reference advantage over every state and date. Maximum locations include stopping-adjacent states. All eighteen historical configurations are retained; no centre-only selection is used.

TABLE XXIV

BOUNDARY-REGION REFINEMENT DIAGNOSTICS

Policy	Grid/dates	Lower region	Upper region	Interior	Above 0.1, %
A11	17/25/20	0.0612	0.0211	0.0176	0.00
A11	25/37/40	0.5360	0.4306	0.4604	11.61
A11	33/49/80	0.3635	0.5851	0.6550	14.27
A29	17/25/20	0.0106	0.0143	0.0092	0.00
A29	25/37/40	0.4473	0.4204	0.4959	10.22
A29	33/49/80	0.2698	0.5630	0.7011	13.21
A47	17/25/20	0.0467	0.0181	0.0141	0.00
A47	25/37/40	0.5371	0.4080	0.4458	11.07
A47	33/49/80	0.3641	0.4891	0.6562	13.13
D11	17/25/20	0.0147	0.0230	0.0114	0.00
D11	25/37/40	0.9249	0.4434	0.7900	18.70
D11	33/49/80	1.1417	0.5408	1.2997	21.20
D29	17/25/20	0.0146	0.0157	0.0090	0.00
D29	25/37/40	1.5662	0.4878	1.1079	18.60
D29	33/49/80	2.1234	0.5859	1.9784	20.80
D47	17/25/20	0.0149	0.0192	0.0231	0.00
D47	25/37/40	0.7631	0.4937	0.6036	16.52
D47	33/49/80	0.9140	0.5868	0.9957	18.73

Each wealth boundary region spans two grid cells. Interior means the remaining wealth nodes. Values are maximum finite-reference advantages, not upper certificates.

TABLE XXV

REFINEMENT ACTION SATURATION RATES, PERCENT

Policy	Grid/dates	c_-	p_-	θ_-	c_+	p_+	θ_+
A11	17/25/20	0.0	0.4	28.4	96.8	53.3	41.0
A11	25/37/40	0.0	0.1	23.4	95.7	47.7	36.0
A11	33/49/80	0.0	0.2	24.2	95.7	48.0	37.6
A29	17/25/20	0.0	0.1	28.1	96.9	54.3	40.7
A29	25/37/40	0.0	0.0	22.6	95.8	49.0	35.5
A29	33/49/80	0.0	0.0	23.4	95.8	49.0	37.2
A47	17/25/20	0.0	0.3	28.1	97.0	54.5	41.1
A47	25/37/40	0.0	0.1	23.2	95.7	49.1	36.3
A47	33/49/80	0.0	0.1	24.0	95.9	49.2	37.7
D11	17/25/20	0.0	0.1	27.3	97.4	53.3	41.8
D11	25/37/40	0.0	0.0	21.6	91.4	46.9	34.5
D11	33/49/80	0.0	0.0	22.8	92.8	47.5	36.4
D29	17/25/20	0.0	0.0	27.5	97.4	54.5	41.5
D29	25/37/40	0.0	0.0	21.7	91.4	48.6	34.2
D29	33/49/80	0.0	0.0	22.9	92.8	48.7	36.2
D47	17/25/20	0.0	0.1	27.5	97.6	54.7	41.8
D47	25/37/40	0.0	0.0	21.8	91.6	48.7	34.5
D47	33/49/80	0.0	0.0	23.0	93.1	48.9	36.5

Rates cover active state-time nodes; an action is counted within 10^{-4} of a bound. The box is an economic primitive, and saturation is not interpreted as an unconstrained first-order optimum.

27. FRESH TRAINING, CORRECTION STORAGE, AND TOTAL COST

The new spatial refinement is genuinely nested: every coarse spatial node is a fine spatial node, and every coarse date is a fine date. A trained fine-grid policy is not substituted into the frozen-policy historical experiment. The guard acts only on the chosen policy’s evaluation target. Its storage and query cost are explicit, and the actor-free comparison receives the same correction.

TABLE XXVI

EVERY FRESH NESTED-GRID TRAINING RUN

Policy	Nodes u /dates	Ref. max.	Value fit	Seconds	Corrected %
A11G	17/20	0.0372	0.0002	26.35	90.5
A29G	17/20	0.0085	0.0003	27.12	90.1
A47G	17/20	0.0372	0.0003	25.72	90.4
S11G	17/20	0.0372	0.0003	13.02	90.6
S29G	17/20	0.0372	0.0003	13.26	89.9
S47G	17/20	0.0371	0.0003	12.81	90.0
A11	33/40	0.0932	0.2043	89.58	–
A29	33/40	0.0625	0.1650	89.60	–
A47	33/40	0.0872	0.1649	85.80	–
S11	33/40	0.0932	0.1605	56.66	–
S29	33/40	0.0626	0.1910	50.84	–
S47	33/40	0.0932	0.1545	45.71	–
A11G	33/40	0.0041	0.0001	88.85	95.0
A29G	33/40	0.0093	0.0001	85.55	95.1
A47G	33/40	0.0155	0.0001	88.86	95.0
S11G	33/40	0.0138	0.0001	47.69	95.0
S29G	33/40	0.0156	0.0001	49.15	95.1
S47G	33/40	0.0156	0.0001	45.36	95.2

G denotes the indexed guard. All eighteen fresh runs appear. Value fit is a policy-value diagnostic, distinct from the one-step target residual. Unguarded failures are not excluded.

TABLE XXVII

INDEPENDENT VERIFICATION AND CORRECTION STORAGE

Policy	Uniform bound	Policy residual	Correction KiB	Evaluation s.
A11	0.0963	0.1100527	0.0	4.22
A29	0.0654	0.0962556	0.0	4.20
A47	0.0903	0.1126844	0.0	4.48
S11	0.0963	0.1103166	0.0	5.01
S29	0.0655	0.0967997	0.0	4.99
S47	0.0963	0.1130800	0.0	5.00
A11G	0.0140	0.0000623	568.5	4.20
A29G	0.0311	0.0000625	568.5	4.24
A47G	0.0356	0.0000624	568.5	4.28
S11G	0.0354	0.0000624	568.5	5.01
S29G	0.0357	0.0000622	568.5	5.01
S47G	0.0349	0.0000624	568.5	5.01

Same fine-grid upper envelope for every row. Correction memory counts saved indexed corrections and masks over all dates, not only one network. Policy residuals use independent outward point-action evaluation. The complete source contains raw arrays and nodewise masks.

TABLE XXVIII

EXPLICIT SHARED-COST ACCOUNTING, SECONDS

Policy	Train	One policy	Nine policies	100 policies	Nine + welfare
A11	15.01	389.04	57.91	20.24	85.32
A29	28.49	402.57	71.44	33.77	98.30
A47	26.46	400.53	69.40	31.73	96.34
D11	32.02	406.18	75.05	37.38	102.10
D29	32.80	406.98	75.84	38.17	104.18
D47	30.52	404.74	73.60	35.93	101.10
S11	13.97	388.27	57.14	19.47	88.45
S29	13.72	388.06	56.92	19.25	88.26
S47	12.87	387.20	56.06	18.39	87.43

Training/reference/policy-evaluation times are archived R8 measurements; the new certificate and welfare measurements are current. This is an auditable mixed-provenance cost account, not a same-machine speed comparison. Each policy receives its own independent payoff interval. Shared-cover cost is amortized only within the identical economy.

28. AN ABSOLUTE BENCHMARK AND PAIRED CAPITAL PAYOFFS

The continuous analytical bound in the article applies to the original unbounded-state capital economy from the specified initial state. The simulations below assess an explicit feasible schedule on the same discretized Brownian paths as the historical learned policies. A narrow Monte Carlo interval cannot remove unknown Euler bias. The analytical bound and the simulation comparison have different, stated roles.

TABLE XXIX

FEASIBLE SCHEDULE ON THE MATCHED BROWNIAN PATHS

Dimension	Euler steps	Mean payoff	95% lower	95% upper
10	40	-1.2555	-1.2694	-1.2416
10	80	-1.2583	-1.2723	-1.2444
10	160	-1.2598	-1.2738	-1.2458
20	40	-1.2565	-1.2697	-1.2433
20	80	-1.2595	-1.2728	-1.2463
20	160	-1.2611	-1.2744	-1.2478

512 paths per dimension. Pointwise Student intervals describe Monte Carlo error only; no continuous-time bias correction is assumed.

29. CONTINUOUS-CAPITAL REPLICATION DETAILS

The new study uses the identical capital flow, terminal payoff, diffusion covariance, primitive action interval, and dense coupling rule as the historical experiment. A regression test compares the numerical functions and matrices directly. Ten and twenty dimensions reuse the existing economies; fifty dimensions extends their declared generator. There are $d + 1$ independent Brownian drivers and a full-rank d -dimensional state covariance. These are different counts.

29.1. *Complete fitted-policy record*

Table XXXI lists every requested fit. All methods receive the same 600 critic updates, two independent two-probe trace banks, batch size, initialization seeds,

TABLE XXX

LEARNED-POLICY MINUS FEASIBLE-SCHEDULE PAYOFFS

Dimension	Policy	Training domain	Mean	95% lower	95% upper
10	N11	Narrow	-0.0544	-0.0572	-0.0517
10	N11	Wide	0.0008	0.0008	0.0008
10	N29	Narrow	-0.0490	-0.0510	-0.0470
10	N29	Wide	0.0008	0.0008	0.0009
10	N47	Narrow	-0.0452	-0.0476	-0.0429
10	N47	Wide	0.0003	0.0002	0.0003
10	D11	Narrow	-0.0211	-0.0218	-0.0203
10	D11	Wide	0.0003	0.0002	0.0003
10	D29	Narrow	-0.0181	-0.0185	-0.0176
10	D29	Wide	0.0009	0.0008	0.0009
10	D47	Narrow	-0.0177	-0.0182	-0.0171
10	D47	Wide	0.0003	0.0002	0.0003
20	N11	Narrow	-0.0822	-0.0854	-0.0791
20	N11	Wide	0.0003	0.0002	0.0003
20	N29	Narrow	-0.0256	-0.0263	-0.0249
20	N29	Wide	0.0003	0.0003	0.0003
20	N47	Narrow	-0.2429	-0.2441	-0.2417
20	N47	Wide	0.0004	0.0004	0.0004
20	D11	Narrow	-0.0192	-0.0197	-0.0187
20	D11	Wide	-0.0006	-0.0006	-0.0005
20	D29	Narrow	-0.0111	-0.0114	-0.0108
20	D29	Wide	-0.0038	-0.0039	-0.0036
20	D47	Narrow	-0.0413	-0.0423	-0.0403
20	D47	Wide	-0.0002	-0.0002	-0.0001

Every stored configuration is retained. Differences are paired on the same 160-step Euler paths; confidence intervals do not cover Euler discretization bias. They are not continuous neural-policy regret bounds.

and sample stream. Direct methods additionally invoke the global Hamiltonian maximizer. Actor training has no maximizing labels. The exact-Hessian held-out

diagnostics are collected only after the fit. Process memory includes library overhead and should not be read as the size of the deployed network.

TABLE XXXI: All fitted policies, without seed exclusion

d	Method	Seed	Seconds	RMS	Gap	MiB	Fail
10	NBO actor	11	5.29	0.0253	0.0019	303.2	0
10	NBO actor	29	5.31	0.0223	0.0014	302.6	0
10	NBO actor	47	5.25	0.0236	0.0040	303.0	0
10	Direct tube	11	6.42	0.0253	0.0012	302.7	0
10	Direct tube	29	6.40	0.0229	0.0002	302.4	0
10	Direct tube	47	6.46	0.0235	0.0006	302.4	0
10	Direct full	11	6.32	0.0257	0.0000	302.9	0
10	Direct full	29	6.40	0.0230	0.0000	302.9	0
10	Direct full	47	6.38	0.0235	0.0000	302.8	0
20	NBO actor	11	5.61	0.0241	0.0037	302.8	0
20	NBO actor	29	5.63	0.0219	0.0016	302.8	0
20	NBO actor	47	5.57	0.0187	0.0017	302.7	0
20	Direct tube	11	6.76	0.0238	0.0004	302.6	0
20	Direct tube	29	6.76	0.0225	0.0002	302.4	0
20	Direct tube	47	6.74	0.0210	0.0000	302.2	0
20	Direct full	11	6.63	0.0221	0.0000	302.6	0
20	Direct full	29	6.65	0.0232	0.0000	302.6	0
20	Direct full	47	6.69	0.0232	0.0000	302.4	0
50	NBO actor	11	6.31	0.0194	0.0016	306.5	0
50	NBO actor	29	6.25	0.0206	0.0023	306.8	0
50	NBO actor	47	6.21	0.0177	0.0023	306.9	0
50	Direct tube	11	7.84	0.0204	0.0002	306.7	0
50	Direct tube	29	7.79	0.0215	0.0005	306.4	0
50	Direct tube	47	7.88	0.0196	0.0007	306.6	0
50	Direct full	11	7.87	0.0207	0.0000	306.6	0

d	Method	Seed	Seconds	RMS	Gap	MiB	Fail
50	Direct full	29	7.94	0.0217	0.0000	306.7	0
50	Direct full	47	7.98	0.0192	0.0000	306.4	0

RMS and gap are diagnostics on 128 held-out states, not uniform certificates. Peak RSS is the separate process high-water mark, including the interpreter and libraries. Every requested fit is retained. All raw optimizer histories, negative trace-product batches, saved weights, sample counts, and bisection counts are committed. The output JSON records negative trace-product batches, all checkpoint losses and elapsed times, state-query and bisection counts, raw and guarded deployment costs, weight hashes, and failures. The initial output layer is zero, so the actor initially equals the common feasible schedule. Its guarantee is structural even before training. The protocol was developed after local pilots and is a pinned reproduction design, not a preregistered confirmatory experiment.

29.2. All certificate settings

Table XXXII includes both quadrature levels, all three radii, and all three initial-dispersion bounds. For the same d , a fixed matrix majorant is reused. Saved pivot intervals permit inspection of the spectral check. Independent replay rebuilds the majorant from the saved matrix and recomputes all 54 integral records. High-precision exponential and logarithmic checks test the interval implementation separately; these checks supplement rather than replace its inclusion argument.

TABLE XXXII: Complete radius, dispersion, and quadrature account

d	s_0	ϵ	Panels	Anchor upper	Policy upper
10	0.00	0.00	4096	0.018945	0.018945
10	0.00	0.00	16384	0.018747	0.018747
10	0.00	0.05	4096	0.018945	0.032498
10	0.00	0.05	16384	0.018747	0.032240

	d	s_0	ϵ	Panels	Anchor upper	Policy upper	
1							1
2							2
3	10	0.00	0.10	4096	0.018945	0.057748	3
4	10	0.00	0.10	16384	0.018747	0.057430	4
5	10	0.25	0.00	4096	0.021736	0.021736	5
6	10	0.25	0.00	16384	0.021519	0.021519	6
7	10	0.25	0.05	4096	0.021736	0.036103	7
8	10	0.25	0.05	16384	0.021519	0.035827	8
9	10	0.25	0.10	4096	0.021736	0.062168	9
10	10	0.25	0.10	16384	0.021519	0.061831	10
11	10	0.50	0.00	4096	0.024777	0.024777	11
12	10	0.50	0.00	16384	0.024541	0.024541	12
13	10	0.50	0.05	4096	0.024777	0.039959	13
14	10	0.50	0.05	16384	0.024541	0.039664	14
15	10	0.50	0.10	4096	0.024777	0.066838	15
16	10	0.50	0.10	16384	0.024541	0.066482	16
17	20	0.00	0.00	4096	0.025799	0.025799	17
18	20	0.00	0.00	16384	0.025526	0.025526	18
19	20	0.00	0.05	4096	0.025799	0.040877	19
20	20	0.00	0.05	16384	0.025526	0.040533	20
21	20	0.00	0.10	4096	0.025799	0.067660	21
22	20	0.00	0.10	16384	0.025526	0.067244	22
23	20	0.25	0.00	4096	0.029103	0.029103	23
24	20	0.25	0.00	16384	0.028807	0.028807	24
25	20	0.25	0.05	4096	0.029103	0.045012	25
26	20	0.25	0.05	16384	0.028807	0.044646	26
27	20	0.25	0.10	4096	0.029103	0.072627	27
28	20	0.25	0.10	16384	0.028807	0.072188	28
29	20	0.50	0.00	4096	0.032667	0.032667	29
30	20	0.50	0.00	16384	0.032348	0.032348	30
31	20	0.50	0.05	4096	0.032667	0.049408	31
32	20	0.50	0.05	16384	0.032348	0.049018	32

d	s_0	ϵ	Panels	Anchor upper	Policy upper
20	0.50	0.10	4096	0.032667	0.077853
20	0.50	0.10	16384	0.032348	0.077392
50	0.00	0.00	4096	0.027836	0.027836
50	0.00	0.00	16384	0.027540	0.027540
50	0.00	0.05	4096	0.027836	0.043327
50	0.00	0.05	16384	0.027540	0.042958
50	0.00	0.10	4096	0.027836	0.070525
50	0.00	0.10	16384	0.027540	0.070081
50	0.25	0.00	4096	0.031283	0.031283
50	0.25	0.00	16384	0.030963	0.030963
50	0.25	0.05	4096	0.031283	0.047610
50	0.25	0.05	16384	0.030963	0.047216
50	0.25	0.10	4096	0.031283	0.075644
50	0.25	0.10	16384	0.030963	0.075175
50	0.50	0.00	4096	0.034992	0.034992
50	0.50	0.00	16384	0.034649	0.034649
50	0.50	0.05	4096	0.034992	0.052155
50	0.50	0.05	16384	0.034649	0.051737
50	0.50	0.10	4096	0.034992	0.081025
50	0.50	0.10	16384	0.034649	0.080532

Quadrature endpoints are outward, not an error tolerance passed to an ordinary numerical integrator. Zero radius denotes the exact feasible schedule. Printed upper endpoints are rounded upward. The same matrix majorant is used for each dimension.

29.3. Deployment and arithmetic contract

The interval schedule guard uses the inherited binary64 arithmetic implementation with outward elementary operations and polynomial transcendental bounds. The additional square-root routine verifies squared endpoints and handles a zero

lower endpoint explicitly. An SVD proposes the spectral majorant, but the interval factorization must verify it. A bad matrix shape, nonfinite matrix, nonpositive pivot, mismatched matrix hash, empty inward tube, or out-of-horizon clock produces an error. There is no fall-through from a failed arithmetic check to an empirical maximum.

The actor’s neural proposal need not be evaluated by validated transcendental routines to maintain its action constraint: its finite output is projected onto the inward interval. The theoretical result concerns the resulting continuous-time feedback rule. Euler paths approximate its payoff, and their bias remains uncomputed. A sampled Euler path is not claimed to be an exact draw of the continuous controlled diffusion. A finite proposal failure is recorded separately from the mathematical fact that the specified fallback remains in the tube.

29.4. Paired path archive and interpretation

For each dimension, a fixed generator with seed $20261004 + d$ produces the 160-step Brownian increments. Summing consecutive increments produces the coarser 40- and 80-step paths. The same increments are used for the schedule, all three actors, and all six direct fits. The archive contains the increments themselves, every payoff vector, and all finest-level terminal states. All reported mean and interval endpoints are recomputed from these arrays by the audit. Seed-averaged comparisons first average the three fixed seed payoffs within a path; the confidence interval then concerns independent paths, not independent training seeds. No sampling interval is subtracted from a continuous upper bound to obtain an unsupported policy certificate.

30. ANALYTICAL AND NONSMOOTH BENCHMARKS

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (2)$$

With $\rho = .04$, $\gamma = 2$, $\mu = .08$, $r = .02$, and $\sigma = .2$, these numbers are $p^* = .75$ and $m^* = .04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule. A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = .01$ with no short sales, the unconstrained share is $-.125$ and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = .03$. The maximized HJB remains an equality. The action KKT condition is an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section 4, rather than this homothetic corner solution, supplies the nonsmooth selection test.

30.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates .025 and .008; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0, 1, 2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that

TABLE XXXIII

CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	.04125	.75000	3	5.6×10^{-16}
No-short-sale corner	.03000	.00000	3	2.3×10^{-16}
Recursive matching-bequest test	.04550	.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

finite-horizon interpretation, not a boundary condition hard-imposed during these particular updates.

Table XXXIII reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within 3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

31. RECURSIVE UTILITY

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (3)$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (4)$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This

distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (5)$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right], \quad K = [\rho(m^*)^{a-1}]^{b/a}. \quad (6)$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = .04$, $\mu = .08$, $r = .02$, $\sigma = .2$, this gives $(m^*, p^*) = (.0455, .3)$.

31.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (6) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[\underline{x}, \bar{x}]$ with $0 < \underline{x} < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping boundary and at T . Consumption shares are in a compact positive interval containing .0455, and portfolio shares are in a compact interval containing .3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the Hamiltonian and matches all stopping payoffs; Theorem 2 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-

complementarity parameters straddle one (Herdegen et al., 2021). The paper retains that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

31.2. Stability and baseline design

The recursive experiment in Table XXXIII tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript's constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this revision. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

32. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \quad (7)$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (8)$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses $[1.2, 3]$, so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative

or transformed diffusion, but that would be a different primitive and must be identified as such.

32.1. Cardinal utility and the objective

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (9)$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (10)$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (11)$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately .84657. This is a derivative of flow utility, not a sign theorem for the endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

32.2. Hamiltonian, constrained policies, and economic implications

The HJB equation is

$$-V_t = \sup_{c,p,\theta} \left\{ U(c,u) - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c]V_X \right. \\ \left. + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}p^2\sigma_X^2 X^2 V_{XX} + \varrho\sigma_u\sigma_X p X V_{uX} \right\}. \quad (12)$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (13)$$

$$\theta^* = V_u / k, \quad (14)$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (15)$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (14) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2}\mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (16)$$

The sign of the hedging component in (15) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

32.3. Independent two-state reference and identification

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[.5, 2]$, and $G(u, X) = X^{1-u}/(1-u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \rho) = (.02, .08, .2, .08, -.3, .04)$. The finite action set is the product of $c/X \in \{.02, .05, .1\}$, $p \in \{0, .375, .75\}$, and $\theta \in \{-.15, 0, .15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (.1, .75, -.15)$ at that point. The approximately .048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section 32.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \theta) + \varepsilon_j$, let $m(\theta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_\theta m$, after accounting for latent preference states, normalization parameters, and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

The next comparison is the retained R8 evidence under its original enclosure. The new signed envelope and fresh nested-grid policies are reported separately in Section 37; historical failure counts are not overwritten.

32.4. *Neural preference policies and the role of the action oracle*

We compute the original preference-adjustment economy with the cardinal utility, shock correlation, reflected preference state, and wealth stopping payoff specified above. The main grid has 17 preference points, 25 log-wealth points, and 20 time steps over a unit horizon. Adjustment cost is $k = 1$. The expanded control box is

$$m = c/X \in [.02, .30], \quad p \in [0, 1.5], \quad \theta \in [-.45, .45].$$

The inherited finite-action comparison uses the 175 triples formed by five consumption shares, five portfolio shares, and seven adjustment efforts, including all endpoints. This is the same constrained economy across the matched finite methods, not a change of utility normalization.

Two experiments distinguish action representation from action optimization. First, the R7 finite study trains two-hidden-layer, width-32 tanh critics and categorical actors. At each backward step it computes the complete 175-action payoff vector using the current continuation critic. Its exact argmax provides a label, and the vector also supplies the actor's expected action-loss objective. Thus *the training oracle is exhaustive*, even when the final actor is accurate. It is a useful neural policy-representation experiment, but not evidence that the actor avoided Bellman maximization.

Second, R8 trains a continuous-output actor directly through its own positive-weight payoff. It never receives an all-action value vector or an exact argmax label. The actor receives 160 Adam updates per time level; the continuation critic receives 320 Adam updates and up to 300 L-BFGS iterations, with failed polishing steps rolled back. The actor-free direct comparator optimizes eight independent bounded controls at every state for the same 160 iterations, followed by the same critic fit. Every restart and rejected action query is charged. Neither procedure invokes a Riccati solution, an analytically invariant policy class, or the independently computed reference during training.

Finite-action neural policies. The raw R7 actor is the primary neural representation result. Its full dynamic policy loss is below the declared .1 target for each fixed seed. A top-four shortlist and a separately reported correction map further reduce the

loss. The correction map is not part of the raw network. The complete finite policy, not only one-step deviations against a fitted critic, is evaluated by independent backward recursion. These results repair the earlier R6 actor failures, which remain

TABLE XXXIV

FINITE PREFERENCE POLICIES: R7 EVIDENCE

Method	Seed	Raw loss	Final loss	Value error	Seconds	Guard (%)
NBO	11	0.0445	0.0261	0.0924	30.59	0.294
Direct	11	0.0181	0.0181	0.0790	24.79	0.000
NBO	29	0.0500	0.0260	0.0814	30.49	0.332
Direct	29	0.0263	0.0263	0.0551	25.19	0.000
NBO	47	0.0500	0.0263	0.1325	29.89	0.230
Direct	47	0.0264	0.0264	0.0824	25.17	0.000

Loss bounds cover every stored state and time of the same 175-action economy. Seconds include training and the three reported policy verifications; classical backward induction averages 0.102 seconds. A raw actor, shortlist, and corrected policy are distinct objects. Utility losses are absolute units.

in the historical numerical record. They do not reverse the cost comparison: the matched direct finite-Bellman fit is faster and more accurate on average, and classical backward induction is substantially cheaper on this small grid. Tables report those comparisons without selecting a favorable seed. The actor's approximation cost is part of the result, not an argument for discarding the economic application.

Continuous-action optimization without exact labels. The initial continuous actor and direct-gradient pilot fail the .1 deviation diagnostic. The actor can saturate at a portfolio bound while a lower portfolio share improves continuation value. Four actor heads with output reinitialization do not cure that failure in the retained pilot. We therefore execute a disclosed local-improvement variant: compare the actor proposal with eight box corners, then perform two coordinate-search sweeps at each of eight decreasing step sizes. The same local search is applied to the direct comparator. An actor-free ablation omits all gradient action updates and starts the identical search from feasible proposals. This ablation measures how

much of the hybrid result comes from the search itself. The table distinguishes

TABLE XXXV

CONTINUOUS-CONTROL TRAINING WITHOUT EXACT MAXIMIZING LABELS

Method	Seed	Raw diagnostic	Hybrid gain	Train (s)	Queries (10^6)
Actor + search	11	0.2596	0.0617	15.01	2.261
Actor + search	29	0.0733	0.0188	28.49	2.261
Actor + search	47	0.2597	0.0475	26.46	2.261
Direct + search	11	0.1554	0.0303	32.02	11.841
Direct + search	29	0.1566	0.0203	32.80	11.841
Direct + search	47	0.1554	0.0251	30.52	11.841
Search only	11	5.5370	0.0617	13.97	0.961
Search only	29	7.2093	0.0616	13.72	0.961
Search only	47	8.0316	0.0616	12.87	0.961

Raw diagnostic is the largest finite-reference payoff minus the raw policy payoff. Hybrid gain is the augmented finite-deviation gain, a lower bound on continuous-action regret, not a certificate. All methods use zero exhaustive training backups and zero exact argmax training labels. Search-only omits action-gradient updates. Training, reference, and full-cover verification costs are separate.

three quantities. The raw-network loss against the finite reference tests the actor alone. The augmented finite-deviation gain permits all 175 inherited actions and the learned continuous action at every state and time; it is a *lower bound* on missed continuous-action gains. The complete action-box upper bound in Proposition 1 is a separate verification result. A small augmented gain is not inserted into that proposition as if it were a uniform upper bound. Local changes occur at nearly every active state-time node in the principal hybrid runs, so this experiment does not describe a sparse correction map or an actor-only controller. The action cover includes binding constraints and wealth exits. Its unresolved boxes remain in the bound when the declared budget is exhausted. The resulting upper array can audit the actor, the direct optimizer, and the local-search ablation without retraining. This produces an operational error account on a multidimensional state grid with a continuous, nonconcave action objective. It does not establish that all runs meet the

TABLE XXXVI

FULL CONTINUOUS-ACTION POLICY BOUNDS ON THE TWO-STATE NODAL ECONOMY

Policy	Seed	Uniform bound	Center bound	Eval. (s)	Bound $\leq .1$
Actor + search	11	0.1294	0.0601	0.22	No
Actor + search	29	0.1227	0.0583	0.22	No
Actor + search	47	0.1301	0.0608	0.25	No
Direct + search	11	0.1153	0.0604	0.23	No
Direct + search	29	0.1156	0.0586	0.22	No
Direct + search	47	0.1203	0.0609	0.25	No
Search only	11	0.1196	0.0647	0.24	No
Search only	29	0.1179	0.0586	0.23	No
Search only	47	0.1209	0.0610	0.22	No

The full action box is covered at all 425 states and 20 times. The common optimal upper envelope costs 1624.59 seconds across the retained implementations and 40.96 million action-box evaluations. Each row adds its own interval policy evaluation. Unresolved leaves retain their upper bounds. No continuous-state/time error is set to zero.

tighter .1 complete-action target, nor that the actor dominates the direct method at matched economic accuracy.

Alternative representations and economic units. Additional same-model comparisons use a gated neural continuation architecture and tensor Chebyshev projection. The gated method uses DGM-style state-dependent gates (Sirignano and Spiliopoulos, 2018), but here it is fitted to finite Bellman targets, not called a mesh-free differential DGM solver. The projection exploits the same two-state geometry and imposes the same wealth-boundary lifting. Degrees, parameter counts, all-action costs, failures, and actual accuracy appear in the supplement. These are executed post-review comparisons, not results attributed to the earlier unexecuted development protocol. A generic semilinear BSDE comparator is not substituted for this controlled-covariance, stopped problem without reformulating its second-order control term.

All losses are in the specified cardinal utility units. The optimal finite-model value at the center is approximately -4.8410 , and the initial value range is approximately

[−9.1599, −.1250]. A loss of .1 is about 2.066 percent of the center's *absolute utility magnitude*; this scaling is not a consumption-equivalent welfare percentage. Endogenous curvature and adjustment costs make a common consumption-equivalent conversion a separate economic calculation.

Action bounds and controlled refinements. The inherited sensitivity study retains both adjustment-cost values, the original 27-action truncation, 175 and 1,053 actions in the expanded box, a wider action box, wider wealth bounds, and finer space–time grids. Consumption and portfolio upper bounds remain frequently active. Expanding the box changes the constrained economy; refining actions inside one box does not. R8 additionally evaluates the *same frozen continuous policy*, bilinearly interpolated in state and held on its original time schedule, on three nested state–time levels. This isolates changes in policy evaluation from retraining. The supplemental differences are observed refinement diagnostics, not an estimated order promoted to a continuous-time error bound. The original continuous control problem, its utility normalization, and its comparative-static questions remain the economic objects of interest.

33. SOPHISTICATED TEMPORAL SELVES

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho+\kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (17)$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (18)$$

Write $V = \beta W_1 + (1 - \beta)W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (19)$$

The current self's spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (20)$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The appendix derives (20) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (19) and one actor for (20). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

33.1. An equilibrium calculation with endogenous continuation values

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$. Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1 - \beta}{\lambda_2(m)} \right]. \quad (21)$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$. Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = .02$, $\rho = .04$, $\gamma = 2$, and $\kappa = .2$, the roots for $\beta = 1, .9, .7, .5$ are respectively .030000, .031372, .034606, and .038760. The code independently optimizes finite-duration deviations and records the gain divided by duration as the duration shrinks. For $\beta = .7$, this quantity declines from approximately 4.60×10^{-5} at duration .1 to 4.66×10^{-9} at duration .001. This is a diagnostic of the derived spike

equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

34. DYNAMIC GAMES

The dynamic Cournot application retains firm-specific capital and investment. Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in \mathcal{A}_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (22)$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

34.1. Player-specific differentiation and deviation bounds

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_\nu \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2V_i)]. \quad (23)$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem 2 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem 3 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is $1/3$ per firm. Joint-profit maximization gives a symmetric allocation $1/4$. At the latter profile, a firm optimally deviates to $3/8$, earning a gain of $1/64$. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

34.2. Symmetry and market scaling

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section 34.3 replaces its evidentiary role with a

fully specified finite dynamic game and independent best-response calculations,
without asserting convergence of unrestricted neural game training.

34.3. *Neural dynamic capital policies and full unilateral deviations*

For two firms, we retain the capital game with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[\cdot 05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. Baseline adjustment coefficients are $b_1 = b_2 = 1$; the earlier asymmetric experiment has $b_2 = 1.2$. The principal neural grid has 17×17 states, 30 time steps, and 11 actions per player. Positive cubature and interpolation define the finite game. Market normalizations $M_2 = 1$ and $M_2 = 2$ are distinct economies, not evidence for an uncomputed large-population limit.

At each time and state, the R7 calculation trains separate critics and actors. It constructs both full 11×11 continuation-payoff matrices. The training labels are a pure one-step equilibrium when one exists and a minimum-maximum-defect pair otherwise. The actor's own objective is differentiated with the rival action and continuation critic fixed. Thus this study actually trains neural strategic policies, but it also pays for an exhaustive joint-action training oracle. Raw policies, three-by-three shortlist policies, and any corrected policies are distinct stored arrays.

After freezing the entire profile, an independent backward optimization permits every own action at every state and time against the fixed rival policy. Its value minus the profile payoff is the unilateral exploitability map. This is not a one-period test with the approximate continuation held fixed. The raw and final profiles meet the .1 absolute-profit target in every principal R7 case; the full-oracle guard is unused in those cases. Five state-time nodes across the six neural runs lack a pure one-step equilibrium for the fitted continuation matrices. Their locations and defects are reported in the supplement. A minimum-defect selection at those nodes is not called an exact equilibrium. Nevertheless, the complete dynamic best-response bound remains valid for the resulting profile: its proof does not assume that every fitted one-step game has a pure solution. Moreover, for fixed Markov rivals, randomized or history-dependent own deviations cannot improve on the finite Markov best-response value, because each conditional action mixture

TABLE XXXVII

R7 NEURAL DYNAMIC-GAME PROFILES AND FULL BEST RESPONSES

Market	Seed	Raw max	Firm 1	Firm 2	Missing	Seconds
1	11	0.0259	0.0164	0.0252	3	52.43
2	11	0.0241	0.0255	0.0163	1	56.13
1	29	0.0300	0.0288	0.0296	0	52.44
2	29	0.0626	0.0624	0.0547	1	57.23
1	47	0.0190	0.0081	0.0183	0	52.53
2	47	0.0101	0.0074	0.0113	0	56.98

All bounds cover every stored finite state, time, and unilateral action, including deviations at the reported missing pure stages. The full-oracle guard is unused in these principal runs. Profit losses are absolute units; market normalizations are different economies.

is bounded by its largest pure-action payoff. The calculation therefore checks such unilateral deviations as well; it does not compute a mixed-strategy equilibrium profile or a continuous-action game.

R8 adds actor-free neural continuation fitting and classical backward selection for both market normalizations. The neural comparator allocates 480 critic Adam updates per time level rather than R7's 320 critic plus 160 actor updates, with the same L-BFGS cap. Counts and timings reveal the difference between nominal update matching and matched wall time. Both methods receive their own independent full dynamic best responses. The earlier six classical game cases, asymmetric costs, alternative tie rules, boundary frequencies, coarse counterexample, and finer-grid results remain in the supplement and raw record. The finite-game result is a numerically evaluated approximate equilibrium with a specified profit-unit bound. It is not inferred from stationarity of a summed actor loss or the static 1/64 diagnostic. State interpolation, continuous investment, diffusion timing, and continuously monitored exits remain separate terms in a continuous-game error account.

35. SCALABILITY AND THE DIFFUSION TRACE

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional com-

TABLE XXXVIII

R8 ACTOR-FREE DYNAMIC-GAME COMPARISONS

Method	Market	Seed	Firm 1	Firm 2	Train (s)	Verify (s)
Classical	1	11	0.0000	0.0000	0.22	6.40
Classical	2	11	0.0000	0.0000	0.23	6.29
Direct	1	11	0.0223	0.0273	37.43	6.09
Direct	1	29	0.0199	0.0282	37.68	6.18
Direct	1	47	0.0223	0.0409	36.03	6.26
Direct	2	11	0.0091	0.0694	40.43	5.75
Direct	2	29	0.0306	0.0185	43.07	6.34
Direct	2	47	0.0206	0.0358	40.47	6.08

The direct method fits two continuation networks with the same class and 480 critic Adam updates, versus R7's 320 critic and 160 actor updates. Classical selection uses no neural approximation. Every method receives a new full dynamic best response. R7 and R8 timings were measured in their recorded environments; cross-environment ratios are not a hardware-controlled speed experiment.

ponents, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (24)$$

The drift is a nonnormal dense matrix $A = -.7I + .06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + .311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction of the original benchmark is unavailable. All methods still exploit the declared quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P)/\rho. \quad (25)$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block

laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

35.1. Trained, coupled nonquadratic capital economies

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [.02, 2]$, and

$$dY_i = \left[.10 + .15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(.15^2 + .10^2) \right] dt + .15 dW_i + .10 dW_0, \quad (26)$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - .10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed. Horizon is one, discount is .04, and terminal utility is $\bar{Y} - .03 d^{-1} \sum_i (Y_i - \bar{Y})^2$. The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense nonlinear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + .2 \bar{m}^*)^{-1}, .02, 2 \right),$$

TABLE XXXIX

ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-.5, .5]^d$. The failure criterion is RMS residual above .025 or queried action gap above .01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20$, 600 Adam updates, and batches of 128 independently sampled state-time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-.5, .5]^d$. A common diagnostic target is residual root mean square at most .025 and maximum queried action gap at most .01. The recorded first crossing is not necessarily sustained; final failures are also reported. The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verifi-

cation box to $[-1.5, .5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

35.2. Method-paired payoff evidence and arithmetic replay

The R7 follow-up evaluates the coupled-capital policies at dimensions ten and twenty, for all three fixed seeds, both fitting domains, and 40, 80, and 160 Euler steps. The same 512 Brownian paths are used by NBO and the direct comparator within every cell. For path j , define $D_j = J_j^{\text{NBO}} - J_j^{\text{direct}}$. We report $\bar{D}$ and the pointwise Student interval $\bar{D} \pm t_{511, .975} s_D / \sqrt{512}$. Pairing removes the irrelevant component of common path variation from the comparison. The complete set of 36 comparisons is reproduced from the pinned raw arrays in the supplement. The old twenty-dimensional fitting box is exited by essentially all recorded paths. Wider-domain fitting eliminates observed exits in the recorded audits and changes both payoff agreement and method differences. Zero observed exits in 512 paths is not a zero exit-probability theorem. Likewise, these intervals quantify pointwise Monte Carlo uncertainty under a specified Euler scheme. They are neither simultaneous confidence statements across all 36 comparisons nor bounds on time bias, optimal value, or the global differential residual. The experiment identifies a consequential coverage failure and its measured correction without redefining the unbounded economic state space as a stopped box.

The one-state consumption certificate has a different evidentiary status. R7 independently replays its previously generated complete leaf cover with `mpmath.iv` at 40 decimal digits, without importing the original interval arithmetic implementation. For the stored seed-11 networks, the NBO policy-loss bound is about .04961 and the direct-HJB bound about .04967. Both meet the stated threshold. This is an *independent interval-arithmetic replay on the previously generated complete cover*, not independent network training, independent cover generation, or a formal machine

TABLE XL

PAIRED PAYOFF DIFFERENCES AT THE FINEST RECORDED EULER LEVEL

Dim.	Seed	Domain	Difference	95% lower	95% upper
10	11	Narrow	-0.03338	-0.03544	-0.03133
10	11	Wide	0.00052	0.00049	0.00054
10	29	Narrow	-0.03094	-0.03250	-0.02938
10	29	Wide	-0.00003	-0.00004	-0.00002
10	47	Narrow	-0.02755	-0.02941	-0.02569
10	47	Wide	0.00000	-0.00001	0.00001
20	11	Narrow	-0.06302	-0.06572	-0.06033
20	11	Wide	0.00084	0.00080	0.00088
20	29	Narrow	-0.01449	-0.01490	-0.01408
20	29	Wide	0.00409	0.00394	0.00424
20	47	Narrow	-0.20161	-0.20211	-0.20111
20	47	Wide	0.00058	0.00053	0.00062

NBO minus direct payoff on the same 512 paths. Full 40/80/160-step comparisons are in the supplement. Confidence intervals quantify Monte Carlo variation only.

proof. Together with the new complete three-control cover on the two-state preference grid, it demonstrates two different ways of operationalizing verification. Neither removes the dimensional cost of covering an unrestricted high-dimensional continuous state space.

35.3. Exact and stochastic trace computation

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (27)$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (28)$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (29)$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics. All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator’s execution without identifying a random training objective with a deterministic certificate.

36. VALIDATED EVALUATION AND ACTION IMPROVEMENT

The neural Bellman map must account for two distinct approximation decisions: which continuation values are passed to the next stage, and how tightly the admissible action supremum is bounded. This section supplies an explicit output rule for the first decision and a signed enclosure for the second. Neither construction presumes that a stationary point of the training loss is economically optimal. Both apply to the original preference economy and leave its controls, state law, and stopping convention unchanged.

36.1. An evaluation rule with a prescribed nodal error

Let T_h^a denote a positive-weight, one-step policy operator on a finite state grid, and suppose its continuation weights have sum at most $q_h \leq 1$. At a given date, let w be the continuation vector, a the selected feasible policy, $\hat{v}$ the fitted neural critic, and $\tilde{T}_h^a w$ the computed selected-policy target. We distinguish the computation of this target from maximization over actions. In particular, it uses one action at each node, not a table of optimizing labels. For a prescribed tolerance $\tau_h > 0$, define

$$C_h(\hat{v}, a, w)_i = \begin{cases} (\tilde{T}_h^a w)_i, & |\hat{v}_i - (\tilde{T}_h^a w)_i| > \tau_h, \\ \hat{v}_i, & \text{otherwise.} \end{cases} \quad (30)$$

The comparison in the implementation uses an outward upper bound for the difference. A selected entry is assigned the target directly; it is not reconstructed by adding two cancelling floating-point numbers. The indexed corrections, their masks, and the uncorrected neural outputs are saved. The deployed continuation is therefore a neural representation with a recorded nodal correction, not a raw network.

PROPOSITION 2—Evaluation guard: Suppose $\|\tilde{T}_h^a w - T_h^a w\|_\infty \leq \eta_h$. Then

$$\|C_h(\hat{v}, a, w) - T_h^a w\|_\infty \leq \tau_h + \eta_h.$$

For a frozen sequence of policies over N stages, with exact terminal data, the guarded backward values and the exact values of those policies differ by at most $\sum_{j=0}^{N-1} q_h^j (\tau_h + \eta_h)$.

If $Nh = T$, $\tau_h = 0.1h^2$, and $\eta_h = o(h)$, this evaluation perturbation vanishes as $h \downarrow 0$. This statement is a property of the output rule, not a rate for the neural optimizer or an assertion of policy optimality.

The guard gives an implementable meaning to the evaluation component of a vanishing-perturbation condition. It does not supply the action-improvement error, the consistency error of a stopped diffusion approximation, or convergence of an uncorrected network away from the nodes. Its cost can be substantial: in the numerical economy it corrects a large fraction of fitted nodal values. We consequently count the correction memory and compare against an actor-free method with the identical guard. The construction is not evidence of neural compression when almost every node is retained.

36.2. Signed enclosures for continuous actions

Fix a state-time node and a continuation vector w . Write $Q_w(a)$ for its one-step payoff. The action set is the full compact box of the preference economy. Let $B = [\ell, r]$ be an action rectangle with centre c . A natural interval calculation supplies an upper bound $U^{\text{nat}}(B)$, while independent feasible evaluations supply a lower incumbent L .

The continuation is bilinear within state cells and continuous across their internal faces. Reflection changes its one-sided slopes. A stopped transition is different: the implementation switches between interpolated continuation and the specified stopping payoff, which need not agree away from boundary grid points. We call B *regular* when every cubature transition stays entirely inside, or entirely outside, each wealth stopping face. Only on such a box do we use derivative information. All intersected interpolation cells and both sides of reflection folds enter the derivative hull.

Suppose $G_j(B) = [g_j^-, g_j^+]$ encloses every one-sided partial derivative with respect to action component j on a regular box. The mean-value upper enclosure is

$$U^{\text{mv}}(B) = \bar{Q}_w(c) + \sum_j \max\{g_j^-(\ell_j - c_j), g_j^-(r_j - c_j), g_j^+(\ell_j - c_j), g_j^+(r_j - c_j)\}, \quad (31)$$

where $\overline{Q}_w(c)$ is an outward point upper bound. The algorithm uses the smaller of this bound and $U^{\text{nat}}(B)$. If $g_j^- > 0$, it restricts the supremum to $a_j = r_j$; if $g_j^+ < 0$, it restricts it to $a_j = \ell_j$. These are verified reductions to a maximizing face, not decisions based on a sampled derivative. For boxes crossing a stopping switch, the natural union enclosure is retained without this reduction.

THEOREM 3—Signed action cover: *Assume that the natural enclosures contain their respective payoffs, that the derivative hulls contain all one-sided derivatives on each regular box, and that feasible point values and arithmetic bounds are outward. Start from a cover of the action box and apply subdivision, reduction to a verified maximizing face, and removal of a box only when its upper bound is at most $L + \varepsilon$. At any interruption, retain the upper bound of every unresolved box. Then*

$$\sup_{a \in \mathcal{A}} Q_w(a) \leq \max\{L + \varepsilon, \max_{B \text{ unresolved}} U(B)\}. \quad (32)$$

The bound remains valid when a depth, work, or time budget is exhausted. If r action coordinates have verified strict derivative signs throughout a regular parent box, maximization over that box reduces to a face of dimension at most $d_a - r$.

The last statement describes a source of computational improvement, not a dimension-free worst-case bound. Under a K -Lipschitz gradient, a regular stationary box of diameter s has mean-value excess of order Ks^2 when its point evaluation error is negligible. Boxes that cross stopping switches, have ambiguous derivative signs, or contain many interpolation cells can remain expensive. Our experiment measures their actual cost and keeps unresolved bounds; it does not assume that every box has favorable structure.

The operational NBO output is now explicit: the frozen policy representation, its guarded continuation vector when used, an independently propagated policy interval, and a complete-action optimal upper envelope. A nodal lookup policy is the object certified in the preference experiment. An off-grid interpolation of that policy is a different deployment rule and is evaluated separately. No online search is charged to a stored lookup table; constructing that table includes every actor, critic, and local-search query.

37. COST, REFINEMENT, AND ECONOMIC ACCURACY

The preceding applications define the original economic scope of NBO. We now evaluate the guarded map and signed cover on the same preference economy, and connect the high-dimensional experiment to an absolute economic benchmark. The comparison retains every earlier failure. In particular, a new upper envelope does not turn an actor-plus-search policy into an actor-only policy, and retraining on a finer grid does not repair the off-grid performance of a previously frozen policy.

37.1. *A tighter certificate for identical frozen policies*

The reference specification has 17×25 state nodes, twenty dates, and the continuous action box $[0.02, 0.30] \times [0, 1.5] \times [-0.45, 0.45]$. The ratios constrain consumption, leverage, and preference adjustment in the declared economy. They are economic primitives, not bounds on an unconstrained optimum. Active consumption or portfolio limits remain meaningful features of this constrained specification; changing them changes the economy.

We freeze the nine previously computed actor-plus-search, direct-plus-search, and search-only policies, including their original seeds. The complete-action upper envelope is recomputed independently of these policies' lower payoff envelopes. The new cover uses a one-step tolerance of 0.0002 and retains every unresolved upper bound. Its output can be shared across policies only because their action set, transition rule, domain, and payoff normalization coincide. Table [XLI](#) reports the resulting all-date, all-node loss bounds. The improvement relative to the historical table concerns the tightness of the verification calculation, not additional training or a change in preferences. The raw actors remain separately reported in the machine-readable common accounts. Search remains consequential, and these results do not establish an actor advantage over the actor-free comparator. They do show that the earlier failure of a 0.1 upper-bound target need not imply that the frozen hybrid policies themselves are more than 0.1 from the nodal optimum.

For a controlled cost comparison, Table [XLII](#) gives one-step corner-only and signed covers at three dates, with identical continuation arrays, initial feasible actions, tolerance, depth limit, and work budget. Wall time includes derivative hull

TABLE XLI

COMPLETE-ACTION BOUNDS FOR THE SAME FROZEN POLICIES

Policy	Seed	Uniform bound	Centre bound	≤ 0.1
Actor + search	11	0.0687	0.0068	Yes
Actor + search	29	0.0553	0.0050	Yes
Actor + search	47	0.0552	0.0075	Yes
Direct + search	11	0.0465	0.0071	Yes
Direct + search	29	0.0536	0.0052	Yes
Direct + search	47	0.0466	0.0075	Yes
Search only	11	0.0633	0.0113	Yes
Search only	29	0.0655	0.0053	Yes
Search only	47	0.0631	0.0076	Yes

Full 17×25 state grid, twenty dates, unchanged three-control box. The common signed cover took 372.53 seconds; 20 of twenty one-step covers closed at the requested tolerance. Other levels retain all unresolved upper bounds. Bounds are conditional on the stated outward-arithmetic execution model; continuous-state/time transfer is not included.

construction for the signed cover. The point of this experiment is not to compare two historical machines. Historical training costs are retained in the supplementary total-cost accounts with that qualification. Shared verification is amortized under explicitly stated populations of one, nine, and one hundred policies, and the welfare conversion has a separately counted cost. No claim of neural speed advantage is inferred from this accounting.

37.2. *The whole-domain refinement failure and its repair*

A frozen policy has to be distinguished from a policy refitted on a new grid. The complete historical refinement arrays reveal large losses away from the centre. Their maximum positive reference advantages, locations, quantiles, boundary-region summaries, and action saturation rates are now included in the supplement. The largest historical discrepancy is approximately 2.1234 at high risk aversion and wealth close to the lower stopping boundary. Centre values alone do not summarize that experiment. These discrepancies are feasible-reference lower bounds on policy regret, not upper bounds for the continuous economy.

TABLE XLII

MATCHED ONE-STEP ACTION-COVER COMPUTATION

Cover	Date	Seconds	Q boxes, m.	Bracket	Budget hit
Corner	19	18.67	0.414	0.01336	Yes
Signed	19	9.50	0.066	0.00020	No
Corner	10	19.89	0.440	0.02230	Yes
Signed	10	14.60	0.112	0.00020	No
Corner	0	18.32	0.402	0.03430	Yes
Signed	0	25.84	0.446	0.00173	Yes

Continuation, incumbent, tolerance, and budget are identical within each date. Wall time includes gradient work; a query count is not a hardware-independent speed measure.

The new refinement comparison uses genuinely nested spatial grids, 17×25 and 33×49 , with twenty and forty dates, respectively. Each declared actor and actor-free search case is trained afresh. The network width, activation, actor budget, critic budget, and seed set are unchanged across the relevant ablation. We compare the unguarded fitted critic with the indexed guard in (30). For the forty-date case the target tolerance is 0.0000625. Both the fraction of corrected nodes and their storage cost are reported, rather than being attributed to an uncorrected neural approximation. A new complete-action envelope is computed on the fine grid, not

TABLE XLIII

FRESH POLICIES ON THE NESTED FINE GRID

Method	Uniform bound range	Ref. advantage	Mean train s.	Passes
Actor	0.0654–0.0963	0.0932	88.32	3/3
Actor + guard	0.0140–0.0356	0.0155	87.75	3/3
Search	0.0655–0.0963	0.0932	51.07	3/3
Search + guard	0.0349–0.0357	0.0156	47.40	3/3

Full 33×49 state grid, forty dates; seeds 11, 29, and 47. Reference advantage is a lower regret diagnostic against the feasible 175-action reference, not an upper bound. Uniform bounds use a separately computed complete-action fine-grid envelope. The guard is a saved indexed correction, not an uncorrected network.

interpolated from the coarse certificate. It is applied to all twelve fine-grid policies, including the six unguarded runs. This separates three facts: the failure of a frozen coarse-grid policy; the accuracy achieved by retraining within a declared nodal economy; and the additional error required to transfer a result to a continuously monitored diffusion. The first remains in the record, the second is measured, and the third is not assigned a zero value. Two levels alone do not establish an asymptotic convergence rate.

The deployment specification is a stored state-time action table with its saved neural proposal and continuation correction. Evaluating this table at its nodes requires no online maximizing search. Interpolating the table at off-grid states invokes a different policy rule; the historical experiment shows why this distinction matters. A raw actor without a table or local improvement has a different memory requirement and is not represented by the hybrid loss bounds.

37.3. *A model-specific welfare account*

An absolute utility gap is not a consumption-equivalent percentage when the utility exponent is itself a controlled state. We therefore compute a defined compensating transfer. Hold the saved policy and its state law fixed, leave preference-adjustment costs and the stopping bequest unchanged, and add externally financed consumption so that flow consumption becomes $(1 + e)c_t X_t$. This intervention changes felicity but does not charge its financing to wealth. It is a compensation experiment, not another admissible self-financed policy in the original box.

Let J_e^π be the compensated payoff and U the common optimal upper envelope. A verified lower envelope $\underline{J}_e^\pi \geq U$ at every active starting node and date implies that this transfer compensates the policy's loss everywhere in the declared nodal economy. We search for a common e using monotonicity of CRRA felicity in consumption and verify the final inequality by outward propagation. Table XLIV gives the resulting upper percentages. These conservative transfers can be economically substantial even when the absolute utility bound is below 0.1; the threshold is not described as a universally small welfare loss.

TABLE XLIV

SUFFICIENT FINANCED FLOW-CONSUMPTION SUPPLEMENTS

Proposal	Seed	Supplement, percent	Verification seconds
Actor	11	20.93	27.42
Actor	29	19.17	26.86
Actor	47	18.87	26.94
Direct	11	15.33	27.06
Direct	29	18.06	28.34
Direct	47	15.33	27.50
Search	11	20.44	31.32
Search	29	22.46	31.35
Search	47	19.42	31.37

Uniform sufficient compensation over every active starting node and date. State dynamics, the original policy, terminal bequest, and adjustment cost stay fixed. Consumption is externally financed. These are not feasible self-financed policy improvements or common homothetic consumption equivalents.

37.4. *A global benchmark for the coupled capital economy*

A comparison of two frozen policies does not indicate how far either is from the economic optimum. We supplement the paired simulations by a bound on the original, continuous-time, dense nonlinear capital economy. It does not rely on a Riccati equation, separability of the optimal policy, a state mesh, or a sampled residual.

In the notation of the capital experiment, let Y_i be log capital and $m_i \in [\underline{m}, \bar{m}]$ the consumption ratio. The flow payoff is $\log \bar{m} + \bar{Y} - (\zeta/2)\bar{m}^2$, and the terminal payoff is $\bar{Y} - \chi(Y - \bar{Y})^2$, with $\chi = 0.03$. The drift of Y_i is $a + \kappa \tanh((BY)_i) - m_i - (\sigma_I^2 + \sigma_C^2)/2$. Common noise affects the mean but cancels from cross-sectional dispersion. Define

$$b_{\pm} = a \pm \kappa - \frac{1}{2}(\sigma_I^2 + \sigma_C^2), \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)},$$

and let

$$m_0(t) = \text{clip}_{[\underline{m}, \bar{m}]} \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}.$$

This common, deterministic schedule is feasible in the unchanged economy.

PROPOSITION 4—Continuous-economy performance benchmark: *Starting from $Y_i(0) = 0$ for every i , let V^* be the optimal expected payoff of the original d -dimensional capital economy. Define*

$$U = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_+ - m_0(t)] \} dt,$$

$$D_d = \{ \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \}^2,$$

$$L_d = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_- - m_0(t)] \} dt - \chi e^{-\rho T} D_d.$$

Then $L_d \leq J(m_0) \leq V^ \leq U$ for every admissible dense coupling matrix B in the stated model. The inequalities apply to the continuous economy on $\mathbb{R}^d$, not merely to an Euler simulation.*

The bound exploits bounded nonlinear drift, concavity of consumption utility, and an explicit feasible policy. Its numerical integration uses outward rectangle sums, interval exponential and logarithm evaluations, and square roots checked by interval squaring. Table XLV reports the computed brackets. Their width is economically material and is not presented as a small regret bound for a learned policy. They nevertheless provide an absolute benchmark absent from a paired-policy comparison. Simulations of m_0 use the same Brownian paths as the saved neural policies. The resulting differences and confidence intervals concern their common Euler discretization; an unknown simulation bias is not subtracted from the analytical bound to create a neural-policy certificate.

38. TRAINING, DEPLOYMENT, AND CERTIFIED CAPITAL ACCURACY

We execute the preceding construction on the unchanged capital economy in dimensions ten, twenty, and fifty. Each dimension has three fixed initialization

TABLE XLV

GLOBAL BOUNDS IN THE ORIGINAL CONTINUOUS CAPITAL ECONOMY

Dimension	Time panels	Feasible lower	Optimal upper	Width
10	4096	-1.489064	-1.051803	0.437261
10	16384	-1.488883	-1.051998	0.436884
20	4096	-1.489130	-1.051803	0.437327
20	16384	-1.488949	-1.051998	0.436950

Initial log capital is zero in every coordinate; horizon one. The bounds cover every admissible control on $\mathbb{R}^d$. They are not small loss bounds for neural policies. The same feasible schedule is evaluated on the saved Brownian paths in the supplement.

seeds, 11, 29, and 47. The actor and critic have two hyperbolic-tangent hidden layers of width 32. The actor radius is $\epsilon = .1$. The critic incorporates the terminal payoff exactly and includes a generic neural correction to the discounted mean and dispersion terms. This is an approximation architecture, not an exact invariant solution class.

38.1. A fixed protocol with strong comparators

Each fit uses 600 critic updates, batch size 128, and the same state draws and initialization seeds across methods. The critic learning rate is .0015; the actor learning rate is .002. Two independent Gaussian trace banks, with two probes each, are actually used in the product residual objective. Negative sample losses are retained. Training states have $t \in [0, 1]$ and each log capital in $[-1.5, .5]$. Exact-Hessian diagnostics at 128 held-out points do not enter the continuous certificate.

The direct-tube comparator removes the actor and globally maximizes the same critic Hamiltonian on the identical time-dependent tube. The direct-full comparator instead maximizes on the full primitive box $[.02, 2]^d$. Both exploit the scalar monotone mean-consumption equation and use 44 bisection steps. They receive the same critic architecture, sample stream, trace banks, and update count. Their additional maximization work is counted. This is a matched-update design, not a claim that unequal wall times are equal compute budgets. The untrained feasible schedule m_0 is a fourth comparator.

There are 27 fitted policies, including all seeds and both direct alternatives. Every saved actor is deployed without local action search. The full deployment cost includes the inward scalar schedule guard; direct deployment also includes critic differentiation and bisection. Separate process runs record peak resident memory, training time, query counts, and optimizer histories. The raw proposals, guarded deployment routines, and the scalar benchmark are not treated as interchangeable policies.

38.2. Continuous-time accuracy and its economic scale

Table XLVI reports outward bounds for the anchor and for every policy in the declared tube. The initial cross-sectional standard deviation is at most the stated s_0 , while the initial mean is unrestricted. No state-space covering or unknown diffusion-transfer term enters these numbers. The complete-action comparison class remains the original primitive box, not the actor tube.

TABLE XLVI

CONTINUOUS-ECONOMY BOUNDS FOR EVERY DEPLOYED TUBE POLICY

d	s_0	Anchor bound	Tube bound	Flow supplement (%)
10	0.00	0.01875	0.05743	6.04
10	0.25	0.02152	0.06184	6.52
10	0.50	0.02455	0.06649	7.02
20	0.00	0.02553	0.06725	7.11
20	0.25	0.02881	0.07219	7.65
20	0.50	0.03235	0.07740	8.22
50	0.00	0.02754	0.07009	7.42
50	0.25	0.03097	0.07518	7.98
50	0.50	0.03465	0.08054	8.57

$\epsilon = .1$; 16,384 outward quadrature panels. Initial mean is unrestricted. The optimum allows all adapted controls in $[\cdot, .02, 2]^d$. Supplement is externally financed and uses an outward logarithmic-utility conversion. Printed upper endpoints are rounded upward. Bounds are conditional on the documented arithmetic contract.

The certificate is recomputed at 4,096 and 16,384 quadrature panels. The finer enclosure, verified matrix majorants, and all initial-dispersion and radius variants are saved. This refinement concerns the numerical integration of a proved continuous-economy bound, not a grid approximation to the state process. The spectral verification and scalar integration costs are charged explicitly. The certificate is common to the actor and direct-tube policy class; it is not a seed-specific estimate of achieved regret.

For an interpretable utility scale, consider an externally financed proportional supplement to every consumption flow, holding the policy-induced state law, adjustment cost, and terminal payoff fixed. A supplement e raises discounted payoff by

$$\log(1 + e) \frac{1 - e^{-\rho T}}{\rho}.$$

Thus an outward upper value of $\exp\{\rho(G_d + D_{d,\epsilon}) / (1 - e^{-\rho T})\} - 1$ is a sufficient flow-consumption compensation. This is a defined externally financed welfare comparison, not a claim that the same transfer is self-financing or that all economic models share a common utility percentage.

38.3. *What is learned, and what is guaranteed*

Table XLVII reports the actual cost of training and deploying the policies. The supplement reports every seed, rather than only an average of successful cases. The structural certificate already holds for the initial zero-output actor; therefore the time to its first certificate is not its training time. The anchor has a tighter guarantee and no neural training cost. These facts are central to the interpretation, not omitted ablations.

We separately compare frozen policies on common Brownian paths with 40, 80, and 160 Euler steps and 512 paths per dimension. All fine increments, terminal states, and pathwise payoffs are stored. Table XLVIII gives paired actor-minus-comparator differences at the finest level. The reported intervals cover Monte Carlo variation only; they exclude time-discretization bias and are not used to sharpen

TABLE XLVII

MEASURED COST OF THE TRAINED AND DEPLOYED POLICIES

d	Method	Training (s)	Training + bound (s)	Deployment (ms)
10	Schedule	0.000	0.113	0.053
10	NBO actor	5.284	5.397	1.703
10	Direct tube	6.426	6.539	5.724
10	Direct full	6.366	–	4.140
20	Schedule	0.000	0.115	0.058
20	NBO actor	5.600	5.715	1.766
20	Direct tube	6.753	6.868	6.143
20	Direct full	6.658	–	4.550
50	Schedule	0.000	0.129	0.054
50	NBO actor	6.259	6.387	1.840
50	Direct tube	7.837	7.966	7.711
50	Direct full	7.929	–	6.073

Training is the mean of three separate seed runs; deployment evaluates 256 states and includes the inward schedule guard where applicable. Matrix verification plus scalar quadrature is charged in full to each certified method; no shared-cost amortization is used. The schedule needs no training and has the tighter guarantee. Direct-full training time is recorded, but no tube certificate is assigned to it. Detailed per-run memory and failures appear in the supplement.

Theorem 7. Multiple seeds and resolutions are retained without a multiplicity-adjusted discovery claim.

This experiment supplies a trained multilayer NBO policy, a strong direct comparator, a deployed-policy specification, and an independent continuous-economy guarantee in one coupled nonlinear model. It does not establish that the actor is universally preferable to a structured feasible schedule, that constant-step optimization converges, or that the same certificate applies to endogenous preference formation. The latter economy retains the separate nodal and continuous-transfer accounts developed above. Likewise, the dynamic-game deviation bounds retain their stated finite-game scope.

TABLE XLVIII

PAIRED ACTOR-MINUS-COMPARATOR PAYOFF DIFFERENCES

d	Comparator	Mean difference	Lower 95%	Upper 95%
10	Schedule	0.001299	0.001280	0.001319
10	Direct tube	0.000861	0.000821	0.000902
10	Direct full	0.000965	0.000916	0.001014
20	Schedule	-0.001954	-0.002043	-0.001865
20	Direct tube	0.007563	0.007408	0.007719
20	Direct full	0.071241	0.067724	0.074758
50	Schedule	-0.000647	-0.000707	-0.000587
50	Direct tube	0.012037	0.011900	0.012173
50	Direct full	0.412308	0.402341	0.422276

512 common paths, 160 Euler steps. Each row averages the pathwise payoff differences of the three fixed seed fits before computing a Monte Carlo interval. These are not intervals over a population of training seeds and do not include Euler bias. No simulation result is substituted into the continuous-time proof.

39. PRIOR CONCLUDING DISCUSSION (R10)

Neural Bellman Operators organize computation around policy evaluation, feasible improvement, and independent economic verification. Recursive utility, endogenous preferences, sophisticated temporal selves, and dynamic strategic interaction remain part of that economic formulation. The representation is neural, but an economic claim concerns the policy actually deployed and the error that has been bounded.

The guarded map specifies how a nodal evaluation tolerance is enforced and records the indexed corrections it requires. Signed derivative enclosures reduce complete-action verification work without dropping unresolved boxes or differentiating across a stopping discontinuity. Their numerical contribution is identified using identical frozen policies and matched one-step comparisons. Fresh training and recertification on genuinely nested grids are kept distinct from the full-domain failure of interpolated coarse policies. The compensation account translates util-

ity accuracy into a stated financed-transfer experiment rather than a misleading common percentage.

The dense capital experiment now has a feasible-policy lower benchmark and a continuous-time optimal upper benchmark on the original state space. Those historical absolute brackets are distinct from the new continuous-time neural-policy bound in Theorem 7. The latter uses the original capital economy and full adapted comparison class, with a schedule-centred actor architecture. Paired simulations remain discretized policy comparisons. Likewise, the complete-action preference certificates do not yet bound all continuous-state, continuous-time, and stopping-monitoring defects. The evidence does not establish that actors outperform the strongest low-dimensional search and projection alternatives; those comparisons remain in the record. The contribution is a specified neural Bellman computation with verified evaluation and improvement outputs, measured costs, and separate economic domains for each conclusion.

APPENDIX A: PROOFS

A.1. Proof of Theorem 2

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candidate residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (15).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (14). The triangle inequality gives (16), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$

specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

A.2. Exact policy improvement

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise. The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

A.3. Proof of Theorem 3

For a fixed state and action, let z and $\tilde{z}$ solve (17) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . Q.E.D.

A.4. Proof of Theorem 4

First, (18) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \longrightarrow 0.$$

1 It remains to identify the limit of the exact monotone scheme. Stability makes the up- 1
 2 per and lower half-relaxed limits finite. At a strict local maximum of the upper limit 2
 3 minus a smooth test function, select nearby maximizing grid points. Monotonicity 3
 4 compares the neighboring scheme values to that test function plus the vanishing 4
 5 contact offset. Local consistency then gives the viscosity subsolution inequality. 5
 6 The corresponding minima give the supersolution inequality for the lower limit. 6
 7 Consistent boundary data and the assumed barriers give the required boundary 7
 8 inequalities. Comparison orders the upper limit below the lower limit. The reverse 8
 9 order follows by definition, so the two limits coincide with the unique continuous 9
 10 viscosity solution. The vanishing distance between v_h and v_h^* transfers that conver- 10
 11 gence to the neural arrays. The discrete policy-value conclusion follows from the 11
 12 third bound. Interpreting a discrete policy as a continuous-time implementation 12
 13 additionally requires convergence of its controlled interpolation and payoff; the 13
 14 theorem does not infer this merely from convergence of its state arrays. This is the 14
 15 usual monotone-scheme selection argument, with the neural approximation error 15
 16 made explicit; see [Barles and Souganidis \(1991\)](#), [Crandall et al. \(1992\)](#). Q.E.D. 16

19 A.5. Finite-horizon Bellman error accumulation 19

20 Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz 20
 21 constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and 21
 22 $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives 22
 23

$$24 \quad \|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k). \quad 25$$

26
 27
 28 The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty 28
 29 products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for 29
 30 additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consis- 30
 31 tent terminal data suffice on a fixed finite horizon, without asserting a stationary 31
 32 contraction in an inapplicable utility regime. 32

A.6. Derivation of the temporal-self condition

A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then returns to a^* . For each exponential component, the deviation value is

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t + \mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evaluation equations under a^* leaves

$$\epsilon\{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

Nonpositive first-order gain for every feasible action is exactly (20). This derivation assumes the smoothness and local integrability needed for the expansion; it is an equilibrium deviation condition, not an assertion of optimal commitment over a finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. *Q.E.D.*

APPENDIX B: PROOFS FOR THE GUARDED AND SIGNED MAP

B.1. Proof of Proposition 2

At a replaced node the output equals the computed target. At any other node its distance from that target is at most τ_h . The triangle inequality gives the one-step conclusion. For a frozen policy sequence, let e_n be the sup-norm difference between the guarded and exact policy values at date n . Positivity of the transition weights and their mass bound imply $e_n \leq \tau_h + \eta_h + q_h e_{n+1}$, with $e_N = 0$ for exact terminal data. Backward induction gives the geometric sum. As $N = T/h$ and $q_h \leq 1$, the bound is at most $T(0.1h + \eta_h/h)$ under the stated schedule. No property of the optimizer was used. The assertion concerns the guarded nodal values. An off-node neural interpolant, an improved policy, and a diffusion approximation each require their separate error account. In fixed-precision arithmetic, $\eta_h = o(h)$ is a maintained refinement condition, not a fact that can be inferred as h tends to zero indefinitely. $\square$

B.2. *Proof of Theorem 3*

On a regular box, the payoff restricted to a line segment is continuous and piecewise continuously differentiable, with finitely many interpolation and reflection knots, or is the limit of such segments. Integration of its almost-everywhere derivative yields

$$Q_w(a) - Q_w(c) = \int_0^1 \nabla Q_w(c + s(a - c)) \cdot (a - c) ds.$$

Every partial derivative is included in its interval hull. Maximizing the resulting interval products proves (31). Taking the minimum with another valid upper enclosure preserves validity. If a partial derivative is strictly positive throughout the box, integration along that coordinate shows that replacing it by its upper endpoint cannot lower the payoff; the negative case is analogous. Repeating the argument proves face domination and the dimension statement.

A subdivision replaces a parent by children whose union contains it. A face reduction replaces a set by a subset with the same supremum. Removal occurs only after that supremum has a bound no larger than the incumbent plus tolerance. Feasible point evaluations can only increase the incumbent, so an earlier removed box stays bounded by the final incumbent plus tolerance. The remaining boxes keep their upper bounds without any success inference from an interrupted search. Induction over all operations proves (32). At a stopping switch the line-segment continuity premise can fail; the implementation therefore uses the natural union enclosure and does not invoke the derivative argument there.

For the local quadratic observation, a K -Lipschitz gradient with a zero-gradient point in the box has norm at most Ks throughout a box of diameter s . Integration over a segment of length at most s bounds the variation by Ks^2 . This conditional observation does not extend to an arbitrary switching box, nor does it assert that all nodes possess such a box. $\square$

B.3. Derivative hulls for the preference economy

Write $y = \log X$, and let F be the continuation interpolant evaluated at a cubature transition. On an interior leg, its action derivatives are

$$\partial_c F = -hF_y, \quad \partial_p F = F_y \{h(0.06 - 0.04p) + s_z\}, \quad \partial_\theta F = hF_u r_u,$$

where s_z is the stored portfolio shock coefficient and $r_u \in \{-1, 1\}$ is the reflection slope; both slopes enter the hull at a fold. The flow derivatives are

$$h \exp\{(1-u)y - u \log c\}, \quad 0, \quad -hk\theta.$$

Outside a wealth stopping face, clipped wealth is constant and F is the exact stopping payoff. Its y derivative with respect to the transition is zero, while

$$F_u = \frac{\exp\{(1-u)y\} \{1 - (1-u)y\}}{(1-u)^2}.$$

Every crossed bilinear cell contributes its one-sided derivative range, restricted by the other coordinate's interval. Interval evaluation includes these ranges and the flow terms. A leg whose wealth interval meets both sides of a stopping switch disables the signed bound for that box. Point and interval evaluations share the stated nodal primitives but are computed independently of the actor's derivative graph.

B.4. Proof of Proposition 4

Let $\bar{m} = d^{-1} \sum_i m_i$. Jensen's inequality gives $d^{-1} \sum_i \log m_i \leq \log \bar{m}$. Moreover, $d^{-1} \sum_i \tanh((BY)_i) \leq 1$. The mean-state drift is consequently no larger than $b_+ - \bar{m}$. The dispersion penalty at the terminal date is nonnegative and can be dropped for an upper bound. Fubini's theorem applied to the integrated expected mean state and terminal mean state yields the kernel $w(t)$. The Brownian terms have zero mean; bounded controls and bounded nonlinear drift ensure the required integrability. Thus every admissible policy has expected payoff at most

$$\int_0^T e^{-\rho t} \max_{m \in [\underline{m}, \bar{m}]} \{\log m - \zeta m^2 / 2 + w(t)(b_+ - m)\} dt.$$

The integrand is strictly concave in m . Its derivative is $1/m - \zeta m - w(t)$, whose positive root gives $m_0(t)$ before clipping. This proves the upper bound.

For the feasible common schedule, the expected mean-state drift is at least $b_- - m_0(t)$. Define the normalized dispersion norm $\|Z\|_{2,d}^2 = \mathbb{E}[d^{-1} \sum_i Z_i^2]$. Centred common controls and common noise vanish from the dispersion equation. For numbers in $[-1, 1]$, their cross-sectional variance is at most one. Hence Minkowski's inequality bounds the norm of the integrated centred nonlinear drift by κT . The centred idiosyncratic Brownian part has norm $\sigma_I \sqrt{(1 - 1/d)T}$. A second application of Minkowski and squaring give the upper bound D_d on expected terminal dispersion. Subtracting $\chi e^{-\rho T} D_d$ from the lower mean-payoff calculation proves $J(m_0) \geq L_d$. Feasibility gives $J(m_0) \leq V^*$. The derivation never restricts the dense matrix B beyond the stated bounded-tanh form. $\square$

B.5. Compensation and arithmetic scope

For each realized state and positive consumption, CRRA felicity is increasing in consumption, including when $u > 1$ and its level is negative. The financed supplement leaves the state law and every other payoff term unchanged. Its expected payoff is therefore nondecreasing in e . Positivity of the nodal transition operator propagates a lower interval payoff at each date. If its lower endpoint exceeds the optimal upper envelope at every active starting node and date, the corresponding financed supplement compensates the loss throughout that nodal economy. Bisection searches for a sufficient supplement; only the final verified inequality provides the stated upper bound.

All implemented interval conclusions use the inherited binary64 execution contract, with outward elementary operations and separately bounded transcendental evaluations. They concern exact real interpretations of stored primitive coefficients and grid nodes. They are not a formal verification of the interpreter, hardware, or all library internals. In the capital bound, square-root seeds are accepted only after interval squaring verifies both endpoint inequalities, and the time integral is bounded by interval rectangles rather than by an adaptive quadrature tolerance.

Random stress tests are regression evidence, not substitutes for these inclusion arguments.

APPENDIX C: PROOF OF THE CONTINUOUS-CAPITAL GUARANTEE

All norms in this proof use the normalization $d^{-1} \sum_i z_i^2$. This normalization is important: the spectral norm of B remains the ordinary Euclidean operator norm, while the stochastic moment of cross-sectional dispersion is of order one rather than order d .

C.1. Logarithmic tangent remainder

For $a > 0$ and $x > -a$, twice integrating the second derivative gives

$$\frac{x}{a} - \log(1 + x/a) = x^2 \int_0^1 \frac{1-r}{(a+rx)^2} dr.$$

For $x \geq 0$ the integral decreases as x increases. For $x < 0$ it is at least $1/(2a^2)$, its value at zero. Consequently, for every $m \in [\underline{m}, M]$ and interior $a < M$,

$$\log(m/a) - \frac{m-a}{a} \leq -\frac{\mu(a)}{2}(m-a)^2, \quad \mu(a) = \frac{2\{\log(a/M) + (M-a)/a\}}{(M-a)^2}. \quad (33)$$

On the smaller interval $|m-a| \leq \epsilon < a$, the same integral also yields

$$0 \leq \frac{m-a}{a} - \log(m/a) \leq \frac{(m-a)^2}{2(a-\epsilon)^2}. \quad (34)$$

These are global inequalities on the specified intervals, including their endpoints; no empirical concavity check is used.

C.2. Synchronous comparison and baseline moments

Construct Y^m and Y^0 on the same probability space with identical initial state and Brownian drivers, where the second process uses m_0 . With $\Delta Y = Y^m - Y^0$ and $\Delta m = m - m_0 \mathbf{1}$, additive noise cancels. Since componentwise hyperbolic tangent is one-Lipschitz,

$$\|\Delta Y_t\|_d \leq \int_0^t \{\beta \|\Delta Y_s\|_d + \|\Delta m_s\|_d\} ds.$$

Gronwall's inequality followed by Minkowski's integral inequality implies

$$(\mathbb{E}\|\Delta Y_t\|_d^2)^{1/2} \leq \int_0^t e^{\beta(t-s)} r(s) ds, \quad r(s) = (\mathbb{E}\|\Delta m_s\|_d^2)^{1/2}. \quad (35)$$

All terms are integrable because controls and the nonlinear production term are bounded. For the baseline, common consumption, common drift constants, and common noise vanish under Π . Furthermore, $\|\Pi \tanh(BY)\|_d \leq 1$ and $\mathbb{E}\|\Pi W_T\|_d^2 = (1 - 1/d)T$. Minkowski's inequality therefore gives

$$(\mathbb{E}\|\Pi Y_T^0\|_d^2)^{1/2} \leq \|\Pi y\|_d + \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \leq Q_d. \quad (36)$$

This estimate is uniform in the initial mean. It does not suppose that the optimal policy is symmetric or that the dense economy separates.

C.3. An exact payoff-difference identity

Put $h(Y) = \overline{\tanh(BY)}$. Integrating the mean-state equation inside the discounted flow and terminal mean, and interchanging integrable expectations and time integrals, yields

$$\begin{aligned} J_y(m) - J_y(m_0) &= \int_0^T e^{-\rho t} \mathbb{E} \left[\overline{\log(m_t/m_0(t))} - w(t) \overline{\Delta m_t} - \frac{\zeta}{2} \{\bar{m}_t^2 - m_0(t)^2\} \right] dt \\ &\quad + \kappa \int_0^T e^{-\rho t} w(t) \mathbb{E}[h(Y_t^m) - h(Y_t^0)] dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \left[2 \langle \Pi Y_T^0, \Pi \Delta Y_T \rangle_d + \|\Pi \Delta Y_T\|_d^2 \right]. \end{aligned} \quad (37)$$

The martingale mean terms have zero expectation. The schedule satisfies $m_0(t)^{-1} - w(t) - \zeta m_0(t) = 0$. Thus all terms linear in $\overline{\Delta m_t}$ cancel. Equation (33) bounds the first integral above by $-\frac{1}{2} \int e^{-\rho t} \mu(t) r(t)^2 dt$; the remaining mean-adjustment square is nonpositive and may be dropped in an upper bound.

Next, $|h(Y^m) - h(Y^0)| \leq \|B\|_2 \|\Delta Y\|_d$. Equations (35) and (36), Cauchy-Schwarz, and a change in integration order bound the potentially positive production and terminal-dispersion terms by $\int_0^T K(s) r(s) ds$. Dropping the nonpositive terminal

square, we obtain

$$J_y(m) - J_y(m_0) \leq \int_0^T \left[K(s)r(s) - \frac{1}{2}e^{-\rho s}\mu(s)r(s)^2 \right] ds \leq \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds.$$

Taking the supremum over the original adapted controls proves (37). Attainment of the supremum is not needed.

For a tube policy, $r(s) \leq \epsilon$. Reverse the payoff difference in (37). Equation (34) and $\mathbb{E}(\overline{\Delta m})^2 \leq \mathbb{E}\|\Delta m\|_d^2$ bound the consumption and adjustment loss by

$$\frac{\epsilon^2}{2} \int_0^T e^{-\rho s} \{[m_0(s) - \epsilon]^{-2} + \zeta\} ds.$$

The production and terminal cross terms contribute at most $\epsilon \int K(s)ds$. The terminal square now has positive sign and is bounded by $\chi e^{-\rho T} \epsilon^2 A_\beta(T)^2$ using (35). This proves $J_y(m_0) - J_y(m^\phi) \leq D_{d,\epsilon}$. Adding (37) proves (38). Feasibility supplies the nonnegative lower bound. *Q.E.D.*

C.4. Interval spectral and deployment checks

For the stored exact-real matrix B , outward products and sums enclose each entry of $B^\top B$. The Schur complement recurrence for an $LDL^\top$ factorization is enclosed entry by entry. If every pivot interval has strictly positive lower endpoint, every exact pivot is positive, so $\lambda I - B^\top B$ is positive definite. Hence $\|B\|_2 < \sqrt{\lambda}$. The proposed numerical singular value is not part of this proof; an unsuccessful interval factorization supplies no spectral bound.

If $\ell \leq m_0(t) \leq u$, then $\text{up}(u - \epsilon) \geq m_0(t) - \epsilon$ and $\text{down}(\ell + \epsilon) \leq m_0(t) + \epsilon$. Every finite output after projection onto (41) therefore meets the theorem's tube condition under the declared arithmetic contract, regardless of errors in the neural proposal. The proof does not verify the interpreter or processor. The arithmetic assumptions, overflow checks, saved weights, and input-clock convention remain explicit components of the computational claim.

APPENDIX D: REPLICATION AND SCOPE OF NUMERICAL EVIDENCE

Integrated R8 candidate. The present article incorporates the following immutable repository inputs.

1 R7 source: 5bab4649728860fb655c5371f3705787f10b4a78 1
2 R7 evidence: b1409008625f550195ce9ead468391ad2449a240 2
3 R7 advisory report: c63f6817f498e1cfe9a653139f7a25bf44a4418a 3
4 4

5 The preceding main manuscript and supplement are preserved by exact Git blob 5
6 identity in the R8 archive. R8 source, all executed development configurations, raw 6
7 arrays, independent certificates, method-paired comparisons, generated tables, and 7
8 compilation logs belong to the candidate itself. The response answers every B1–B7 8
9 and M1–M9 comment of the latest report. The later unexecuted R7 development 9
10 protocol is not imported as evidence. A table manifest checks each printed number 10
11 against its input file; the execution manifest identifies the exact R8 source and 11
12 candidate workflow. Earlier review and revision branches remain unchanged. 12

13 The revision directory contains the executable block core, the experiment program, 13
14 the regression tests, raw numerical outputs, and a response-to-referee document. 14
15 The reviewed manuscript, supplement, and historical comments are preserved byte- 15
16 for-byte in the archive, and the review branch remains unchanged. The new main 16
17 manuscript is authoritative for the revised equations. The archive is not compiled 17
18 into the revised article and its unsupported numerical statements are not reused as 18
19 evidence. 19

20 The inherited R2 execution contains 29 recorded experiments and eight regression 20
21 tests, all retained and rerun. The R6 record adds nine primary stopped-consumption 21
22 runs, six architecture/update sensitivities, six full-domain certificates, six principal 22
23 NDU neural runs, sixteen NDU reference sensitivities, twenty-four dense-capital 23
24 primary fits, six stochastic-trace fits, twelve coverage follow-ups, and six principal 24
25 dynamic-game solves. Pilot failures and coarse-game counterchecks are retained 25
26 separately rather than counted as successful primary runs. The original referee 26
27 diagnostics continue to concern their pinned original sources. 27

28 The initial development summary and the corresponding script are preserved 28
29 separately. Inspection found a sign error in the *reported* quadratic residual formula 29
30 even though the Lyapunov solutions agreed with Riccati values. The formula was 30
31 corrected, its acceptance test strengthened to include that residual, and the complete 31
32 experiment suite rerun. The wealth stopping boundary was also imposed explicitly 32

at all NDU time levels before the final run. The earlier results are marked superseded diagnostics rather than silently relabelled as final evidence.

The R6 directory contains complete source, generated tables, model and numerical arrays, optimizer histories, certificate covers, tests, and a provenance manifest. The replication workflow materializes the revision source, independently reruns its computations, and commits the resulting evidence only to the designated R6 candidate branch; an exact-source snapshot and raw-result hashes identify each execution. Local execution, remote replication, and independent referee acceptance remain different facts. No historical GPU benchmark, unexecuted deep-BSDE comparison, unrestricted neural Cournot equilibrium, or empirical identification result is inferred from the computations reported here.

4.0.0.1. R9 revision record. The authoritative R9 source descends from the review of the R8 candidate, dated 3 October 2026. Its new material, all raw runs, complete-action arrays, full-domain refinement summaries, proofs, response, and execution identities are stored under `revisions/2026-10-03-r9`. The archived R8 main text and supplement preserve the pre-revision scholarly record. Every inherited section and label is retained. The new response addresses the R8 report’s B1–B7 and M1–M10 comments. Earlier replication and response descriptions in this historical record refer to their respective R6–R8 rounds; historical workflow success is not substituted for a run of the current source.

4.0.0.2. R10 revision record. The current source descends from the complete R9 evidence commit. Section 7 and its proof give the continuous-capital guarantee; Section 38 reports the complete training and deployment comparison. Exact previous roots, all new source and results, the response to the 3 October 2026 R8 advisory report, and the current execution identities are stored under `revisions/2026-10-04-r10`. Earlier numerical accounts and unsuccessful experiments remain unchanged.

5. R11 COMPLETE NUMERICAL RECORD

TABLE I: All primary policies. The utility endpoints include time, sampling, clipping, and arithmetic errors.

d	Method	Seed	Iter.	Mean gain	L_ϕ	Regret upper
10	Direct policy	11	100	0.001885	0.001109	0.017638
10	Direct policy	29	100	0.001888	0.001113	0.017634
10	Direct policy	47	100	0.001890	0.001114	0.017632
10	Direct policy	71	120	0.001886	0.001110	0.017636
10	Direct policy	89	100	0.001884	0.001109	0.017638
10	Direct policy	107	100	0.001881	0.001105	0.017642
10	Direct policy	131	120	0.001882	0.001107	0.017640
10	Direct policy	149	120	0.001886	0.001110	0.017637
10	Direct policy	167	100	0.001893	0.001117	0.017629
10	Direct policy	191	120	0.001887	0.001111	0.017635
10	Affine	11	120	0.001197	0.000425	0.018322
10	Affine	29	120	0.001190	0.000418	0.018328
10	Affine	47	120	0.001180	0.000408	0.018339
10	Affine	71	120	0.001190	0.000418	0.018329
10	Affine	89	120	0.001184	0.000412	0.018335
10	Affine	107	120	0.001188	0.000416	0.018330
10	Affine	131	120	0.001193	0.000421	0.018326
10	Affine	149	120	0.001181	0.000409	0.018338
10	Affine	167	120	0.001189	0.000417	0.018330
10	Affine	191	120	0.001209	0.000436	0.018310
10	NBO	11	60	0.001961	0.001185	0.017562
10	NBO	29	100	0.001960	0.001184	0.017563
10	NBO	47	80	0.001965	0.001189	0.017558
10	NBO	71	80	0.001963	0.001187	0.017560
10	NBO	89	120	0.001969	0.001193	0.017554
10	NBO	107	100	0.001963	0.001187	0.017559
10	NBO	131	120	0.001959	0.001183	0.017564
10	NBO	149	100	0.001960	0.001184	0.017562
10	NBO	167	120	0.001966	0.001190	0.017557
10	NBO	191	80	0.001962	0.001186	0.017561

	d	Method	Seed	Iter.	Mean gain	L_ϕ	Regret upper	
1								1
2	20	Direct policy	11	120	0.001501	0.000639	0.024887	2
3	20	Direct policy	29	120	0.001497	0.000635	0.024890	3
4	20	Direct policy	47	120	0.001495	0.000633	0.024893	4
5	20	Direct policy	71	120	0.001499	0.000638	0.024888	5
6	20	Direct policy	89	120	0.001491	0.000629	0.024896	6
7	20	Direct policy	107	120	0.001496	0.000634	0.024891	7
8	20	Direct policy	131	120	0.001496	0.000634	0.024891	8
9	20	Direct policy	149	120	0.001506	0.000644	0.024881	9
10	20	Direct policy	167	120	0.001499	0.000637	0.024888	10
11	20	Direct policy	191	120	0.001500	0.000638	0.024888	11
12	20	Affine	11	120	0.001019	0.000160	0.025365	12
13	20	Affine	29	120	0.001023	0.000164	0.025361	13
14	20	Affine	47	120	0.001020	0.000161	0.025364	14
15	20	Affine	71	120	0.001027	0.000168	0.025357	15
16	20	Affine	89	120	0.001016	0.000156	0.025369	16
17	20	Affine	107	120	0.001023	0.000164	0.025362	17
18	20	Affine	131	120	0.001010	0.000151	0.025374	18
19	20	Affine	149	120	0.001024	0.000164	0.025361	19
20	20	Affine	167	120	0.001010	0.000151	0.025374	20
21	20	Affine	191	120	0.001028	0.000168	0.025357	21
22	20	NBO	11	80	0.001580	0.000718	0.024807	22
23	20	NBO	29	60	0.001579	0.000717	0.024809	23
24	20	NBO	47	120	0.001586	0.000724	0.024801	24
25	20	NBO	71	120	0.001584	0.000722	0.024804	25
26	20	NBO	89	60	0.001577	0.000716	0.024810	26
27	20	NBO	107	100	0.001581	0.000719	0.024807	27
28	20	NBO	131	100	0.001582	0.000720	0.024805	28
29	20	NBO	149	100	0.001585	0.000723	0.024802	29
30	20	NBO	167	120	0.001583	0.000721	0.024804	30
31	20	NBO	191	100	0.001586	0.000724	0.024801	31
32	50	Direct policy	11	120	0.001645	0.000763	0.026777	32
	50	Direct policy	29	120	0.001643	0.000761	0.026779	
	50	Direct policy	47	120	0.001650	0.000768	0.026772	

	d	Method	Seed	Iter.	Mean gain	L_ϕ	Regret upper	
1								1
2	50	Direct policy	71	120	0.001645	0.000763	0.026777	2
3	50	Direct policy	89	120	0.001648	0.000765	0.026774	3
4	50	Direct policy	107	120	0.001645	0.000763	0.026777	4
5	50	Direct policy	131	120	0.001639	0.000757	0.026783	5
6	50	Direct policy	149	120	0.001642	0.000760	0.026780	6
7	50	Direct policy	167	120	0.001646	0.000764	0.026776	7
8	50	Direct policy	191	120	0.001651	0.000769	0.026771	8
9	50	Affine	11	120	0.001012	0.000133	0.027407	9
10	50	Affine	29	120	0.001033	0.000154	0.027386	10
11	50	Affine	47	120	0.001009	0.000130	0.027410	11
12	50	Affine	71	120	0.001036	0.000156	0.027383	12
13	50	Affine	89	120	0.001030	0.000150	0.027389	13
14	50	Affine	107	120	0.001027	0.000148	0.027392	14
15	50	Affine	131	120	0.001024	0.000145	0.027395	15
16	50	Affine	149	120	0.001030	0.000151	0.027389	16
17	50	Affine	167	120	0.001019	0.000140	0.027400	17
18	50	Affine	191	120	0.001028	0.000149	0.027391	18
19	50	NBO	11	80	0.001751	0.000868	0.026671	19
20	50	NBO	29	100	0.001752	0.000869	0.026671	20
21	50	NBO	47	100	0.001750	0.000868	0.026672	21
22	50	NBO	71	120	0.001752	0.000870	0.026670	22
23	50	NBO	89	120	0.001753	0.000871	0.026669	23
24	50	NBO	107	80	0.001751	0.000868	0.026671	24
25	50	NBO	131	120	0.001752	0.000870	0.026670	25
26	50	NBO	149	120	0.001755	0.000872	0.026667	26
27	50	NBO	167	100	0.001753	0.000870	0.026670	27
28	50	NBO	191	80	0.001751	0.000868	0.026671	28
29								29
30								30
31								31
32								32

TABLE II: Complete checkpoint frontiers; seed 11, 2,048 paths
and 512 cells.

d	Method	Iteration	Mean gain	L_ϕ	Regret upper
10	Direct policy	20	0.001258	-0.001028	0.019775
10	Direct policy	40	0.001767	-0.000524	0.019271
10	Direct policy	80	0.001885	-0.000408	0.019155
10	Direct policy	120	0.001875	-0.000418	0.019165
10	Affine	20	0.000583	-0.001699	0.020446
10	Affine	40	0.000770	-0.001513	0.020260
10	Affine	80	0.000994	-0.001290	0.020036
10	Affine	120	0.001195	-0.001091	0.019838
10	NBO	20	0.001908	-0.000383	0.019130
10	NBO	40	0.001947	-0.000345	0.019092
10	NBO	80	0.001957	-0.000335	0.019082
10	NBO	120	0.001951	-0.000342	0.019088
20	Direct policy	20	0.001011	-0.001480	0.027006
20	Direct policy	40	0.001404	-0.001091	0.026616
20	Direct policy	80	0.001490	-0.001005	0.026531
20	Direct policy	120	0.001498	-0.000998	0.026523
20	Affine	20	0.000489	-0.002000	0.027526
20	Affine	40	0.000595	-0.001892	0.027418
20	Affine	80	0.000846	-0.001643	0.027168
20	Affine	120	0.001019	-0.001472	0.026997
20	NBO	20	0.001549	-0.000946	0.026472
20	NBO	40	0.001568	-0.000928	0.026453
20	NBO	80	0.001574	-0.000922	0.026447
20	NBO	120	0.001578	-0.000918	0.026443
50	Direct policy	20	0.000970	-0.001564	0.029103
50	Direct policy	40	0.001420	-0.001116	0.028655
50	Direct policy	80	0.001611	-0.000928	0.028467
50	Direct policy	120	0.001645	-0.000894	0.028434
50	Affine	20	0.000507	-0.002029	0.029568
50	Affine	40	0.000614	-0.001919	0.029459

d	Method	Iteration	Mean gain	L_ϕ	Regret upper
50	Affine	80	0.000836	-0.001697	0.029236
50	Affine	120	0.001012	-0.001521	0.029061
50	NBO	20	0.001717	-0.000823	0.028362
50	NBO	40	0.001743	-0.000797	0.028337
50	NBO	80	0.001747	-0.000793	0.028332
50	NBO	120	0.001750	-0.000790	0.028330

TABLE III: Initial-state, architecture, extended-budget, and adverse-policy records. Outside-box frequency is measured on internal states and is not a domain guarantee.

d	Configuration	Mean gain	L_ϕ	Outside	Saturated
10	saturated correction	-0.010950	-0.012092	0.000	1.000
10	zero correction	-0.000002	-0.001127	0.000	0.000
10	Direct policy 600	0.001896	0.000759	0.000	0.000
20	Direct policy 600	0.001511	0.000273	0.000	0.000
50	Direct policy 600	0.001670	0.000411	0.000	0.000
10	Affine 600	0.001812	0.000676	0.000	0.000
20	Affine 600	0.001468	0.000231	0.000	0.000
50	Affine 600	0.001584	0.000326	0.000	0.000
10	mean -1, sd 1.5	0.001192	-0.000093	1.000	0.000
10	t3 profile	0.001261	-0.000025	1.000	0.000
10	mean -2, sd 0	0.000107	-0.001023	1.000	0.000
10	mean 0, sd 0.5	0.001063	-0.000121	0.691	0.000
10	mean 0, sd 1	0.000817	-0.000415	1.000	0.000
10	mean 2, sd 0	-0.000862	-0.001996	1.000	0.000
10	width 16	0.001962	0.000826	0.000	0.000
10	width 64	0.001961	0.000824	0.000	0.000
20	mean -1, sd 1.5	0.000869	-0.000528	1.000	0.000
20	t3 profile	0.000509	-0.000886	1.000	0.000
20	mean -2, sd 0	0.000159	-0.001073	1.000	0.000

d	Configuration	Mean gain	L_ϕ	Outside	Saturated
20	mean 0, sd 0.5	0.001240	-0.000051	0.857	0.000
20	mean 0, sd 1	0.000831	-0.000514	1.000	0.000
20	mean 2, sd 0	-0.001010	-0.002246	1.000	0.000
20	width 16	0.001580	0.000342	0.000	0.000
20	width 64	0.001581	0.000344	0.000	0.000
50	mean -1, sd 1.5	0.001072	-0.000349	1.000	0.000
50	t3 profile	0.001035	-0.000387	1.000	0.000
50	mean -2, sd 0	0.000126	-0.001129	1.000	0.000
50	mean 0, sd 0.5	0.001631	0.000321	0.895	0.004
50	mean 0, sd 1	0.001378	0.000013	1.000	0.004
50	mean 2, sd 0	-0.000023	-0.001277	1.000	0.000
50	width 16	0.001745	0.000486	0.000	0.000
50	width 64	0.001755	0.000496	0.000	0.001

TABLE IV
VALIDATION-TUNED CRITIC GREEDIFICATION. FULL+PROJECTION IS EXPLICITLY A CONSERVATIVE HYBRID. BUDGETS AND MIXING FRACTIONS ARE CHOSEN WITHOUT THE FINAL NOISE BANK.

d	Critic family	Steps	Mix	Mean gain	L_ϕ
10	Full+projection	1200	1.00	0.001404	0.000264
20	Full+projection	200	0.00	-0.000001	-0.001229
50	Full+projection	200	0.00	-0.000002	-0.001253
10	Tube	1200	1.00	0.001408	0.000269
20	Tube	200	0.00	-0.000001	-0.001229
50	Tube	200	0.00	-0.000002	-0.001253

6. EARLIER LEARNING AND EXPERIMENTAL RECORD

The following learning specification and study are retained from the reviewed manuscript. Their fitted policies and numerical protocols are distinct from the complete population and direct-method study reported in the main article and at the beginning of this supplement.

7. BELLMAN EVALUATION WITH COSTATE TARGETS

The post-training certificate applies to any admissible candidate; it does not select a numerical method in advance. We therefore change the evaluation experiment as well as the verification step. The purpose is to determine whether a separately trained continuation value and feasible actor produce economically useful state dependence, rather than merely remain inside a safe architectural interval.

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\widehat{V}^{(j)}(t, y)$ and its pathwise derivative $\widehat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\widehat{\mathbb{E}} \left[.05 \{v_\theta - \widehat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \widehat{p}^{(j)})\|_d^2 \right]. \quad (38)$$

The factor d puts normalized marginal values on an order-one scale. The critic has the same two hidden tanh layers and hard terminal component as the R10 implementation. Holding its new derivative fixed, the actor then maximizes

$$\widehat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (39)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. Five critic steps and five actor steps are performed for each fresh rollout batch. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor

an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

7.1. *Comparison and selection*

The primary comparisons include direct policy optimization through the same simulator and an affine state-dependent actor followed by tanh and the same feasibility guard. Direct policy optimization has the same nonlinear actor architecture but no learned critic; the affine method exploits a simpler representation. They receive the same mixture of training states, the same seeds, the same rollout length, and the same number of simulation-state visits. Their optimizer-step counts differ by design and are reported, as are wall time and deployment cost. Equality of simulator work is not described as equality of total computation.

Every twentieth iteration is saved. Each method chooses the checkpoint with the highest return on its own fixed validation bank before any final-test shocks are examined. The archive includes all six checkpoints, not only the selected one. Longer direct-policy and affine runs expose whether a short baseline budget is responsible for a comparison. Cost–accuracy frontiers use the saved checkpoints and the same final statistical accounting. The method is not declared superior merely because one direct greedification of an inaccurate critic performs poorly.

Training and validation use finite Euler rollouts. Verification instead uses the specified sampled controller, an independent fine-grid paired statistic, and Proposition 8. An empirical rollout improvement is therefore not substituted for the continuous-time conclusion. Conversely, no uniform HJB residual is assumed merely because a rollout or costate loss is small.

8. THE ECONOMIC VALUE OF STATE-DEPENDENT CONSUMPTION

The quantitative question is whether conditioning consumption on the evolving, densely coupled capital state improves upon a common feasible consumption plan enough to justify the computational intervention. This question is posed in the

original benchmark economy; the exercise is not a calibration to household or firm data. The relevant comparison is the schedule itself, not just another neural implementation. All reported gains are differences in the original discounted utility criterion.

8.1. *Design and primary evidence*

The final design uses ten independent training seeds in each of dimensions ten, twenty, and fifty. Each seed supplies an NBO actor, a direct-policy actor, and an affine actor. The parameter vector and dense matrix-generation rule are unchanged from R10. The primary candidate architecture has width 32, two hidden layers, radius .1, and 120 outer iterations. Each selected policy is evaluated from $y = 0$ using 8,192 fresh paths and 1,024 time cells. A family allocation of 2,000 one-sided statements at total probability .05 covers the primary comparisons and all declared sensitivity rows. The final noise seed differs from the exploratory bank used during development. This is a design fixed before the final remote computation, not preregistration before all research.

TABLE I

PRIMARY CONTINUOUS-ECONOMY COMPARISON. TEN TRAINING SEEDS PER ROW; ALL POLICIES START AT $y = 0$. GAIN IS IN DISCOUNTED UTILITY UNITS. LOWER ENDPOINTS ARE ROUNDED DOWN AND UPPER ENDPOINTS UP.

d	Method	Mean gain	Seed s.d.	Min. L_ϕ	Max. regret
10	NBO	0.001963	0.000003	0.001183	0.017564
10	Direct policy	0.001886	0.000004	0.001105	0.017642
10	Affine	0.001190	0.000008	0.000408	0.018339
20	NBO	0.001582	0.000003	0.000716	0.024810
20	Direct policy	0.001498	0.000004	0.000629	0.024896
20	Affine	0.001020	0.000006	0.000151	0.025374
50	NBO	0.001752	0.000001	0.000868	0.026672
50	Direct policy	0.001645	0.000004	0.000757	0.026783
50	Affine	0.001025	0.000009	0.000130	0.027410

Of the 30 primary NBO policies, 30 have a strictly positive simultaneous lower improvement endpoint. The lower endpoints range from 0.000716 to 0.001193 in discounted utility units. The number meeting the declared improvement target .0005 is 30. The largest policy-specific NBO regret upper endpoint is 0.026672. These are results for the stated implementation, initial state, and conditional sampling contract. The archive contains 230 two-sided evaluation records and 0 logged training failures; an unmet gain target is recorded separately from a training exception.

The table reports the range across fitted seeds, including the least favourable seed, rather than a confidence interval obtained after averaging fitted actors within each path. The family-wise endpoints describe simulation uncertainty conditional on each fitted object. The accompanying seed standard deviations describe algorithmic variability; they are not asserted to estimate the distribution of every possible training run. The raw seed-by-path outcomes permit both quantities to be recomputed.

8.2. Accuracy, cost, and the role of training

TABLE II
RECORDED CPU COSTS AND THE DECLARED IMPROVEMENT TARGET. VERIFICATION INCLUDES SETUP, SIMULATION, AND ARITHMETIC. DECISION TIME INCLUDES A BATCH OF 256 ACTION AND INTERNAL-DRIFT UPDATES. THE LAST COLUMN COUNTS $L_\phi \geq .0005$ AMONG TEN SEEDS.

d	Method	Train (s)	Verify (s)	Decision (ms)	Target / 10
10	NBO	5.53	17.53	0.447	10
10	Direct policy	3.77	14.56	0.446	10
10	Affine	2.77	9.49	0.259	0
20	NBO	5.91	25.64	0.521	10
20	Direct policy	4.16	22.44	0.523	10
20	Affine	3.20	17.16	0.334	0
50	NBO	7.33	50.32	0.755	10
50	Direct policy	5.37	47.32	0.751	10
50	Affine	4.39	41.76	0.566	0

TABLE III

ANALYTICAL COMPARATORS BEFORE POLICY-SPECIFIC IMPROVEMENT. THE LAST TWO COLUMNS ARE EXTERNALLY FINANCED FLOW-EQUIVALENT PERCENTAGES, NOT EMPIRICAL POPULATION WELFARE EFFECTS.

d	G_d	$G_d + D_{d,1}$	Anchor (%)	Tube (%)
10	0.018747	0.057430	1.931	6.034
20	0.025526	0.067244	2.639	7.101
50	0.027540	0.070081	2.850	7.411

The feasible schedule has zero training cost. Its deterministic regret bound is printed beside the policy-specific endpoints. The reported verification cost includes fine-grid simulation and enclosure construction; it is not hidden in a training-only runtime. Once a policy has been verified, deployment requires the stored actor and an internal state update, but no critic optimization or global action search. Machine timing is a property of the recorded CPU run, not a dimension-free complexity theorem.

The archived R10 actors had negative actor-minus-schedule payoff differences in dimensions twenty and fifty. Those outcomes are retained. They are not relabelled as results of the new rollout-costate implementation. The current comparison changes policy evaluation and specifies sampled deployment; the new continuous-time transfer bound accounts for the latter. It does not isolate every source of the difference from R10, and it does not turn a method-specific experiment into a universal ranking of actor-critic and direct methods.

Checkpoint frontiers and extended-budget baselines make improvement per unit of work visible. A larger actor class has to earn its extra cost relative to the affine policy and the direct-policy method. These comparisons are retained even when a simpler method is competitive. All final policies, losses, failures, selected iterations, and peak memory records appear in the replication archive.

8.3. Economic normalization and stress

Let $A = \int_0^T e^{-\rho t} dt$. We report $100\{\exp(\Delta/A) - 1\}$ as a flow-consumption-equivalent normalization of a utility difference. It describes an externally financed

TABLE IV

CHECKPOINT FRONTIER IN DIMENSION FIFTY, SEED 11. EACH CHECKPOINT HAS 2,048 PATHS AND 512 CELLS; THE EXPLICIT TRANSFER BOUND IS RETAINED. A NEGATIVE LOWER ENDPOINT IS NOT REPLACED BY A PASS.

d	Method	Iteration	Mean gain	L_ϕ	Regret upper
50	Direct policy	20	0.000970	-0.001564	0.029103
50	Direct policy	40	0.001420	-0.001116	0.028655
50	Direct policy	80	0.001611	-0.000928	0.028467
50	Direct policy	120	0.001645	-0.000894	0.028434
50	Affine	20	0.000507	-0.002029	0.029568
50	Affine	40	0.000614	-0.001919	0.029459
50	Affine	80	0.000836	-0.001697	0.029236
50	Affine	120	0.001012	-0.001521	0.029061
50	NBO	20	0.001717	-0.000823	0.028362
50	NBO	40	0.001743	-0.000797	0.028337
50	NBO	80	0.001747	-0.000793	0.028332
50	NBO	120	0.001750	-0.000790	0.028330

log-flow supplement holding the reference resource allocation and other payoff terms fixed. It is not a self-financing change in consumption and is not an estimated welfare percentage for a calibrated population. A positive lower gain also bounds an admissible additional implementation charge in utility units. This interpretation makes both the magnitude of the learned benefit and its practical limits visible.

The validation-tuned critic-greedy comparisons, including the explicitly projected direct-full hybrid, appear in the supplement.

The supplement reports radii .05, .10, and .15; the first and third rescale a frozen fitted correction and are not represented as retraining at those radii. Widths 16 and 64 are separately trained. Initial-state tests include shifted means, greater dispersion, and a combined shift–dispersion stress. The exact initial vectors, action saturation, and fraction of internal states outside the training box are saved. Global feasibility alone is not evidence of useful extrapolation. The statistical statements remain pointwise at these declared states, not uniform over every state between them.

TABLE V

FROZEN-CORRECTION RADIUS COMPARISON, SEED 11. RADIUS .10 USES 8,192 PATHS; THE RESCALED RADII USE 4,096. ALL USE 1,024 CELLS AND THEIR OWN SIMULTANEOUS ERROR ACCOUNT.

d	Radius	Mean gain	L_ϕ	Regret upper
10	0.05	0.001431	0.000796	0.017951
10	0.10	0.001961	0.001185	0.017562
10	0.15	0.001649	-0.000482	0.019228
20	0.05	0.001173	0.000454	0.025071
20	0.10	0.001580	0.000718	0.024807
20	0.15	0.001274	-0.000972	0.026498
50	0.05	0.001268	0.000529	0.027011
50	0.10	0.001751	0.000868	0.026671
50	0.15	0.001488	-0.000781	0.028320

8.4. Connection to the other economic applications

Recursive utility, endogenous preference formation, temporal selves, and dynamic games remain applications of the NBO evaluation–improvement distinction, not consequences of the additive-noise capital theorem. Their full model equations, original calculations, counterexamples, tables, and proofs have been relocated without deletion to the supplement. The recursive problem requires its utility-domain and comparison conditions. The preference certificates refer to their specified nodal and continuous-action objects, with continuous-state, time, and stopping transfer still separately accounted for. The strategic results bound deviations in the saved finite dynamic games. The sophisticated-self problem uses its own equilibrium deviation condition.

This organization preserves the original research program while placing one complete computation and its new policy-specific guarantee at the centre of the article. Neither the broader title nor the retention of those applications enlarges the scope of Theorem 9. The final source records a byte-verified relocation map and exact pre-revision roots, so the change in exposition can be distinguished from a change in scientific content.

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